



CHAPTER 12

Flows and Harmonic Functions

12.1 Harmonic Duals

12.1.1 Dual Flows

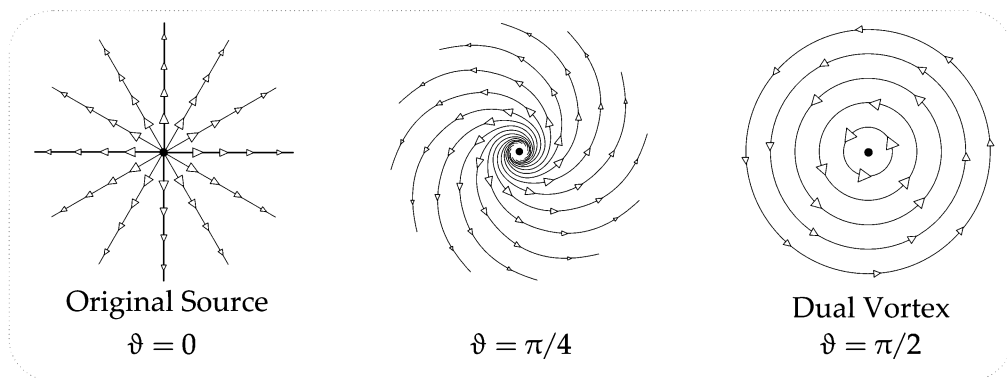
As in the previous chapter, let $\overline{H} = \overline{\Omega}'$ be a steady, sourceless, irrotational vector field with complex potential $\Omega = \Phi + i\Psi$. If at each point we rotate $\overline{H}$ through a fixed angle ϑ then we obtain the Pólya vector field of the analytic function $H_\vartheta \equiv e^{-i\vartheta} H$, namely, $\overline{H}_\vartheta = e^{i\vartheta} \overline{H}$. Thus this rotated vector field is automatically sourceless and irrotational, and its complex potential is $\Omega_\vartheta = e^{-i\vartheta} \Omega$. Writing $\Omega_\vartheta \equiv \Phi_\vartheta + i\Psi_\vartheta$, the potential and stream functions are therefore

$$\Phi_\vartheta = (\cos \vartheta) \Phi + (\sin \vartheta) \Psi \quad \text{and} \quad \Psi_\vartheta = (\cos \vartheta) \Psi - (\sin \vartheta) \Phi.$$

Henceforth we shall concentrate on the particularly simple and important case in which $\vartheta = +(\pi/2)$. After rotating $\overline{H}$ through this right angle we obtain the Pólya vector field of $H_{\pi/2}$, for which we shall use the special symbol $\widehat{H}$. Thus $\widehat{H} \equiv \overline{H}_{\pi/2} = i\overline{H}$. In complex analysis, the standard terminology is to say that $\widehat{H}$ is “conjugate” to the original flow $\overline{H}$. However, I know of no mathematical¹ connection between this concept and the familiar one of complex conjugation. Furthermore, our use of Pólya vector fields (involving genuine complex conjugation) brings these two senses of “conjugate” into direct conflict, for the complex conjugate of the original flow is *not* the “conjugate” flow.

Fortunately, in other areas of mathematics (e.g., topology) there is another term that is commonly used to describe this idea. We therefore propose to call $\widehat{H}$ the *dual* of $\overline{H}$. Similarly, let us call the potential and stream functions of the dual flow the *dual potential* and the *dual stream function*.

¹ Linguistically, the common origin of both terms is the Latin word “conjugatus”, meaning joined together.



[12.1] Evolution of a source into its dual vortex: As ϑ goes from 0 to $(\pi/2)$, the field $\overline{H}_\vartheta \equiv e^{i\vartheta}\overline{H} = e^{i\vartheta}(1/\overline{z})$ rotates from a pure source into its dual pure vortex, $\widehat{H} \equiv \overline{H}_{\pi/2} = i\overline{H} = (i/\overline{z})$.

Later we shall see that the concept of a dual flow is very useful. For example, having found the flow of a fluid round an obstacle, the dual flow represents the electric field which solves an analogous problem in electrostatics.

As interesting examples of dual flows, consider what happens in the vicinity of a singularity. Figure [12.1] illustrates how, as ϑ varies from 0 to $(\pi/2)$, a source gradually evolves into a dual vortex of equal strength. Note (cf. [11.12], p. 556) that the intermediate flow may also be viewed as a superposition of the original flow and its dual. Indeed, this is true quite generally:

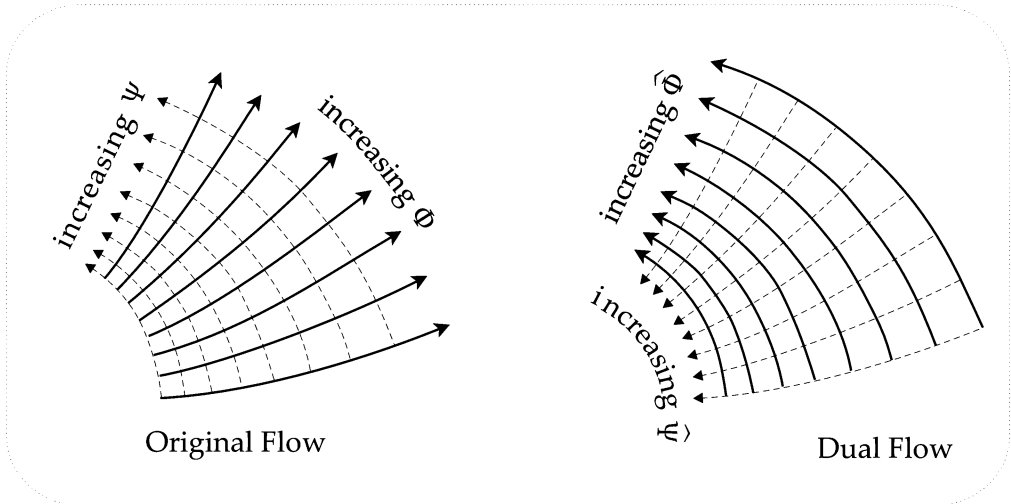
$$\overline{H}_\vartheta = (\cos \vartheta)\overline{H} + (\sin \vartheta)\widehat{H}.$$

Check for yourself that the type of qualitative change of flow exhibited in [12.1] does *not* occur in the case of higher multipoles. For example, the dual of a dipole is just another dipole. As ϑ varies from 0 to $(\pi/2)$, are all the intermediate flows dipoles, as well? See Ex. 10 of the previous chapter.

Observe that in passing from a flow to its dual the roles of the streamlines and equipotentials are interchanged: the streamlines of the dual flow are the equipotentials of the original, while the equipotentials of the dual flow are the streamlines of the original. Symbolically, this interchange of roles is manifested in the fact that the dual potential and stream functions are

$$\widehat{\Phi} = +\Psi \quad \text{and} \quad \widehat{\Psi} = -\Phi.$$

The difference of sign in these two equations is easily understood when we look at [12.2], which depicts a typical flow and its dual. [Streamlines are solid and equipotentials are dashed.] Recall that if we think of these pictures as force fields, work is done by the field when a particle moves along a line of force, so the original and dual potentials increase in the illustrated directions. Similarly, when thought of as a fluid flow, the flux across a directed segment of curve is positive when the



[12.2] In passing from a flow to its dual, the streamlines and equipotentials are interchanged.

fluid crosses it from left to right, so the original and dual stream functions increase in the illustrated directions. We now see clearly that $\hat{\Phi}$ and Ψ increase in the same direction, while $\hat{\Psi}$ and Φ increase in opposite directions.

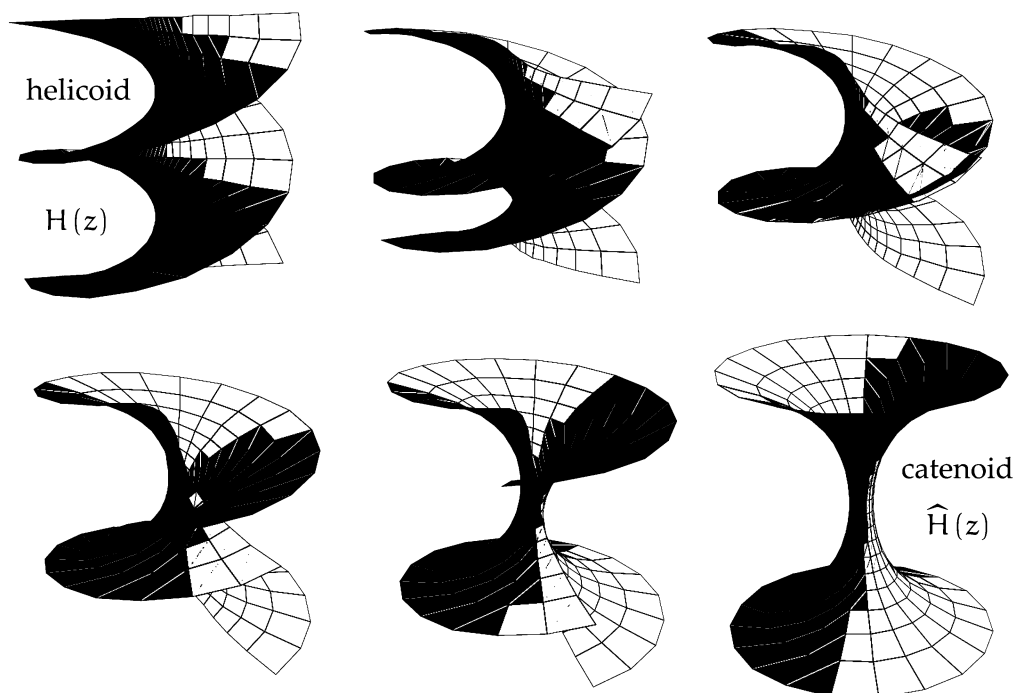
Given a complex potential $\Omega = \Phi + i\Psi$, we may thus think of Ψ as either the stream function, or as the dual of the potential function. Likewise, Φ may be thought of as either the potential function, or as minus the dual of the stream function. Since any analytic function $f = u + iv$ may be thought of as a complex potential, we may extend this language and say that v is dual to u , and that $-u$ is dual to v .

Finally, we cannot resist at least mentioning two miraculous connections between the above ideas and the study of soap films, also known as *minimal surfaces*, characterized by *vanishing mean curvature*.

Imagine yourself standing on the tangent plane to a saddle-shaped surface. In the case of a soap film, as you turn around, the average (or “mean”) curvature vanishes: the surface bends away from the tangent plane *equally and oppositely* as you turn around.

First miracle: Each complex analytic function $H(z)$ describes the shape of a minimal surface, and vice versa. **Second miracle:** Varying ϑ causes the minimal surface corresponding to $H_\vartheta(z)$ to undergo stretch-free bending: *all these minimal surfaces have identical intrinsic geometry*. For example, if H corresponds to the so-called *helicoid*, then $\hat{H}$ corresponds to the so-called *catenoid*, and [12.3] illustrates the stretch-free bending of one into the other, each intermediate surface itself being a minimal surface.

For an elementary introduction to the fascinating subject of minimal surfaces, see [Hildebrandt and Tromba \(1985\)](#); for the mathematical details, see [Nitsche \(1989\)](#).



[12.3] Miraculously, complex analytic functions correspond to minimal surfaces. *In particular, the helicoid and catenoid correspond to dual complex functions, and the evolution as ϑ goes from 0 to $(\pi/2)$ is isometric: the intrinsic geometry does not change throughout the evolution, and all the intermediary surfaces are minimal, too!*

NOTE added in this 25th Anniversary Edition: An especially good undergraduate introduction to this subject has now appeared in [Woodward and Bolton \(2019\)](#). The discussion includes the 1982 breakthrough discovery by Celso Costa of new minimal surfaces (beyond planes, catenoids, and helicoids) that are closed and without self-intersections. Furthermore, unusually, the authors provide a thorough discussion of the generalization to *surfaces of constant mean curvature* (CMC surfaces), in which the average curvature is the same at every point of the surface, but its value does *not* vanish (as it must do for soap films).

12.1.2 Harmonic Duals

We know that both the real and the imaginary parts of an analytic function are automatically harmonic. It is therefore natural to wonder if, conversely, every harmonic function is the real (or imaginary) part of some analytic function. As we shall see, this is indeed the case. That is, given a harmonic function u we can always find another harmonic function v , the *harmonic dual* of u , such that $f = u + iv$ is analytic. [Again, the standard terminology is that v is the “*harmonic conjugate*” of u .]

We make two remarks before proceeding. First, if v is an harmonic dual then so is $v + \text{const.}$, and consequently v will only be uniquely determined if we impose additional conditions, such as v vanishing at a particular point. Second, the harmonic dual of a single-valued function may itself be a multifunction. Witness the case $u = \ln |z|$, illustrated in [12.1], for which $v = \arg(z)$.

Given an irrotational vector field, we know how to construct a potential function. But, conversely, if we are given a real function $\Phi(z)$ then we may *construct* an irrotational vector field $\bar{H}$ for which Φ is the potential function, namely,

$$\bar{H} = \nabla \Phi.$$

If Φ is harmonic then we know [(11.30), p. 569] that $\bar{H}$ will be sourceless, and so it will possess a stream function Ψ . Since $\bar{H}$ is irrotational, Ψ is harmonic. The complex potential $\Omega = \Phi + i\Psi$ will then be an analytic function having as its real part the given harmonic function Φ . In other words we have shown that

The harmonic dual of a given harmonic function Φ is the stream function of the vector field $\nabla \Phi$.

Alternatively, Ψ is the potential function for the dual of $\nabla \Phi$.

This result means that facts about analytic functions can sometimes be recast as facts about harmonic functions. For example, in Chapter 9 we saw that if $f = u + iv$ is analytic then $\langle f \rangle = f(p)$, where $\langle f \rangle$ denotes the average of f over any circle centred at p . It follows that the harmonic real part of f obeys the law $\langle u \rangle = u(p)$. But we now know that if u is any *given* harmonic function then we may construct an analytic function for which it is the real part. We thus obtain *Gauss's Mean Value Theorem*:

The average value of a harmonic function on a circle is equal to the value of the function at the centre of the circle.

We obtain another example by reconsidering [7.14], p. 406, in which we saw that if $f = u + iv$ is analytic in some region whose boundary is Γ , then the maximum of u occurs on Γ . The existence of harmonic duals therefore implies that

If a function is harmonic in some region, its maximum occurs on the boundary of that region.

The same goes for a (nonzero) minimum of a harmonic function.

Next we give explicit formulae for the construction of harmonic duals. To make Ψ unique, let us demand that it vanish at some point a . Then if K is any path from a to p , we have the flux formula

$$\Psi(p) = \int_K (\nabla \Phi) \cdot \mathbf{N} \, ds.$$

Alternatively, in terms of complex integration, we have

$$\Psi(p) = \text{Im} \left[\int_K (\bar{\nabla} \Phi) \, dz \right].$$

As we have seen, if we restrict ourselves to a simply connected region throughout which Φ is harmonic, these integrals are single-valued. However, if the region is not simply connected, or if Φ has singularities, then (in general) Ψ will be a multifunction.

We illustrate these formulae with the example $\Phi = x^3 - 3xy^2$, which is easily seen to be harmonic. Choose $a = 0$, let $p = X + iY$, and choose K to be the line-segment between them, which may be represented parametrically as $z = x + iy = (X + iY)t$, where $0 \leq t \leq 1$. Since

$$\nabla\Phi = \begin{pmatrix} 3x^2 - 3y^2 \\ -6yx \end{pmatrix} = \begin{pmatrix} 3X^2 - 3Y^2 \\ -6YX \end{pmatrix} t^2, \quad \mathbf{N} = \frac{1}{\sqrt{X^2 + Y^2}} \begin{pmatrix} Y \\ -X \end{pmatrix},$$

and $ds = \sqrt{X^2 + Y^2} dt$, the first formula yields [exercise]

$$\Psi = 3X^2Y - Y^3.$$

Alternatively, since

$$\overline{\nabla\Phi} = (3x^2 - 3y^2) + i6xy = 3z^2,$$

the second formula yields

$$\Psi = \operatorname{Im} \left[\int_0^{X+iY} 3z^2 dz \right] = \operatorname{Im} (X + iY)^3 = 3X^2Y - Y^3.$$

The simplicity of the second method depended crucially on our ability to express $\overline{\nabla\Phi}(x, y)$ as a function $\Omega'(z)$ of z , but it is not always so obvious how to do this. However, there does exist a systematic method of doing this in the case where Φ is defined in a region containing a segment of the real axis.

Let $V(x)$ be the vector field evaluated on the real x -axis, i.e., $V(x) = \overline{\nabla\Phi}(x, 0)$. If $\Phi(x, y)$ is an explicit formula in terms of the familiar functions (powers, trigonometric, exponential) that possess complex analytic generalizations, then $V(x)$ is such a formula also. Hence if in the formula for $V(x)$ we now replace the symbol x with the complex variable z then we obtain an analytic function $V(z)$ which agrees with $\Omega'(z)$ when z is real. But, as we saw in Chapter 5, this implies that the two functions must continue to agree when z becomes *complex*.

Thus our recipe for finding $\Omega'(z)$ as an explicit formula in z is to calculate $\overline{\nabla\Phi}(x, y)$, set $y = 0$, then substitute z for x :

$$\Omega'(z) = \overline{\nabla\Phi}(z, 0). \quad (12.1)$$

For example, if $\Phi = \cos[\cos x \sinh y] e^{\sin x \cosh y}$ then [exercise]

$$\overline{\nabla\Phi}(x, y) = \cos[\cos x \sinh y] e^{\sin x \cosh y} \cos x \cosh y + F,$$

where F stands for three terms which vanish when $y = 0$. Using (12.1) we get $\Omega'(z) = e^{\sin z} \cos z$, and hence $\Psi = \operatorname{Im} e^{\sin z}$.

12.2 Conformal Invariance

12.2.1 Conformal Invariance of Harmonicity

Let $w = f(z)$ be a complex analytic function of z , which we will think of as a conformal mapping (rather than as a vector field) from the z -plane to the w -plane. Using f , any real function $\Phi(z)$ in the z -plane may be copied over (or “transplanted”) to a function $\tilde{\Phi}(w)$ in the w -plane by defining

$$\tilde{\Phi}[f(z)] \equiv \Phi(z). \quad (12.2)$$

In other words, corresponding points in the two planes are assigned equal function values. We will now show (first symbolically then geometrically) that

$$\text{Harmonicity is conformally invariant: } \tilde{\Phi}(w) \text{ is harmonic if and only if } \Phi(z) \text{ is harmonic.} \quad (12.3)$$

As before, think of $\tilde{\Phi}(w)$ as the potential of the vector field $\tilde{V} \equiv \nabla \tilde{\Phi}$. If and only if $\tilde{\Phi}$ is harmonic, $\tilde{V}$ possesses an analytic complex potential $\tilde{\Omega}(w) = \tilde{\Phi}(w) + i\tilde{\Psi}(w)$, where the stream function $\tilde{\Psi}$ is the harmonic dual of $\tilde{\Phi}$. Since f is analytic, so is its composition with $\tilde{\Omega}$:

$$\Omega(z) \equiv \tilde{\Omega}[f(z)] = \Phi(z) + i\Psi(z).$$

Thus $\Phi(z)$ is the real part of an analytic function, and so it is harmonic.

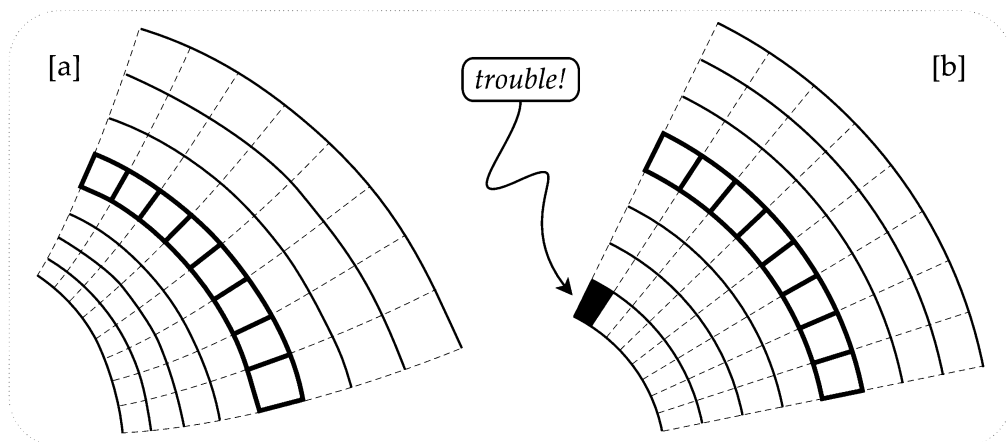
There is a very simple geometric idea behind this important result. Figure [12.4] illustrates a visual means of checking whether or not a given real function Φ is harmonic. Once again, think of Φ as the potential of the force field $V = \nabla \Phi$. We know that Φ is harmonic if and only if V admits a complex potential. This we know occurs if and only if the field may be divided into a grid of (infinitesimal) square k -cells.

To check this we should therefore construct a “test grid”:

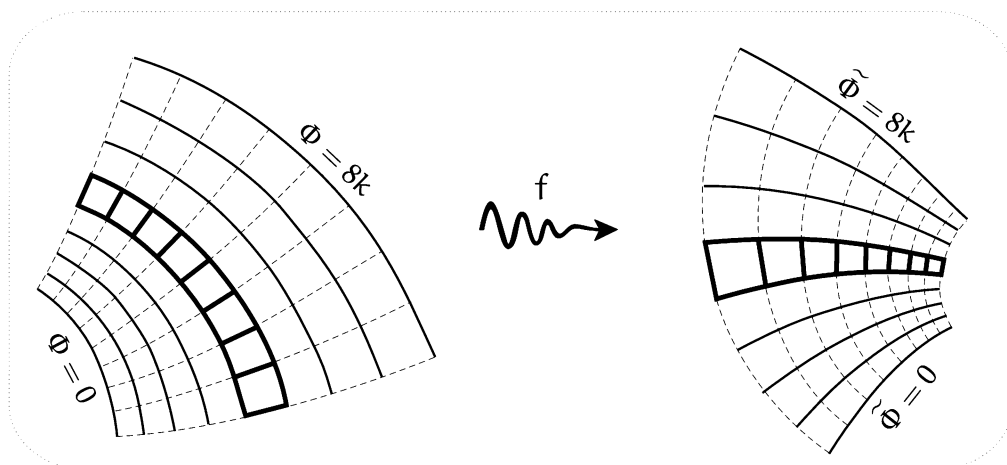
- (i) With a small value of k , draw the equipotentials $\Phi = 0, \pm k, \pm 2k, \pm 3k, \dots$
- (ii) Choose one of the resulting k -work tubes [shaded in the figure] and draw line-segments across it in such a way that the tube is divided into squares.
- (iii) Extend these line-segments into lines of force [dashed] of V , i.e., orthogonal trajectories through the equipotentials.

Then Φ is harmonic if and only if these lines of force divide each k -work tube into squares. Figure [12.4a] illustrates this test for a Φ that is harmonic, while [12.4b] illustrates it for one that is not. The result (12.3) can now be seen as nothing more than a statement of the conformal invariance of this geometric test. Let us spell this out.

Equation (12.2) defines the potential of each point in the z -plane to be the same as its image point (under f) in the w -plane. Thus f maps the k -work tubes of Φ to



[12.4] Geometric test for harmonicity of Φ . *STEP ONE:* With a small value of k , draw the equipotentials $\Phi = 0, \pm k, \pm 2k, \pm 3k, \dots$ *STEP TWO:* Divide a k -work tube [shaded] into squares. *STEP THREE:* Extend the edges of the squares into lines of force. Then Φ is harmonic if and only if these lines of force divide each k -work tube into squares. So [a] is harmonic, and [b] is not.



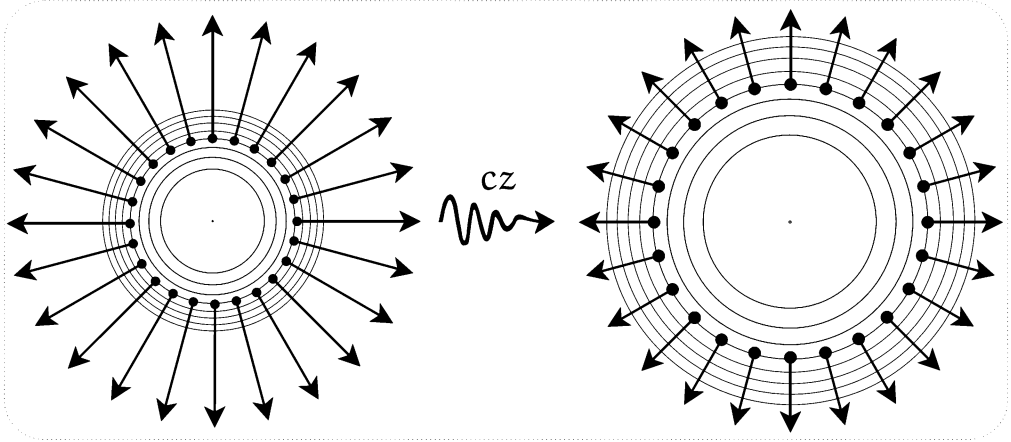
[12.5] Conformal invariance of the harmonicity test for Φ . If f is conformal, then the image test grid will pass the harmonicity test if and only if the original test grid passes the test. Here, the initial grid passes, so its image does, too.

the k -work tubes of $\tilde{\Phi}$. See [12.5]. Finally, since f is conformal, the constructed test grid for Φ will be composed of squares if and only if the image grid is composed of squares. Figure [12.5] illustrates the case where the potentials are harmonic.

12.2.2 Conformal Invariance of the Laplacian

The result (12.3) is merely a special case of the following more general result on the conformal invariance of the Laplacian operator Δ :

$$\Delta \tilde{\Phi}(w) = \frac{1}{|f'(z)|^2} \Delta \Phi(z). \quad (12.4)$$



[12.6] Geometric explanation of $\nabla \cdot \tilde{\mathbf{V}}(w) = [1/|f'(z)|^2] \nabla \cdot \mathbf{V}(z)$. Let us explain this for $f(z) = cz$; in the general case, we need only then replace c with the local amplitude, $f'(z)$. Compare the flux leaving the shaded disc on the left with the flux leaving the shaded image disc on the right. Since the separation of adjacent equipotentials is scaled up by $|c|$, the strength of the field on the rim of the image disc is scaled down by $|c|$, while the circumference of the rim is scaled up by $|c|$. The net effect is that the flux of $\tilde{\mathbf{V}}$ out of the image disc is the same as the flux of $\mathbf{V}$ out of the original disc. Finally, since the area of the disc is scaled up by $|c|^2$, the flux density is scaled down by $|c|^2$. Done.

We will give two explanations of this result.

For the first explanation, we rephrase the result in terms of flux densities. Just as we did for $\tilde{\Phi}$ in the w -plane, let us construct the vector field $\mathbf{V} \equiv \nabla \Phi$ in the z -plane. We wish to understand this:

$$\nabla \cdot \tilde{\mathbf{V}}(w) = \frac{1}{|f'(z)|^2} \nabla \cdot \mathbf{V}(z). \quad (12.5)$$

Now consider [12.6], which illustrates a toy model of the phenomenon. The potential $\Phi(z) = (S/4)|z|^2$ generates a vector field $\mathbf{V} = (S/2)z$ of uniform divergence $\nabla \cdot \mathbf{V} = S$. With a small value of k , the LHS of [12.6] shows the special equipotentials $\Phi = 0, \pm k, \pm 2k, \pm 3k, \dots$, for which the strength of the field is inversely proportional to the separation of the curves. Now apply the mapping $w = f(z) = cz$, which is a rotation and an expansion by $|c|$. By definition, these expanded circles are equal-valued equipotentials of $\tilde{\Phi}(w)$, so that

$$\tilde{\Phi}(w) = \Phi(z) = \frac{1}{4}S|z|^2 = \frac{1}{4}\frac{S}{|c|^2}|w|^2.$$

The field $\tilde{\mathbf{V}} = \nabla \tilde{\Phi}$ therefore has uniform flux density $\nabla \cdot \tilde{\mathbf{V}} = S/|c|^2$, proving (12.5) for this case.

More intuitively still, in [12.6] compare the flux leaving the shaded disc on the left with the flux leaving the shaded image disc on the right. Since the separation of adjacent equipotentials is scaled up by $|c|$, the strength of the field on the rim of

the image disc is scaled down by $|c|$, while the circumference of the rim is scaled up by $|c|$. The net effect is that the flux of $\tilde{V}$ out of the image disc is the same as the flux of V out of the original disc. Finally, since the area of the disc is scaled up by $|c|^2$, the flux *density* is scaled down by $|c|^2$.

To employ this idea in the general setting, it is only necessary to recognize that the *local* behaviour of a general potential is very similar to our toy potential, and that the *local* effect of a general analytic mapping f is very similar to our toy mapping, with $|f'|$ playing the role of $|c|$.

We will not spell this out completely because we will shortly be able to give a second explanation which is even simpler. However, according to (11.17), p. 553, the behaviour of V very near to z_0 is expressible as

$$V(z) = [\nabla \cdot \mathbf{V}(z_0)] \frac{(z - z_0)^2}{2} + Y(z),$$

where Y is sourceless and, of course, irrotational. Correspondingly, the local behaviour of the potential is²

$$\Phi(z) = \frac{1}{4} [\nabla \cdot \mathbf{V}(z_0)] r^2 + \Upsilon(z), \quad (12.6)$$

where Υ is harmonic, and $r = |z - z_0|$ is the small distance from z_0 . Having made explicit the connection with the toy model, we leave the remaining details to the interested reader.

12.2.3 The Meaning of the Laplacian

Given a real function $\Phi(z)$ in the z -plane, we have seen that its gradient vector field $\nabla \Phi$ is a geometric quantity, independent of the coordinates used to describe z . We have also seen that the divergence of a vector field measures its flux-density, so it too is geometrically defined. It follows that the Laplacian $\Delta \Phi = \nabla \cdot \nabla \Phi$ must possess a coordinate-independent interpretation.

In order to state this interpretation, recall that if C is a circle centred at p then $\langle \Phi \rangle$ denotes the average value of Φ on C . We will show that

The Laplacian of Φ at p measures the amount by which the average value of Φ on an infinitesimal circle centred at p exceeds the value of Φ at p itself. More precisely, if r is the infinitesimal radius of this circle then (12.7)

$$\langle \Phi \rangle - \Phi(p) = \frac{1}{4} r^2 \Delta \Phi.$$

Note that this result is in accord with Gauss's Mean Value Theorem, which says that if Φ is harmonic then $\langle \Phi \rangle - \Phi(p) = 0$ for circles of any size, not just infinitesimal ones. In fact if you have already convinced yourself of (12.6) then [exercise] you

² This may also be derived directly by taking the Taylor series for Φ and rewriting it in a rather unobvious way.

may derive (12.7) by using the fact that the harmonic function Υ obeys Gauss's Mean Value Theorem.

Before giving a more direct derivation of (12.7), let us return to Gauss's Mean Value Theorem itself and rederive it without appealing to complex analysis³. Let $\mathbf{V} = \nabla\Phi$ be the vector field of the potential function Φ . The flux of $\mathbf{V}$ out of a (non-infinitesimal) circle C of radius r is then [exercise]

$$\mathfrak{F}[\mathbf{V}, C] = 2\pi r \partial_r \langle \Phi \rangle.$$

Thus if Φ is harmonic, so that $\mathbf{V}$ is sourceless, then $\mathfrak{F} = 0$, by (11.7), p. 547. Since $\partial_r \langle \Phi \rangle = 0$, we see that $\langle \Phi \rangle$ is independent of the radius of C . Shrinking C down to p , we deduce that this radius-independent value must be $\Phi(p)$. Done.

Now suppose that $\mathbf{V}$ is not sourceless, but that its flux-density $\nabla \cdot \mathbf{V} = \Delta\Phi$ is constant. Gauss's Divergence Theorem [(11.7), p. 547] then yields $\mathfrak{F}[\mathbf{V}, C] = \pi r^2 \Delta\Phi$. Inserting this into the previous result, we find that

$$\partial_r \langle \Phi \rangle = \frac{1}{2} r \Delta\Phi,$$

which may be integrated to yield the formula in (12.7). To complete the explanation of (12.7) it is only necessary to observe that the Laplacian of an arbitrary Φ is constant within an infinitesimal circle.

Knowing the meaning of the Laplacian, it is a simple matter to understand its conformal invariance as expressed in (12.4). The analytic mapping f amplitwists an infinitesimal circle C centred at p to an infinitesimal circle $\tilde{C}$ centred at $\tilde{p}$, the new radius being $\tilde{r} = |f'(p)|r$. By definition, $\tilde{\Phi}(\tilde{p}) = \Phi(p)$. Likewise, the values of $\tilde{\Phi}$ at points of $\tilde{C}$ are the same as those of Φ at the preimages on C , so $\langle \tilde{\Phi} \rangle$ on $\tilde{C}$ equals $\langle \Phi \rangle$ on C . Thus (12.7) implies

$$r^2 \Delta\Phi(p) = \tilde{r}^2 \Delta\tilde{\Phi}(\tilde{p}) = |f'(p)|^2 r^2 \Delta\tilde{\Phi}(\tilde{p}),$$

from which (12.4) follows immediately.

12.3 A Powerful Computational Tool

A zealot might wish for an ideal world in which calculation would always be relegated to the confirmation of insights provided by geometry. Alas, even this author must confess to occasional lapses in which calculation has preceded understanding! We now describe a powerful computational tool which in many areas of complex analysis provides a considerable saving of labour. In the next section the study of the “complex curvature” [cf. Chapter 5] will provide a good showcase for its simplicity and elegance.

³ Previously we got it from $\langle f \rangle = f(p)$, which in turn came from Cauchy's formula.

The gradient operator ∇ of vector calculus acts on a real function $R(x, y)$ to produce the gradient vector field

$$\nabla R = \begin{pmatrix} \partial_x \\ \partial_y \end{pmatrix} R = \begin{pmatrix} \partial_x R \\ \partial_y R \end{pmatrix},$$

and we are free (as we have previously done) to think of this as a complex function

$$\nabla R = \partial_x R + i \partial_y R.$$

From this we may abstract the *complex gradient operator* ∇ , together with the conjugate operator $\bar{\nabla}$:

$$\nabla = \partial_x + i \partial_y \quad \text{and} \quad \bar{\nabla} = \partial_x - i \partial_y.$$

These two operators open the way to an exciting new method of calculation.

Given a vector field

$$\mathbf{f} = \begin{pmatrix} u \\ v \end{pmatrix},$$

we have seen how the real version of ∇ may be formally dotted or crossed with $\mathbf{f}$ to yield its divergence $\nabla \cdot \mathbf{f}$ or its curl $\nabla \times \mathbf{f}$. The interpretations of these quantities as flux and work densities shows them to be truly geometric, that is to say, coordinate-independent. However, there would seem to be no natural way of applying ∇ directly to $\mathbf{f}$ to obtain a new vector field $\nabla \mathbf{f}$. However, if we replace ∇ by its complex version ∇ , and replace the vector field $\mathbf{f}$ by the complex function $f = u + iv$, then there *is* a natural definition:

$$\nabla f = (\partial_x + i \partial_y)(u + iv) = (\partial_x u - \partial_y v) + i(\partial_x v + \partial_y u).$$

The equivalent expression

$$\nabla f = \nabla u + i \nabla v = \nabla u + (\nabla v \text{ rotated } \frac{\pi}{2})$$

helps to see that ∇f is geometrically meaningful (because ∇u and ∇v both are).

The power of the complex gradient derives from the following fundamental result [exercise]:

A complex function f is analytic if and only if $\nabla f = 0$, in which case we also have $\bar{\nabla} f = 2f'$.

You may easily verify the following useful properties of ∇ :

- $\nabla(f + g) = \nabla f + \nabla g$.
- $\nabla(fg) = f \nabla g + g \nabla f$.
- If f is analytic then $\nabla f[g(z)] = f'[g(z)] \nabla g$. For example,
 $\nabla e^{g(z)} = e^{g(z)} \nabla g$.

- The concepts of divergence and curl are neatly subsumed by ∇ :

$$\bar{\nabla} f = \nabla \cdot f + i \nabla \times f.$$

Similarly,

$$\nabla f = \nabla \cdot \bar{f} - i \nabla \times \bar{f},$$

which shows, once again, that a vector field is sourceless and irrotational if and only if it is the Pólya vector field of an analytic function.

- The Laplacian operator Δ can be expressed neatly as $\nabla \bar{\nabla} = \Delta = \bar{\nabla} \nabla$.

In the next section, and in the exercises at the end of the chapter, you will see the strength of the new technique. You may also find that exercises from previous chapters are solved more readily by this method. For the moment, here are just two examples of the use of the complex gradient.

The first is simply to observe how neatly the theorems of Gauss and Stokes [p. 547] may be combined into a single complex result: If C is the boundary curve of a simply connected region R then

$$\oint_C f dz = i \iint_R \nabla f dA.$$

If f is analytic ($\nabla f = 0$) we immediately obtain Cauchy's Theorem. This is not a new explanation, of course, merely a mathematically streamlined version of our previous physical one.

Our second example is another derivation of the result (12.4). Let $z = x + iy$ and $w = u + iv$, so that the complex gradient operators in the two planes are $\nabla_z = \partial_x + i \partial_y$ and $\nabla_w = \partial_u + i \partial_v$. The result we wish to prove can then be expressed (less ambiguously than before) as

$$\nabla_z \bar{\nabla}_z \tilde{\Phi}[f(z)] = |f'|^2 \nabla_w \bar{\nabla}_w \tilde{\Phi}(w).$$

Since $w = f(z)$, a straightforward application of the chain rule yields [exercise]

$$\nabla_z = \bar{f}' \nabla_w \quad \text{and} \quad \bar{\nabla}_z = f' \bar{\nabla}_w. \quad (12.8)$$

For example,

$$\bar{\nabla}_z f(z) = f' \bar{\nabla}_w w = f' (\partial_u - i \partial_v)(u + iv) = 2f',$$

as it should. Returning to the problem, we easily obtain

$$\begin{aligned} \nabla_z \bar{\nabla}_z \tilde{\Phi}[f(z)] &= \nabla_z \left\{ f' \bar{\nabla}_w \tilde{\Phi}(w) \right\} \\ &= (\nabla_z f') \bar{\nabla}_w \tilde{\Phi}(w) + f' \nabla_z \bar{\nabla}_w \tilde{\Phi}(w) \\ &= |f'|^2 \nabla_w \bar{\nabla}_w \tilde{\Phi}(w). \end{aligned}$$

12.4 The Complex Curvature Revisited*

12.4.1 Some Geometry of Harmonic Equipotentials

Given a real function $\Phi(x, y)$, figure [12.4] provided a geometric test for harmonicity. However, the first step of the test is not purely geometric in that it uses the *values* of Φ , not just the geometry of the curves $\Phi = \text{const.}$ This leads us to pose the following more subtle problem. *Given a family of curves $\mathcal{E}$ filling a region of the plane, how may we decide whether or not there exists a harmonic function Φ such that $\mathcal{E}$ is its family of equipotentials?*

For example, let $\mathcal{E}$ be the set of origin-centred circles. If we assign potentials to these curves according to the rule $\Phi(z) = |z|$ then our previous test yields figure [12.4b], which shows that this potential is not harmonic. But the question we are now asking is whether these same curves can be assigned potentials according to a different rule that *is* harmonic? In fact they can: let $\Phi(z) = \ln |z|$.

When the family does admit a harmonic assignment of potential in this way, we shall simply extend our use of the word “harmonic” to the family of curves itself. Thus we would say that the family of origin-centred circles is *harmonic*. The opening question may then be rephrased succinctly:

What geometric property of $\mathcal{E}$ determines whether it is harmonic?

One way to answer this question is to generalize the test in [12.4] to the one in [12.7]:

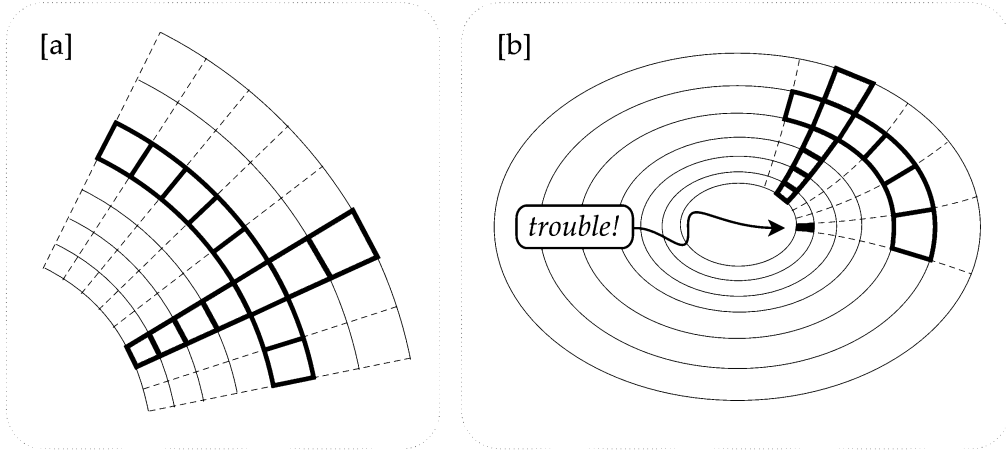
- (i) Choose two members of $\mathcal{E}$ that are very close together.
- (ii) Draw line-segments across the region between them [shaded] so as to divide it into squares.
- (iii) Extend these line-segments into streamlines [dashed], i.e., orthogonal trajectories through $\mathcal{E}$.
- (iv) Choose one of the resulting flux tubes [darkly shaded] and draw line-segments across it so as to divide it into squares.
- (v) Extend these line-segments into members of $\mathcal{E}$.

Then $\mathcal{E}$ is harmonic if and only if the resulting grid is composed of squares.

We already know that a family of concentric circles is harmonic, and [12.7a] shows how symmetry guarantees that it passes the test. Figure [12.7b] shows that a family of similar, concentric ellipses is not harmonic.

12.4.2 The Curvature of Harmonic Equipotentials

We now turn to a second, more elegant answer to our question. Let $\mathcal{S}$ be the family of curves (streamlines) orthogonal to $\mathcal{E}$. When one looks at [12.7] one gets the



[12.7] Geometric harmonicity test for a family of curves, $\mathcal{E}$. *STEP ONE:* Choose two members of $\mathcal{E}$ that are very close together. *STEP TWO:* Draw line-segments across the region between them [shaded] so as to divide it into squares. *STEP THREE:* Extend these line-segments into streamlines [dashed], i.e., orthogonal trajectories through $\mathcal{E}$. *STEP FOUR:* Choose one of the resulting flux tubes [darkly shaded] and draw line-segments across it so as to divide it into squares. *STEP FIVE:* Extend these line-segments into members of $\mathcal{E}$. Then $\mathcal{E}$ is harmonic if and only if the resulting grid is composed of squares. **[a]** Concentric circles are harmonic. **[b]** Concentric ellipses are not harmonic.

feeling that in order for a grid of squares to form, the bending of the curves of $\mathcal{E}$ must be connected in some special way with the bending of the curves of $\mathcal{S}$. By examining the curvatures of the two types of curve we shall see that this is indeed the case.

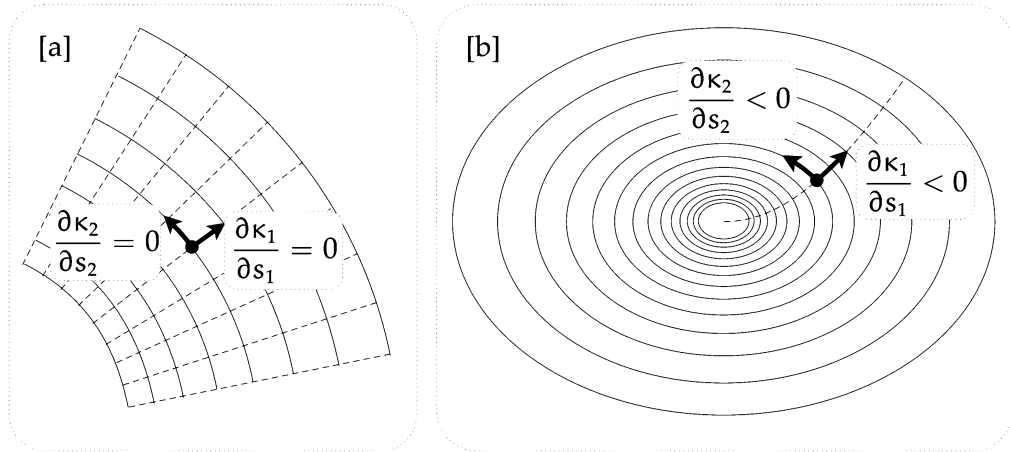
First let us attach directions to the curves of $\mathcal{E}$ and $\mathcal{S}$, so that their curvatures will have well-defined signs. Choose the direction for $\mathcal{S}$ arbitrarily, but then define the direction of $\mathcal{E}$ to be that of a tangent vector to $\mathcal{S}$ rotated through a *positive* right angle. Through any given point w_0 there passes one member $\mathcal{C}_1$ of $\mathcal{S}$ and one member $\mathcal{C}_2$ of $\mathcal{E}$. Let the curvatures of $\mathcal{C}_1$ and $\mathcal{C}_2$ be κ_1 and κ_2 , and let s_1 and s_2 be the arc lengths along $\mathcal{C}_1$ and $\mathcal{C}_2$. We then have the following striking result:

$\mathcal{E}$ is harmonic if and only if

$$\frac{\partial \kappa_1}{\partial s_1} + \frac{\partial \kappa_2}{\partial s_2} = 0. \quad (12.9)$$

In other words, if the rates of change of the curvatures along the two types of curve are exactly equal and opposite. Note that both these rates of change are well-defined even in the absence of a choice of direction for the curves, for reversing such a choice changes the sign of both κ and ∂s .

Figure [12.8] illustrates the new test for the concentric circles and ellipses we considered in [12.7]. Since the circles and lines of [12.8a] have constant curvature,



[12.8] Geometric curvature test for harmonicity: $\mathcal{E}$ is harmonic if and only if $\frac{\partial \kappa_1}{\partial s_1} + \frac{\partial \kappa_2}{\partial s_2} = 0$. [a] Concentric circles are harmonic. [b] Concentric ellipses are not harmonic.

(12.9) is trivially satisfied. In [12.8b] it is clear that both curvatures are decreasing at the indicated point, so (12.9) is violated there.

The result (12.9) seems to have been first published by Bivens (1992). We had also hit upon the same result, but as a natural consequence of investigating our *complex curvature* concept, which we introduced in Chapter 5. Here is the pertinent result:

The complex curvature vector field of an analytic mapping f is automatically sourceless, and its stream function is $\Psi = 1/|f'|$. (12.10)

To see the connection between the two results, first observe that a family $\mathcal{E}$ in the w -plane is harmonic if and only if it is the image set of vertical lines in the z -plane under an analytic mapping $w = f(z)$. For if $\mathcal{E}$ is harmonic then its potential function $\Phi(w)$ is the real part of a complex potential $z = \Omega(w)$ which maps $\mathcal{E}$ to vertical lines, and so $f(z) \equiv \Omega^{-1}(z)$ has the required property. Conversely, if $\mathcal{E}$ is the image set of vertical lines under an analytic f then it is harmonic, indeed its harmonic potential is the real part of the complex potential $\Omega(w) \equiv f^{-1}(w)$.

Thus if $\mathcal{E}$ is harmonic then the curves $\mathcal{C}_1$ and $\mathcal{C}_2$ through $w_0 = f(z_0)$ are the images under f of the horizontal and vertical lines through z_0 . But referring to [5.20], p. 272, we see that curvatures occurring in (12.9) are simply the real and imaginary parts of the complex curvature of f at z_0 :

$$\mathcal{K}(z_0) = \kappa_1 + i\kappa_2.$$

Since infinitesimal horizontal and vertical movements dx and dy are amplified by $|f'(z_0)|$ to movements ds_1 and ds_2 along $\mathcal{C}_1$ and $\mathcal{C}_2$, the flux-density of the complex

curvature is

$$\nabla \cdot \mathcal{K} = \frac{\partial \kappa_1}{\partial x} + \frac{\partial \kappa_2}{\partial y} = |f'(z_0)| \left[\frac{\partial \kappa_1}{\partial s_1} + \frac{\partial \kappa_2}{\partial s_2} \right].$$

The result (12.10) therefore implies that equation (12.9) is a necessary condition for $\mathcal{E}$ to be harmonic; the question of sufficiency will be addressed shortly.

To prove (12.10) we will use the complex gradient technique of the previous section; later we will give a proper geometric explanation. We must show that $\Psi = 1/|f'|$ is the stream function for the complex curvature

$$\mathcal{K} = \frac{i \bar{f}''}{\bar{f}' |f'|},$$

in other words [cf. (11.24), p. 566], $\mathcal{K} = -i \nabla \Psi$.

Since

$$-i \nabla \Psi = -i \nabla \left(\frac{1}{|f'|} \right) = \frac{i}{|f'|^2} \nabla |f'|,$$

we need to know $\nabla |f'|$. Because f is analytic, so is f' , and this implies $\nabla f' = 0$ and $\bar{\nabla} f' = 2f''$. Hence

$$\begin{aligned} 2|f'| \nabla |f'| &= \nabla |f'|^2 = \nabla (f' \bar{f}') = f' \nabla \bar{f}' + \bar{f}' \nabla f' = 2f' \bar{f}'' \\ \implies \nabla |f'| &= \frac{f' \bar{f}''}{|f'|}. \end{aligned}$$

Substituting this into the previous equation, we obtain the desired result:

$$-i \nabla \Psi = \frac{i f' \bar{f}''}{|f'|^3} = \frac{i \bar{f}''}{\bar{f}' |f'|} = \mathcal{K}.$$

The fact that harmonicity implies (12.9) can also be understood without appealing to ideas from complex analysis. Let $\mathcal{S}$ and $\mathcal{E}$ be the streamlines and orthogonal trajectories of a vector field $\mathbf{X}$. With the present notation, the results (11.5) and (11.6) from the previous chapter [see p. 544] become

$$\nabla \cdot \mathbf{X} = \frac{\partial |\mathbf{X}|}{\partial s_1} + \kappa_2 |\mathbf{X}| \quad \text{and} \quad \nabla \times \mathbf{X} = -\frac{\partial |\mathbf{X}|}{\partial s_2} + \kappa_1 |\mathbf{X}|.$$

In order for $\mathcal{E}$ (or $\mathcal{S}$) to be harmonic, $\mathbf{X}$ must be divergence-free and curl-free, so

$$\kappa_1 = \frac{\partial}{\partial s_2} \ln |\mathbf{X}| \quad \text{and} \quad \kappa_2 = -\frac{\partial}{\partial s_1} \ln |\mathbf{X}|,$$

from which (12.9) immediately follows.

We conclude this section by establishing the converse result that equation (12.9) is a sufficient condition for $\mathcal{E}$ to be harmonic. In the w -plane, let $\Theta(w)$ be the angle that the curve $\mathcal{C}_1$ through w makes with the horizontal. The angle of the curve $\mathcal{C}_2$ through w is therefore $\Theta + (\pi/2)$. Since the curvatures of $\mathcal{C}_1$ and $\mathcal{C}_2$ are the rates of

change of these two angles with respect to the distances along the curves, we then have

$$\kappa_1 = \frac{\partial \Theta}{\partial s_1} \quad \text{and} \quad \kappa_2 = \frac{\partial \Theta}{\partial s_2}.$$

Next we calculate the Laplacian of Θ ; the reason will be clear in a moment. Because the Laplacian is coordinate-independent, we may choose our coordinate directions tangent to $\mathcal{C}_1$ and $\mathcal{C}_2$, obtaining

$$\Delta \Theta = \frac{\partial}{\partial s_1} \left(\frac{\partial \Theta}{\partial s_1} \right) + \frac{\partial}{\partial s_2} \left(\frac{\partial \Theta}{\partial s_2} \right) = \frac{\partial \kappa_1}{\partial s_1} + \frac{\partial \kappa_2}{\partial s_2}.$$

Thus equation (12.9) implies that $\Theta(w)$ is harmonic, in which case it is the real part of an analytic function, say, $G(w)$. We may now define an analytic function $H(w) = e^{-iG} \propto e^{-i\Theta}$ such that $\bar{H} \propto e^{i\Theta}$ is everywhere tangent to $\mathcal{S}$. Thus $\mathcal{S}$ and $\mathcal{E}$ are the streamlines and equipotentials of the Pólya vector field of an analytic function. Done.

12.4.3 Further Complex Curvature Calculations

Suppose once again that $\mathcal{S}$ and $\mathcal{E}$ are the images in the w -plane of horizontal and vertical lines in the z -plane under an analytic mapping $w = f(z)$. They are then the streamlines and equipotentials of the vector field $\overline{\Omega}'$ in the w -plane, where Ω is the complex potential $z = \Omega(w) = f^{-1}(w)$ mapping $\mathcal{S}$ and $\mathcal{E}$ in the w -plane back to horizontal and vertical lines in the z -plane.

At present, the curvatures κ_1 and κ_2 at a point $w_0 = f(z_0)$ in the w -plane are represented as the components of the complex curvature of f at z_0 in the z -plane: $\mathcal{K}_f(z_0) = \kappa_1 + i\kappa_2$. [We have added the subscript f because we will shortly be considering the complex curvature of more than one mapping.] But suppose we think of the complex potential $\Omega(w)$ as fundamental, not merely the inverse of f ; how can we express the curvatures directly in terms of Ω ?

We shall answer this question by deriving a remarkably simple relationship between the complex curvature $\mathcal{K}_f(z)$ of f at z , and the complex curvature $\mathcal{K}_\Omega(w)$ of Ω at the image point w . Since $f'(z) = 1/\Omega'(w)$, equation (12.8) yields

$$\begin{aligned} \mathcal{K}_f(z) &= -i \nabla_z (1/|f'|) \\ &= -\frac{i}{\Omega'} \nabla_w |\Omega'| \\ &= -\frac{|\Omega'|^2}{\Omega'} [-i \nabla_w (1/|\Omega'|)] \\ &= -\Omega'(w) \mathcal{K}_\Omega(w). \end{aligned} \tag{12.11}$$

This is interesting. Recall that an infinitesimal complex number ϵ emanating from z is amplitwisted to yield the image complex number $f'(z) \epsilon$ emanating from

w. Thus, since (12.11) may also be written as

$$\mathcal{K}_\Omega(w) = -f'(z) \mathcal{K}_f(z), \quad (12.12)$$

we see that transforming $\mathcal{K}_f(z)$ as if it were an infinitesimal vector yields an image vector at w which is simply the negative of $\mathcal{K}_\Omega(w)$.

In the next section we shall shed some geometric light on this result, but for the moment (12.11) yields the desired formula (also known to Bivens (1992)) for the curvature of the streamlines and equipotentials of a sourceless, irrotational vector field in terms of its complex potential:

$$\kappa_1 + i\kappa_2 = -i \frac{|\Omega'| \overline{\Omega''}}{(\Omega')^2}.$$

Next we turn to the curl of $\mathcal{K}$. The complex curvature is the Pólya vector field of the function $(-if''/f'|f'|)$, and the presence of $|f'|$ in the denominator prevents this from being analytic. Thus while $\mathcal{K}$ has been shown to be divergence-free, it cannot also be curl-free. What then is its curl?

To find it, recall that

$$\overline{\nabla} \mathcal{K} = \nabla \cdot \mathcal{K} + i \nabla \times \mathcal{K}.$$

Since $\overline{\nabla} f' = 0 = \overline{\nabla} f''$,

$$\overline{\nabla} \mathcal{K} = \overline{\nabla} \left\{ \frac{i \overline{f''}}{\overline{f'} |f'|} \right\} = \frac{\overline{f''}}{\overline{f'}} \left\{ i \overline{\nabla} \left(\frac{1}{|f'|} \right) \right\} = \frac{\overline{f''}}{\overline{f'}} \{ \mathcal{K} \} = -i \frac{|f''|^2}{|f'|^3}.$$

Thus $\nabla \cdot \mathcal{K} = 0$, which we already knew, and

$$\nabla \times \mathcal{K} = -\frac{|f''|^2}{|f'|^3}.$$

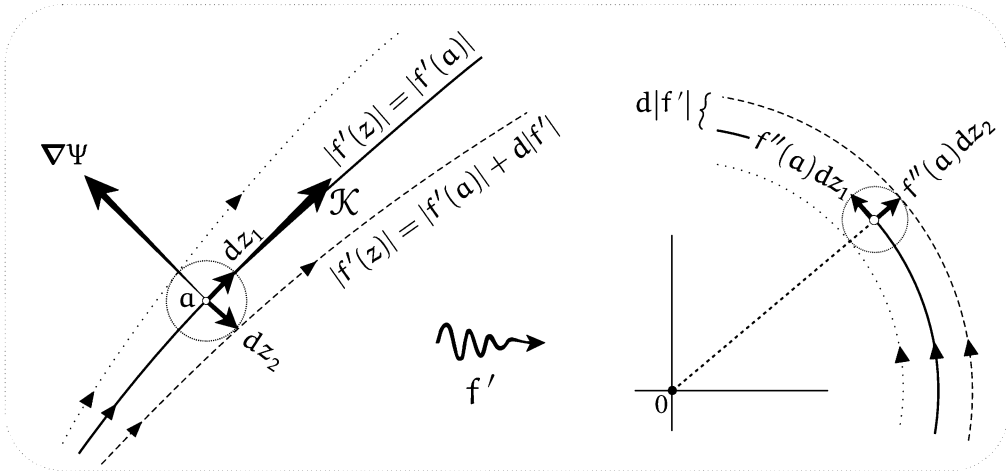
Although we have just differentiated $\mathcal{K}$, note that the result does not depend on any higher derivatives of f than occur in $\mathcal{K}$ itself. Indeed, the result may be re-expressed as

$$\nabla \times \mathcal{K} = -|f'| |\mathcal{K}|^2. \quad (12.13)$$

Since $\mathcal{K}$ is geometrically defined by f , and since the curl operator is also geometric, the curl of $\mathcal{K}$ must encode some (presumably simple) geometric information about the mapping f . Unfortunately, we have not yet succeeded in decoding this information.

12.4.4 Further Geometry of the Complex Curvature

The above results were derived by pure calculation in order to illustrate the complex gradient technique. We now revert to form and seek more geometrical explanations, beginning with (12.10).



[12.9] Geometric proof that the complex curvature $\mathcal{K}$ has stream function $\Psi = 1/|f'|$. We know from (5.32) that the streamlines of the complex curvature are the level curves of the amplification, $|f'|$. It follows readily that $-i\nabla\Psi$ points in the direction of $\mathcal{K}$. Next, observe that under the mapping f' , the movement dz_1 along $\mathcal{K}$ is mapped to a movement tangent to the circle $f' = \text{const.}$, and therefore dz_2 is mapped to a radial movement of length $d|f'| = |f''(a)| dz_2$. Finally, $|\nabla\Psi| = -\frac{d\Psi}{|dz_2|} = \frac{1}{|f'|^2} \frac{d|f'|}{|dz_2|} = \frac{|f''|}{|f'|^2} = |\mathcal{K}|$. Done.

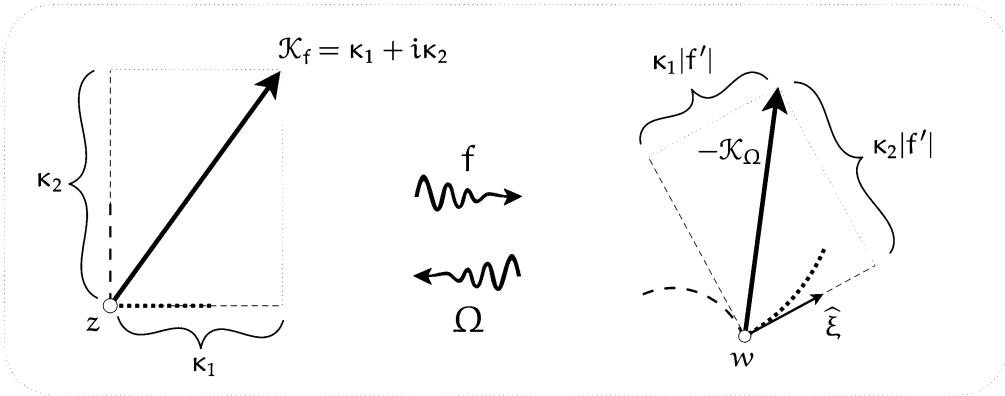
The geometrically derived result (5.32), p. 273, says, in part, that *the streamlines of the complex curvature are the level curves of the amplification, $|f'|$* . Figure [12.9] illustrates this: sections of adjacent streamlines are mapped by f' to arcs of origin-centred circles. It also shows how the infinitesimal complex number dz_1 along the streamline through the point a is amplitwisted by $f''(a)$ to a complex number at $f'(a)$ that points counterclockwise along the circle $|f'(z)| = |f'(a)| = \text{const.}$ Correspondingly, the orthogonal number $dz_2 = -i dz_1$ is amplitwisted to a complex number at $f'(a)$ that points radially outwards.

In order for $\mathcal{K}$ to be sourceless it must be of the form $\mathcal{K} = -i\nabla\Psi$, so that $\nabla\Psi$ (the direction of maximum increase of Ψ) must be directed as shown, and Ψ must be a function of $|f'|$. Since $|f'|$ increases in the direction dz_2 , while $\nabla\Psi$ is in the opposite direction, Ψ must be a decreasing function of $|f'|$. Thus if Ψ is any decreasing function of $|f'|$ then $-i\nabla\Psi$ will be a sourceless vector field in the direction of $\mathcal{K}$. It only remains to show that for the particular function $\Psi = 1/|f'|$, the *magnitudes* also agree, i.e., $|\nabla\Psi| = |\mathcal{K}|$.

Let $d|f'|$ and $d\Psi$ be the changes in $|f'|$ and Ψ that result from the movement dz_2 . From the picture we see that $d|f'| = |f''(a)| dz_2$, so that with $\Psi = 1/|f'|$,

$$|\nabla\Psi| = -\frac{d\Psi}{|dz_2|} = \frac{1}{|f'|^2} \frac{d|f'|}{|dz_2|} = \frac{|f''|}{|f'|^2} = |\mathcal{K}|,$$

as was to be shown.



[12.10] Geometric proof that $-\mathcal{K}_\Omega(w) = f'(z) \mathcal{K}_f(z)$. The horizontal line-segment at z is mapped by f to a segment of curvature κ_1 , with unit tangent $\hat{\xi}$. The inverse mapping Ω unbends this piece of curve, so, by the general transformation law of curvature, (5.31), $\mathcal{K}_\Omega \cdot \hat{\xi} + \kappa_1/|\Omega'| = 0$. Therefore, the component of $-\mathcal{K}_\Omega$ in the direction of $\hat{\xi}$ is $\kappa_1/|\Omega'| = \kappa_1|f'|$. Likewise, the orthogonal component is $\kappa_2|f'|$. Thus $-\mathcal{K}_\Omega$ is obtained by expanding $\mathcal{K}_f$ by $|f'|$, and rotating it by $\arg(\hat{\xi})$. But $\arg(\hat{\xi}) = \arg(f')$, so $-\mathcal{K}_\Omega(w) = f'(z) \mathcal{K}_f(z)$. Done.

Next we give a more geometric derivation of (12.12). Figure [12.10] shows a horizontal line-segment at z being mapped by f to a segment of curve at w whose curvature is κ_1 and whose unit tangent is $\hat{\xi}$. The inverse mapping Ω unbends this piece of curve and sends it back to the straight line-segment at z . Hence, by the general transformation law of curvature, (5.31), page 272,

$$\mathcal{K}_\Omega \cdot \hat{\xi} + \kappa_1/|\Omega'| = 0.$$

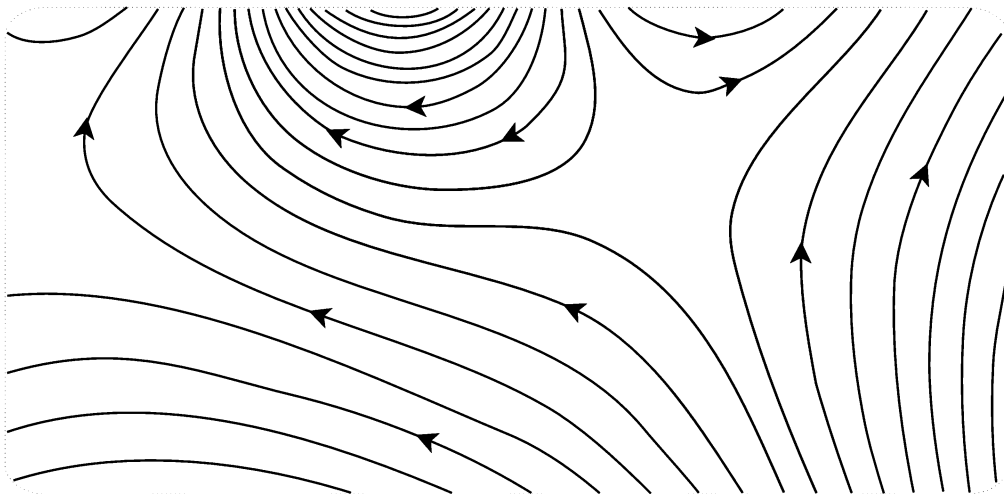
In other words, as illustrated, the component of $-\mathcal{K}_\Omega$ in the direction of $\hat{\xi}$ is $\kappa_1/|\Omega'| = \kappa_1|f'|$. Likewise, the orthogonal component is $\kappa_2|f'|$.

We now see that $-\mathcal{K}_\Omega$ is obtained by expanding $\mathcal{K}_f$ by $|f'|$, and rotating it by $\arg(\hat{\xi})$. But since an infinitesimal real number ϵ at z is amplitwisted to a complex number $\epsilon f'(z)$ at w which points in the direction of $\hat{\xi}$, we see that $\arg(\hat{\xi}) = \arg(f')$. Hence

$$-\mathcal{K}_\Omega(w) = f'(z) \mathcal{K}_f(z),$$

as was to be shown.

It is also possible to give a more geometric derivation of the result (12.13). However, we shall not bother to do this since we have not yet been able to establish the significance of that result.



[12.11] Finding the flow around a given obstacle, B. Given a frictionless flow, imagine that we instantly freeze the shaded region between streamlines, turning it into an obstacle, B, around which the flow continues, undisturbed. The image of the boundary under the conformal complex potential is a horizontal line, and, conversely, the problem of finding the flow around a given B therefore amounts to finding a conformal mapping Ω with this property.

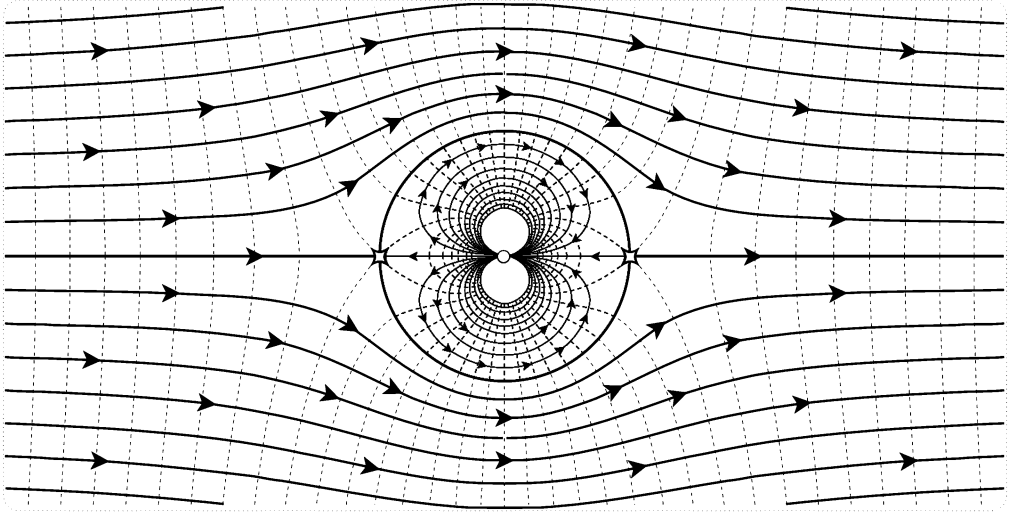
12.5 Flow Around an Obstacle

12.5.1 Introduction

Consider a typical fluid flow such as [12.11]. From our assumption that the fluid has no viscosity [this is most certainly an idealization] it follows that if we were to suddenly freeze the fluid within a flux tube, such as the shaded region in [12.11], then the unfrozen fluid would continue to flow in exactly the same way as before. The same idea applies if [12.11] instead represents flowing heat: if the shaded region of the metal plate were suddenly replaced with material which did not conduct heat then there would be no disturbance to the flow of heat in the remainder of the plate. This is much less of an idealization than in the case of fluid flow.

Conversely, if we insert an obstacle into a flow then the new disturbed flow must be such that the boundary B of this obstacle is a streamline, or is made up of segments of streamlines. If we think of the complex potential Ω of this disturbed flow as a mapping, this means that Ω maps B to a horizontal line, or to segments of a horizontal line.

The problem of finding flows around a given B therefore amounts to finding conformal mappings Ω with this property. In fact, since the complex potential of a given flow is only defined up to a constant, we may further demand that the horizontal image line be the real axis. Alternatively, this characterization may be



[12.12] The flow around a disc can be realized as the previously considered superposition of a uniform flow with a dipole, with complex potential $\Omega(z) = z + (1/z)$.

restated in terms of harmonic functions: we seek harmonic stream functions Ψ which vanish on B . There are of course infinitely many different flows which satisfy this requirement; we only obtain a unique solution if we add other requirements, such as demanding that the flow become uniform far from the obstacle. Indeed, the superposition of any two flows round B yields a third such flow.

12.5.2 An Example

As our first example of a flow in the presence of a barrier, reconsider the case of $\Omega(z) = z + (1/z)$. See [12.12]. As we previously discussed, this represents a uniform flow to the right into which has been inserted a dipole at the origin. On the other hand it certainly appears that the picture can be interpreted in a second way, as the flow around a circular obstacle (the shaded unit disc) inserted into a uniform flow. [Later we shall see that it is no accident that the flow may be interpreted in these two ways.] That both the top and bottom halves of the unit circle are indeed segments of the streamline $\Psi = 0$ is easily seen from the fact that Ω maps them to the segment $-2 \leq x \leq 2$ of the real axis:

$$\Omega(e^{i\theta}) = e^{i\theta} + e^{-i\theta} = 2 \cos \theta.$$

Note the breakdown of the grid at ± 1 and check that this corresponds to the fact that these are the stagnation points of the flow.

While this is a *possible* flow round C that is uniform at infinity, it is not the only such flow. First let us discuss this fact in terms of the flow round an obstacle of arbitrary shape.

For any flow round an obstacle, the boundary curve B is composed of streamlines, so B has zero flux. If the obstacle has been inserted into a uniform flow then there are no singularities outside B (except at infinity), and it follows from the deformation theorem for flux [(11.9), p. 547] that if B is deformed into any loop round the obstacle then the flux through it continues to vanish. Loops that do not enclose the obstacle may be contracted to a point without crossing any singularities, so their fluxes also vanish. Since all loops have vanishing flux, the flow is not merely locally sourceless, it is totally sourceless. Put differently, this says that the flux crossing any curve between two points in the flow is independent of the path.

It is a different story for the circulation/work round B , for there is no *a priori* reason for this to vanish. Let S denote the value of this circulation. By the Deformation Theorem (11.10), p. 547, the circulation round any simple loop enclosing the obstacle will also equal S , while the circulation round loops that do not enclose the obstacle will vanish. Put differently, this says that the circulation along a path between two points in the flow is dependent on the choice of that path. The circulation $\Phi(z)$ along a path from a fixed point to a variable point z is a multifunction of z : if the difference of two paths is a loop winding round the obstacle n times, the difference of the two values of $\Phi(z)$ will be nS .

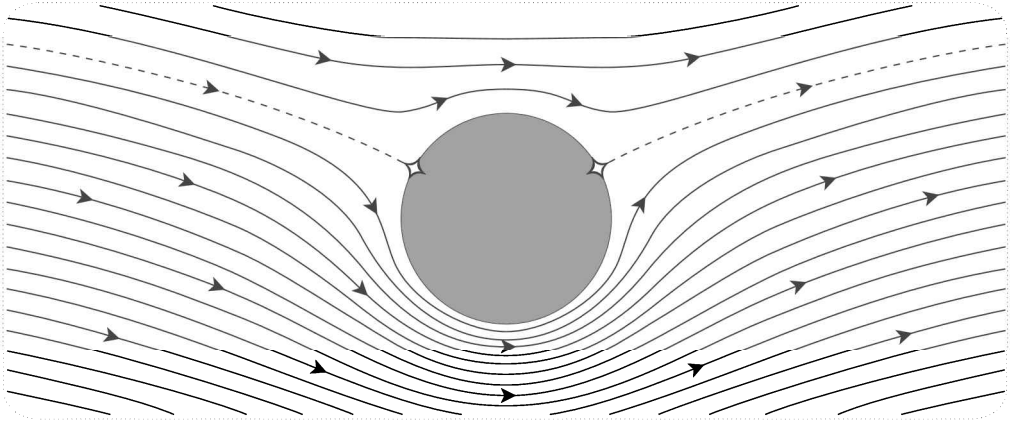
We may now state (without proof) the uniqueness property of flows round a given obstacle when it is inserted into a uniform flow of given velocity: for each value of the circulation S there is precisely one flow. In particular, there is a unique flow that is totally irrotational, i.e., for which $S = 0$. For a circular obstacle, this is the one shown in [12.12].

In the case of the disc it is easy enough to construct each of the flows for which $S \neq 0$. We need only superpose the totally irrotational flow in [12.12] with the flow of a vortex of strength S at the origin. Figure [12.13] illustrates this flow for a small value of S . The reader is strongly encouraged to use a computer to verify this figure.

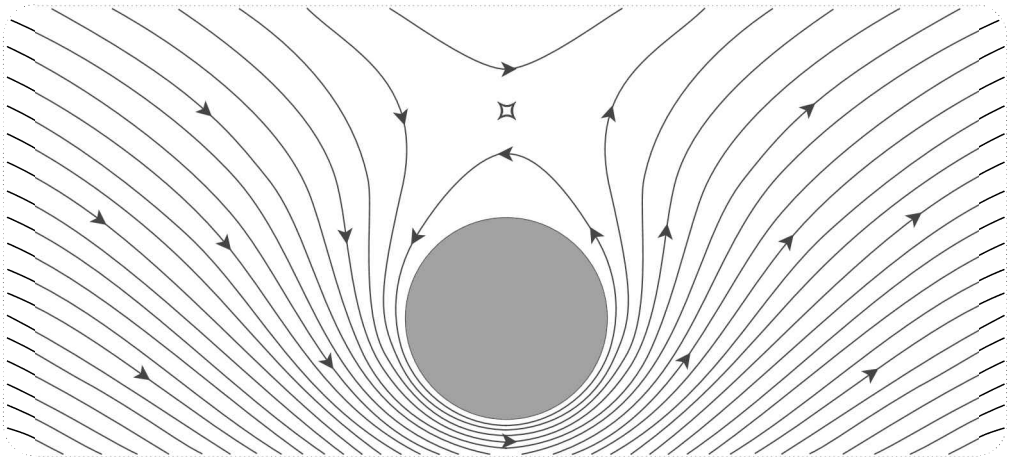
As you gradually increase the value of S , notice how the stagnation points on the circle move towards each other and finally coalesce at i . At what value of S does this occur? Verify your empirical answer with an exact calculation. Increasing the value of S still further, this single stagnation point moves off the circle and up the imaginary axis, and we obtain the qualitatively different flow shown in [12.14].

Returning to the totally irrotational flow in [12.12], figure [12.15] is an attempt to illustrate the geometry of its complex potential in greater detail than before. The top half is essentially just a copy of [12.12], while the bottom half illustrates the image under the complex potential mapping. Though the figure is intended to be largely self-explanatory, we make the following observations:

- If we choose to measure circulation and flux along paths emanating from p , then $\Omega(p) = 0$.

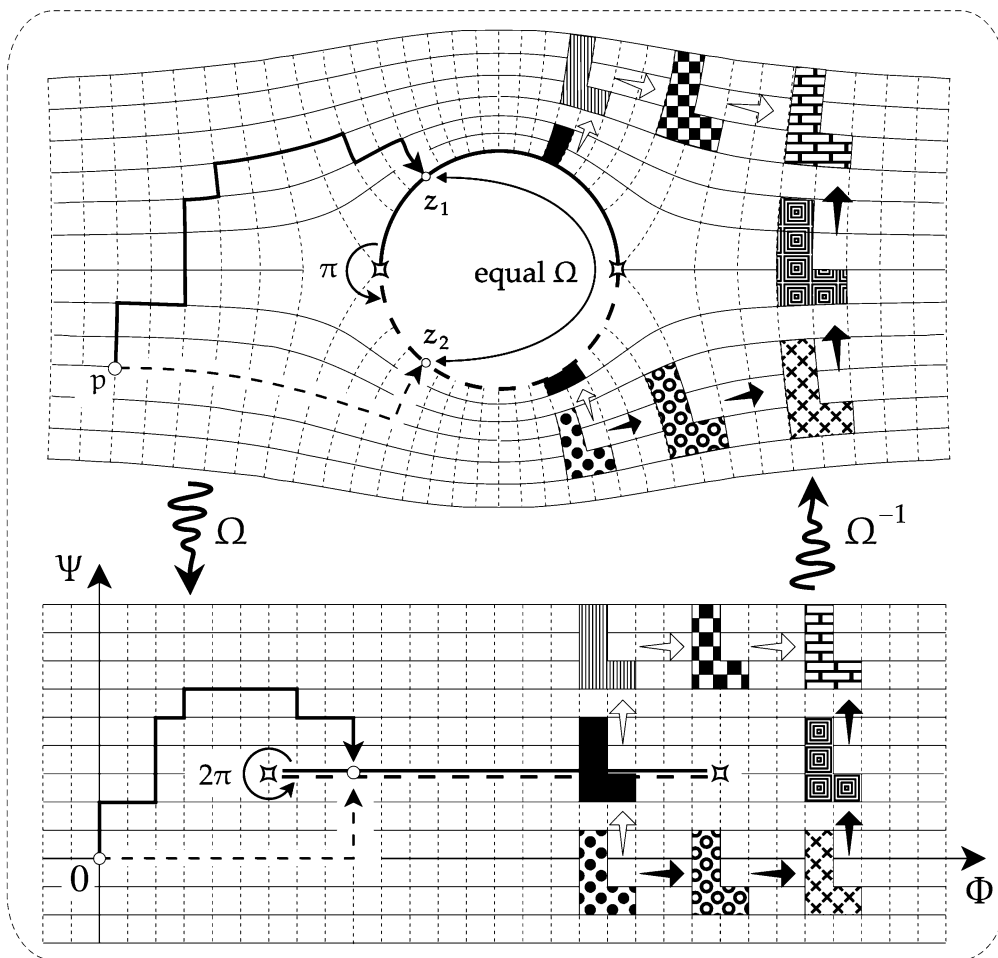


[12.13] For each value of the circulation S around an obstacle, there exists a *unique* flow. If $S = 0$ then the flow is given by the totally irrotational flow in the previous figure. If $S \neq 0$, the unique flow is the superposition of the $S = 0$ flow with an origin-centred vortex of circulation S . This figure depicts the case of a small value of S .



[12.14] Increasing S eventually results in a qualitatively different flow. Gradually increasing the value of S from that of the previous figure, the stagnation points on the unit circle move towards each other and finally coalesce at i . Increasing the value of S still further, this single stagnation point moves off the circle and up the imaginary axis, and we obtain the qualitatively different flow shown here.

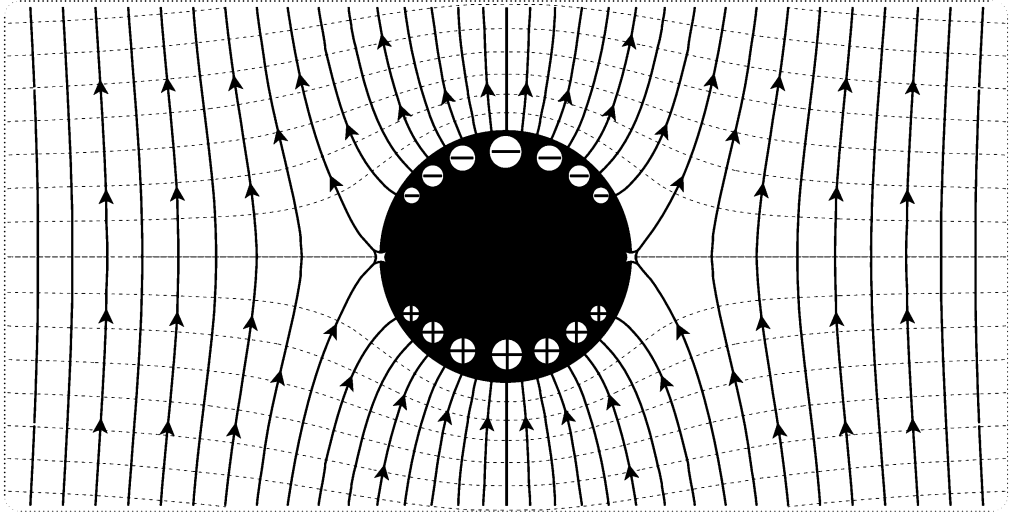
- Here the flow is totally sourceless and irrotational, so $\Omega(z)$ is a single-valued function of z .
- The paths from p to z_1 and z_2 both have the same circulation and flux: Ω is two-to-one on the boundary of the disc.
- However, no two points lying strictly outside the disc have the same Φ and Ψ , so we have a one-to-one mapping (with inverse Ω^{-1}) between the exterior of the



[12.15] A more detailed view of the complex potential mapping of the irrotational flow [12.12].

disc and the exterior of the line-segment connecting $\Omega(\pm 1)$. [The latter region is often described as a plane with a *cut* along this segment.]

- The bottom half of the figure illustrates two routes [black/white arrows] by which the dot-filled shape may be moved to the brick-filled shape. The top illustrates their images under Ω^{-1} (as defined above). Both routes yield the same image for the brick-filled shape, but the price we pay is that Ω^{-1} is not even continuous (let alone analytic) on the cut: witness the fate of the black shape as we follow the white route. However, consider the following alternative. Again following the white arrows, sketch what happens if we *analytically continue* Ω^{-1} as we cross the cut. Hint: look at the flow inside the unit circle in [12.12].
- The fact that we have ended up with two different images for the brick-filled shape reflects the fact that the two routes enclose a branch point at $\Omega(1)$. This



[12.16] Electrostatic interpretation of the dual of the irrotational flow around the disc. A cross section of a copper cylinder inserted into a uniform electric field: the former streamlines now become equipotentials. The surface of such a conductor must itself be an equipotential, with the static electric field running perpendicular to it, or else charges would experience a force acting within the surface, causing them to move.

is implied by the existence at ± 1 of stagnation points, i.e., critical points of Ω . Geometrically, this is evident from the doubling of angles at these points, e.g., the angle π at -1 is doubled to 2π at $\Omega(-1)$.

We now turn to two more physical interpretations of [12.12]. First, we may view the two-dimensional flow in [12.12] as being a cross section of a genuine three-dimensional flow. Suppose we were to make a few thousand photocopies of this figure and stack them neatly one on top of the other. The shaded discs would fit together to form a cylinder perpendicular to the stack, and the streamlines would represent the flow round this cylinder. Of course a real cylinder has ends, and, when it is inserted into a uniform flow perpendicular to its axis, the flow on a plane that is close to one of these ends will no longer look like [12.12].

To give the second interpretation we must explain a previous remark, namely, that the dual of a flow round an obstacle is the electric field which solves an analogous problem in electrostatics. The dual of the flow in [12.12] is shown in [12.16]. Since C was a streamline of the original flow, it is now an equipotential of the dual flow.

Suppose we insert a long copper cylinder into a uniform electric field perpendicular to its axis. Almost instantly, the free electric charges within the cylinder will settle themselves into an equilibrium distribution such that the electric field becomes steady (“electrostatic”). Figure [12.16] then represents a cross section of

this electrostatic field on a plane that cuts perpendicularly through the conductor somewhere in the middle. [Again, the field near the ends will be different.]

Here's why. Just as we require Ψ to be constant on the boundary of an obstacle in a fluid flow, so we require Φ to be constant on the boundary of a conductor in an electrostatic field. This is equivalent to demanding that the electric field be perpendicular to the boundary of the conductor, and it is not hard to see why the latter must be true. For if the electric field $E = \nabla\Phi$ were *not* perpendicular, it would have a nonzero component within the surface of the conductor. But this means that the free charges there would experience a force and would *move*, contradicting our assumption that the field has settled into an *electrostatic* one.

The figure also sketches the equilibrium distribution of negative $\ominus$ and positive $\oplus$ charge on the surface of the cylinder. This distribution is such that the charge density is proportional to the (signed) strength of the electric field at the surface. If a phase portrait has been divided into squares, as this one has, this implies [why?] that the charge density is proportional to the density of field lines (per unit length) leaving the conductor's surface. To learn more of the physics of electrostatics, see [Feynman \(1963\)](#); for a geometric approach, see [Maxwell \(1881\)](#).

12.5.3 The Method of Images

To illustrate the different kinds of flow which are possible in the presence of one and the same obstacle, suppose that we instead inserted the unit disc into the flow of a dipole located at 2, having dipole moment $-(1 + i)$, and thus represented by the (undisturbed) complex potential

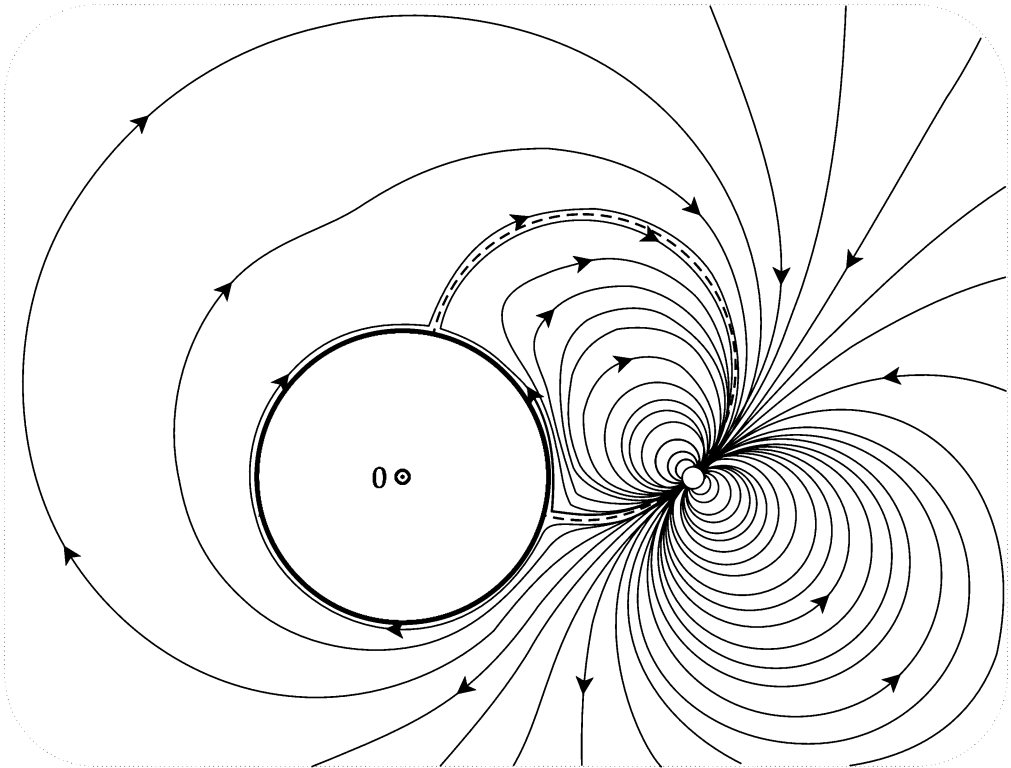
$$\Omega_u(z) = \frac{(1 + i)}{z - 2}. \quad (12.14)$$

Clearly the disturbed flow must look something like [12.17]: the streamlines behave like those of Ω_u both close to 2 and far from the obstacle, and the unit circle is composed of streamlines. Later we will use the so-called *method of images* to show that (modulo artistic error) this figure is in fact the exact flow. To explain this method we begin with a much simpler example.

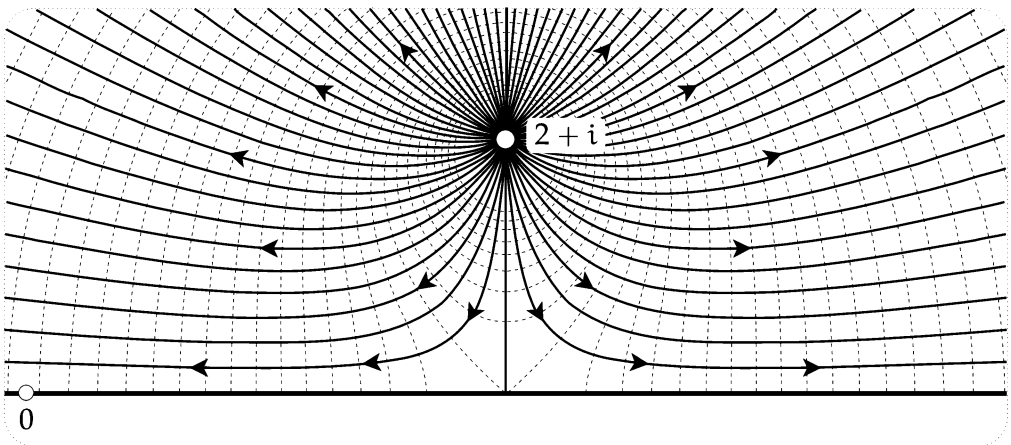
With the lower half-plane as the obstacle, consider the flow of a source of strength 2π located at $(2 + i)$, for which the undisturbed complex potential is

$$\Omega_u(z) = \log(z - 2 - i).$$

Clearly the disturbed complex potential Ω_d (which we seek) must have streamlines which look something like those in [12.18]: they closely resemble those of Ω_u near the singularity, and the barrier is a streamline. That [12.18] is actually the *exact* solution follows from the illustrated fact that equipotentials [dashed] may be drawn so as to divide this flow into squares. We now describe how this Ω_d may be found by the method of images.

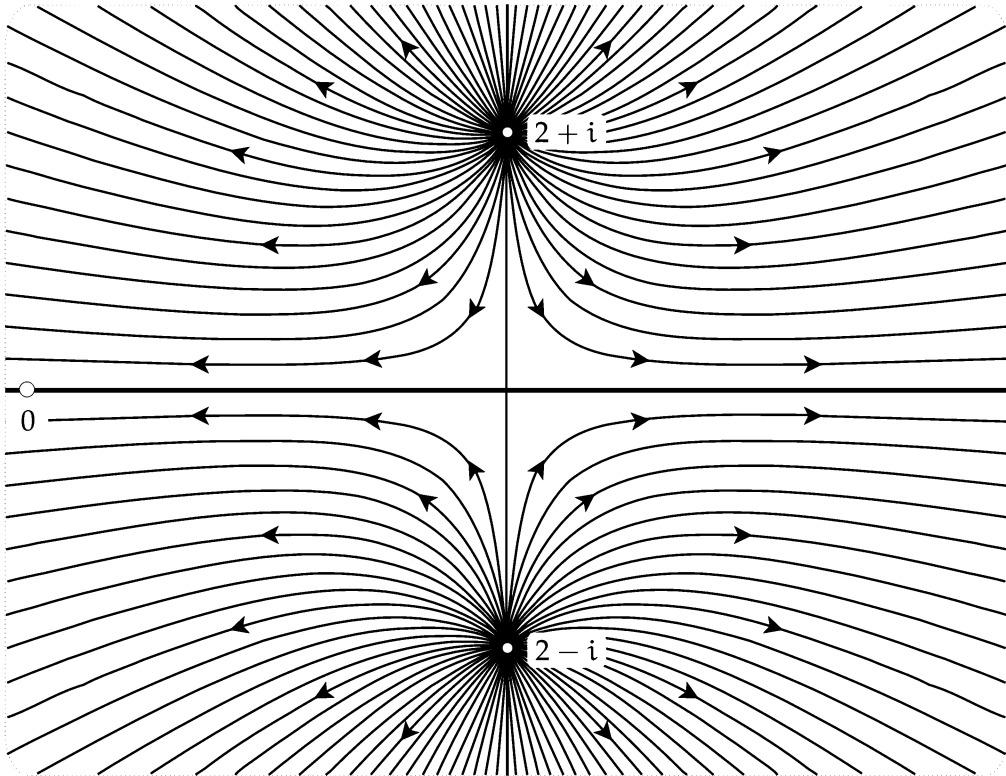


[12.17] The flow resulting from inserting the unit disc into a dipole field.



[12.18] The flow of a source in the presence of a barrier along the real axis.

As we saw in Chapter 5, Schwarz's Symmetry Principle (p. 286) says that a function which is analytic on one side of the real line, and which takes real values on that line (e.g., Ω_d), may be analytically continued to the other side by taking conjugate points to have conjugate images. This is vividly clear in [12.18]: the grid of squares



[12.19] The Method of Images with a linear boundary. To find the flow depicted in the previous figure, we superpose the original source with its image (of equal strength) after reflection across the real axis barrier. More generally, if $\Omega_u(z)$ is the undisturbed complex potential of a set of multipoles in the upper half-plane, without any barrier, then $\Omega_u^*(z) \equiv \overline{\Omega_u(\bar{z})}$ represents the undisturbed flow of the mirror-image multipoles in the lower half-plane. Then the method of images tells us that in the presence of the real axis barrier, **the disturbed flow is the superposition of the original flow and its mirror image:** $\Omega_d(z) = \Omega_u(z) + \Omega_u^*(z)$.

in the upper half-plane may be continued to the lower half-plane by reflecting the grid in the real axis. The complete flow is thus given by [12.19].

Intuitively, we have found that the flow in the presence of the barrier may be obtained by removing the barrier and instead inserting another source of the same strength as the original, but located at the mirror image in the real axis. Thus Ω_d is the superposition of these two sources:

$$\Omega_d(z) = \log(z - 2 - i) + \log(z - 2 + i).$$

Use a computer to verify that this formula produces [12.19].

More generally, if Ω_u is the undisturbed complex potential of a superposition of multipoles (sources, vortices, dipoles, etc.) in the upper half-plane, then Ω_d will be

the superposition of the undisturbed flow of this set of multipoles and the undisturbed flow of their mirror images. Note that while the mirror image multipoles will be of the same type and strength as the originals, the direction of their multipole moments will be different. For example, a dipole in the direction $(3 - 2i)$ reflects to one in the conjugate direction $(3 + 2i)$. More generally, a multipole with multipole moment Q reflects to one with moment $\bar{Q}$.

Now let us turn this method into a formula. In Chapter 5 we showed that from a given analytic mapping $f(z)$ we can produce a new analytic mapping $f^*(z)$ according to the recipe

$$f^*(z) \equiv \overline{f(\bar{z})}. \quad (12.15)$$

The physical significance of this new analytic function is easy to see: if Ω_u again represents the complex potential of a superposition of multipoles in the upper half-plane, then Ω_u^* will be the undisturbed complex potential of their mirror images in the lower half-plane. The formula for the disturbed complex potential is thus

$$\Omega_d(z) = \Omega_u(z) + \Omega_u^*(z). \quad (12.16)$$

Note that this formula does indeed satisfy Schwarz's Symmetry Principle: $\Omega_d^*(z) = \Omega_d(z)$, so the real axis is a streamline. Naturally, if Ω_u instead represents a collection of multipoles in the lower half-plane, then (12.16) is again the solution in the presence of the barrier.

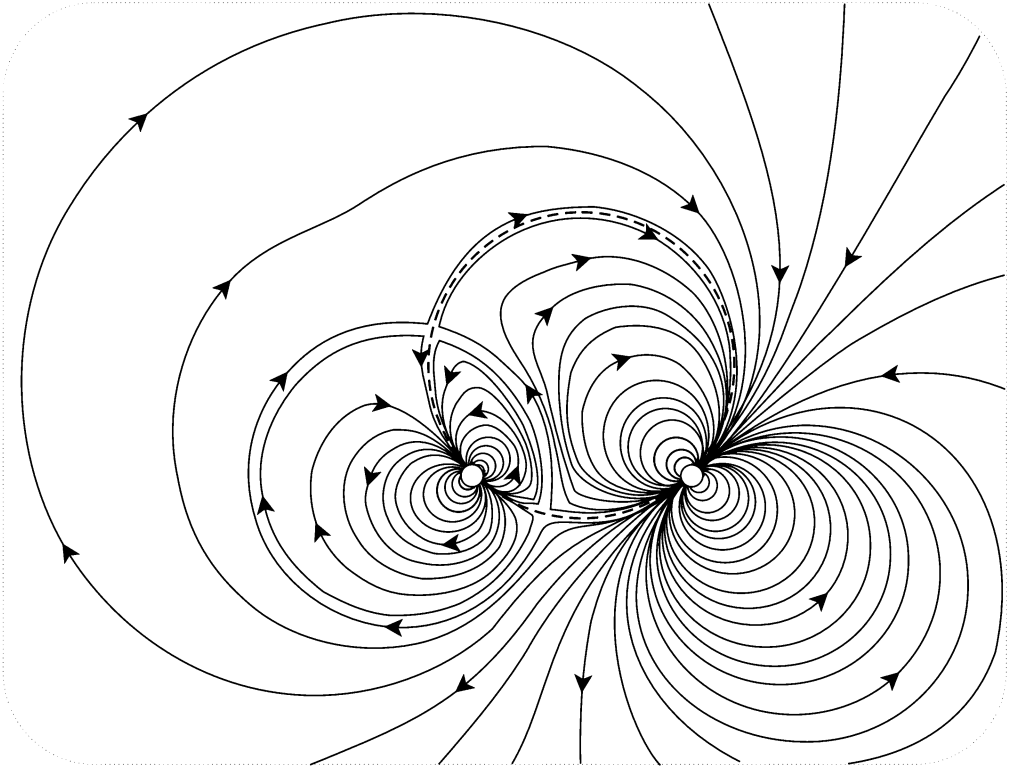
Essentially the same method may be used to find the disturbed flow in [12.17], in which the barrier is now a circle rather than a line. Reconsider that figure. We have drawn the streamlines $\Psi = \text{const.}$ at random, but had we instead chosen the values of Ψ in arithmetic progression (as we did in [12.12]) then it would have been possible to divide this flow into a grid of infinitesimal squares, with the unit circle being comprised of edges of these squares. As we saw in Chapter 5, it is again possible to extend this grid across the barrier (cf. [12.12]), but to do so we must replace reflection in a line by its analogue for circles, namely, *inversion*.

Performing this inversion in the unit circle we obtain [12.20], the dipole at 2 inverting to another dipole at $(1/2)$. It is now clear that to find Ω_d we should remove the barrier and superpose the undisturbed flows of these two dipoles. But what is the undisturbed complex potential of this new dipole at $(1/2)$?

As we saw in Chapter 5, if reflection in the real line is replaced by inversion in the unit circle then the recipe (12.15) may be modified to generate a new analytic function $f^\dagger$ given by

$$f^\dagger(z) \equiv \overline{f\left(\frac{1}{\bar{z}}\right)}. \quad (12.17)$$

The physical significance of this new analytic function is much as before: if Ω_u represents the complex potential of a superposition of multipoles outside the unit



[12.20] The Method of Images with a circular boundary: Milne–Thomson Circle Theorem. To find the flow depicted in [12.17], we superpose the original dipole with its image under inversion in the unit circle. More generally, if $\Omega_u(z)$ is the undisturbed complex potential of a set of multipoles outside the unit circle, without any barrier, then $\Omega^\dagger(z) \equiv \overline{\Omega(\frac{1}{\bar{z}})}$ represents the undisturbed flow of the mirror-image multipoles inside the unit disc, and vice versa. Then the method of images tells us that in the presence of the unit circle barrier, **the disturbed flow is the superposition of the original flow and its mirror image:** $\Omega_d(z) = \Omega_u(z) + \Omega_u^\dagger(z)$.

circle, then $\Omega_u^\dagger$ will be the undisturbed complex potential of their images under inversion. The analogue of (12.16) is now

$$\Omega_d(z) = \Omega_u(z) + \Omega_u^\dagger(z), \quad (12.18)$$

which automatically satisfies the symmetry requirement $\Omega_d^\dagger(z) = \Omega_d(z)$, so that the unit circle is a streamline. This result is known as the *Milne–Thomson Circle Theorem*. If Ω_u instead represents a collection of multipoles *all* of which lie inside the unit circle, then this Ω_d is again the solution in the presence of the barrier. The reason for the emphasis on “all” will be explained later.

Let us apply this method to find the disturbed complex potential of [12.20] (and hence of [12.17]). If Ω_u is given by (12.14) then

$$\Omega_u^\dagger(z) = \frac{(1-i)}{(1/z)-2} = \frac{(1-i)z}{1-2z}.$$

That this is indeed a dipole at $(1/2)$ may be seen by rewriting it as

$$\Omega_u^\dagger(z) = -\frac{(1-i)}{4} \left[\frac{1}{z-(1/2)} \right] - \frac{(1-i)}{2}.$$

Because constants have no effect on the flow, this is a dipole at $(1/2)$ with dipole moment $(1-i)/4$. Unlike the case of reflection across a line, note that it is not only the direction of the dipole moment which is affected by the inversion, but also its magnitude: here the strength of the inverted dipole is one quarter the strength of the original. Superposing the two dipoles we obtain

$$\Omega_d(z) = \Omega_u(z) + \Omega_u^\dagger(z) = \frac{(1+i)}{z-2} + \frac{(1-i)z}{1-2z},$$

and you may use a computer to verify that this formula does yield the flow in [12.20].

Like [12.19], figure [12.20] has symmetry, but it is of a more subtle kind than before: by construction, the figure reproduces itself under inversion in the unit circle. This symmetry can be made to leap from the page by projecting [12.20] onto the Riemann sphere. As we learnt in Chapter 3, inverting in the unit circle is equivalent to reflecting the Riemann sphere in its equatorial plane. The flow on the northern and southern hemispheres should therefore be mirror images of each other in this plane. Behold figure [12.21]! Note that on the sphere the strengths of the two dipoles become equal.

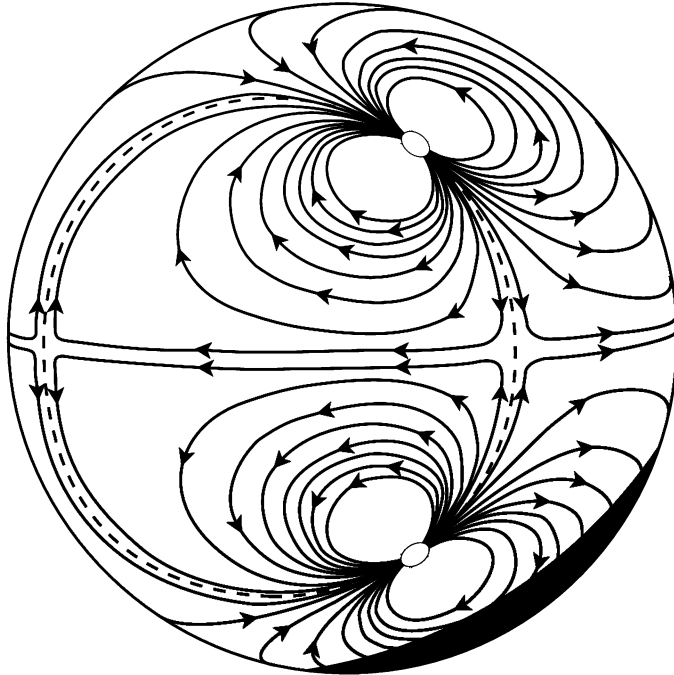
We can now see that [12.12] is simply a limiting case of [12.21], for as the southern dipole moves towards the south pole (0) , its reflection moves towards the north pole (∞) , and a solitary dipole at ∞ projects to a uniform flow in the plane.

In Chapter 5 we saw that both reflection in a line and inversion in a circle were special cases of Schwarzian reflection in a curve; see [5.30], page 290. We now use this fact to generalize the method of images. For example, suppose that we insert an ellipse E into a uniform flow. Clearly the flow will look something like [12.22], and that this is in fact the exact flow is once again apparent from its divisibility into squares. How did we find it?

The generalization of the formulae (12.15) and (12.17) to the case of Schwarzian reflection $\mathfrak{R}_K$ in a curve K is

$$f^\dagger(z) \equiv \overline{f[\mathfrak{R}_K(z)]}.$$

For example, if K is the real line (so that $\mathfrak{R}_K(z) = \bar{z}$) then $f^\dagger = f^*$, while if K is the unit circle (so that $\mathfrak{R}_K(z) = 1/\bar{z}$) then $f^\dagger = f^\ddagger$. Thus if Ω_u is the undisturbed



[12.21] Projection of the previous figure onto the Riemann sphere. By construction, the previous figure is symmetric under inversion in the unit circle, which induces reflection of the Riemann sphere in its equatorial plane. So the flow on the northern and southern hemispheres should therefore be mirror images of each other, and indeed we see that they are! Note that the unequal strengths of the original dipoles in the plane project to dipoles of equal strength on the sphere. Also observe that [12.12] may now be understood as a limiting case of this figure, for as the southern dipole moves towards the south pole (0), its reflection moves towards the north pole (∞), and a solitary dipole at ∞ projects to a uniform flow in the plane.

complex potential of a collection of multipoles on one side of K then the disturbed complex potential with K as barrier will be

$$\Omega_d(z) = \Omega_u(z) + \Omega_u^\dagger(z).$$

This formula automatically satisfies $\Omega_d^\dagger = \Omega_d$, so that K is a streamline.

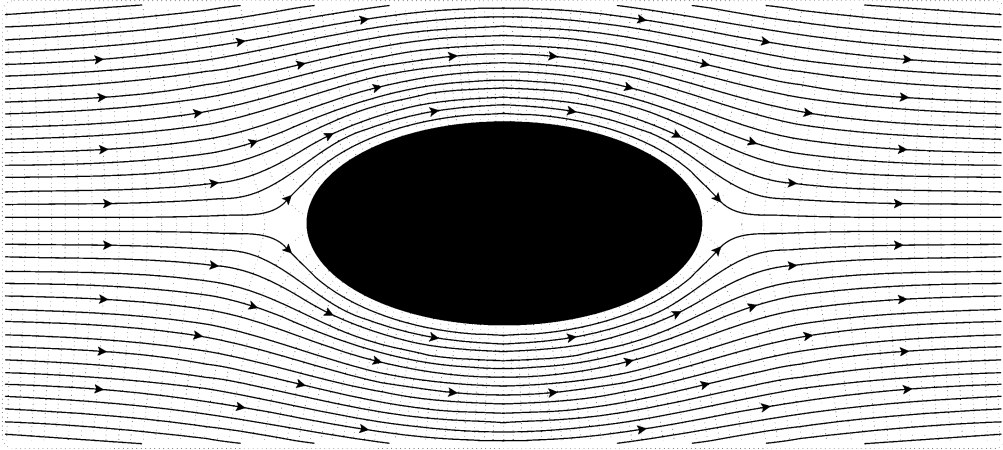
For example, take the ellipse E in [12.22] to have equation $(x/2)^2 + y^2 = 1$. In this case we have [see p. 291]

$$\Re_E(z) = \frac{1}{3} \left[5\bar{z} - 4\sqrt{z^2 - 3} \right].$$

Thus, with $\Omega_u(z) = z$,

$$\Omega_d(z) = \frac{4}{3} \left[2z - \sqrt{z^2 - 3} \right],$$

and you may use a computer to verify that this formula yields figure [12.22]. Note, however, that everything is not quite as it seems. While Ω_u is a uniform



[12.22] The Method of Images with a general boundary via Schwarzian reflection.

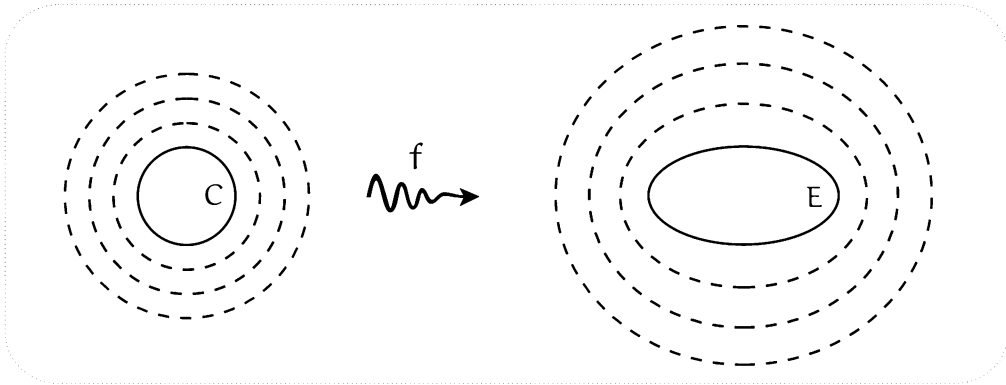
To find the illustrated flow around the ellipse, E , we superpose a steady eastward flow with its Schwarzian reflection $\Re_E$ across E ; see [5.31]. More generally, if $\Omega_u(z)$ is the undisturbed complex potential of a set of multipoles on one side of a curve K , without any barrier, then $\Omega_u^\dagger(z) \equiv \overline{\Omega_u[\Re_K(z)]}$ represents the undisturbed flow of the Schwarzian reflections of the multipoles on the other side of K , and vice versa. Then the method of images tells us that in the presence of the barrier, K , **the disturbed flow is the superposition of the original flow and its mirror image:** $\Omega_d(z) = \Omega_u(z) + \Omega_u^\dagger(z)$. This subsumes both previous methods of images as special cases, for if K is the real axis then $\Re_K(z) = \bar{z}$ and $f^\dagger = f^*$, while if K is the unit circle then $\Re_K(z) = 1/\bar{z}$ and $f^\dagger = f^\dagger$.

flow to the right with speed 1, the behaviour of Ω_d for large values of $|z|$ is a uniform flow to the right with speed $(4/3)$. Of course if we wish the flow to have unit speed far from the ellipse then we need only multiply this Ω_d by $(3/4)$.

12.5.4 Mapping One Flow Onto Another

We previously established the fact that a conformal mapping sends the streamlines and equipotentials of steady, sourceless, irrotational flow to the streamlines and equipotentials of another such flow. This idea has many theoretical and practical uses.

A theoretical benefit is a fresh insight into the very concept of a complex potential. Reconsider figure [11.21], p. 570. Applying any conformal mapping f to the flow on the left yields another steady, sourceless, irrotational flow on the right. The complex potential may now be defined as the special mapping $f = \Omega$ for which the image flow is uniform with velocity 1. For example, what is the complex potential of a uniform flow with velocity 1? From the new point of view it is the conformal mapping which sends this flow to a uniform flow with velocity 1. Thus it is the identity mapping $\Omega(z) = z$, as it should be!



[12.23] Conformally mapping concentric circles to confocal ellipses. Recall from [5.22], on page 276, that as $z = e^{it}$ describes the unit circle, C , $w = p e^{it} + q e^{-it}$ describes the ellipse $(x/a)^2 + (y/b)^2 = 1$, where $p = (a+b)/2$ and $q = (a-b)/2$. Thus, in the specific example of $(x/2)^2 + y^2 = 1$, the conformal mapping we seek of C to E is $z \mapsto w = f(z) = \frac{1}{2} (3z + \frac{1}{z})$. Circles concentric to C are mapped to ellipses confocal to E . Far from C , the mapping behaves like $\frac{3}{2}z$.

To demonstrate the practicality of mapping one flow onto another, let us return to the problem of finding the flow round an obstacle, restricting our attention to the case where the obstacle is inserted into a *uniform* flow. For example, let us rederive the flow round the ellipse $(x/2)^2 + y^2 = 1$ in [12.22]. Suppose that we knew of a one-to-one conformal mapping $w = f(z)$ from the exterior of the unit circle C in the z -plane to the exterior of this ellipse E in the w -plane. We already know the flow round C when it is inserted into a uniform flow of velocity 1, so applying f to these streamlines yields *some* flow round E . If we want the flow round E to be uniform far from E then we must further demand that $f(z)$ behave like a multiple of the identity far from C : $f(z) \approx cz$ if $|z|$ is large. Assigning equal values of Ψ to the original and image streamlines, the image flow far from E will then be uniform, with speed $(1/|c|)$ and direction c [why?].

Recall figure [5.22], p. 276. As $z = e^{it}$ describes C , $w = p e^{it} + q e^{-it}$ describes the ellipse $(x/a)^2 + (y/b)^2 = 1$, where $p = (a+b)/2$ and $q = (a-b)/2$. Thus in the case of $(x/2)^2 + y^2 = 1$, the mapping we seek is

$$w = f(z) = \frac{1}{2} (3z + \frac{1}{z}). \quad (12.19)$$

As illustrated in [12.23], C is mapped to E and circles concentric to C are mapped to ellipses confocal to E . Far from C the mapping behaves like $(3/2)z$: you can see that as the circles on the left grow, their images approach circles that are $(3/2)$ as big. Thus f should map the flow round C to the flow that results from inserting E into a uniform flow of velocity $(2/3)$.

Let's go through the details and check that we recover the flow in [12.22]. Since the flow round C has complex potential $\Omega(z) = z + (1/z)$, and since the complex potential $\tilde{\Omega}(w)$ at the image $w = f(z)$ of z is defined to be the same as Ω at z , we have $\tilde{\Omega}(w) = z + (1/z)$. To express this as an explicit function of w we solve (12.19) for z and obtain [why the choice of +?]

$$z = \frac{1}{3} \left(w + \sqrt{w^2 - 3} \right).$$

Although we could immediately insert this into the formula for $\tilde{\Omega}(w)$, we may save ourselves a little algebra by first noting that (12.19) implies $(1/z) = 2w - 3z$. Thus

$$\tilde{\Omega}(w) = z + (1/z) = 2(w - z) = \frac{2}{3} \left(2w - \sqrt{w^2 - 3} \right).$$

Apart from the factor of $(2/3)$, signifying that the velocity is $(2/3)$ far from E (as anticipated), this is the same formula we previously obtained by Schwarzian reflection.

As another illustration of this idea, suppose that E were instead inserted into a uniform flow in the direction of $e^{i\phi}$. To find the flow round E we need only find the flow round C when it is inserted into such a flow, and then apply f . Since the undisturbed complex potential is $\Omega_u(z) = e^{-i\phi}z$, the method of images says that the flow round C is

$$\Omega_d(z) = \Omega_u(z) + \Omega_u^\dagger(z) = e^{-i\phi}z + \frac{e^{i\phi}}{z}.$$

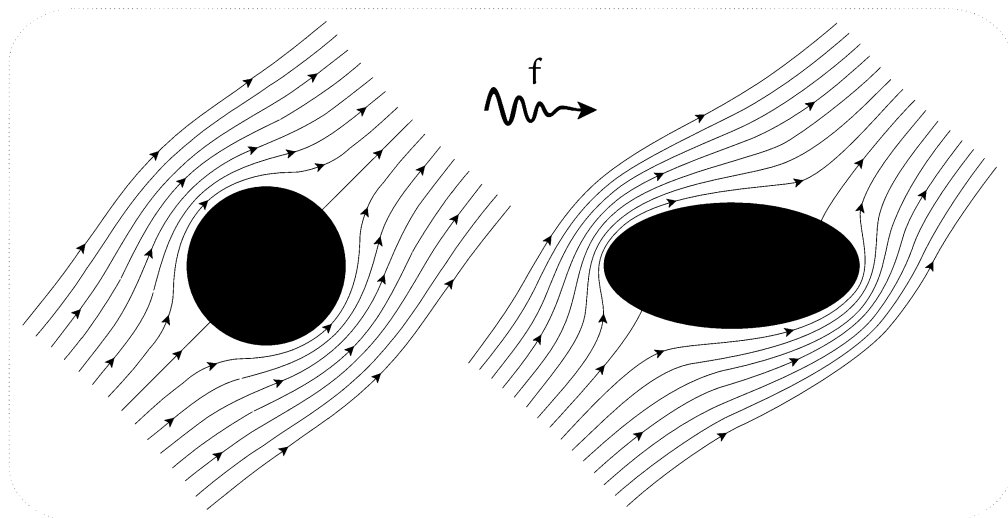
With $\phi = (\pi/4)$ this flow is illustrated on the left of [12.24]. On the right is the desired flow round E obtained by applying f to the flow round C . Use your computer to verify this figure.

In this manner, we may derive the flow round an infinite variety of obstacles: choose any analytic mapping $w = f(z)$ which is one-to-one outside C and which behaves like cz for large $|z|$, then $\tilde{\Omega}(w) = f^{-1}(w) + [1/f^{-1}(w)]$ is the flow round an obstacle whose boundary curve B is $f(C)$.

12.6 The Physics of Riemann's Mapping Theorem

12.6.1 Introduction

Recall that Riemann's Mapping Theorem [p. 204] asserts that any simply connected region R (other than the entire plane) may be mapped one-to-one and conformally to any other such region S . Granted this, we saw that no loss of generality results from taking S to be the unit disc D , and that there must be as many mappings from R to S as there are automorphisms of D . Later [p. 407] we showed that these automorphisms are the hyperbolic rigid motions, which have three degrees of freedom.



[12.24] Conformally mapping one flow onto another. Using the mapping obtained in the previous figure, the known flow on the left around the unit disc may be conformally mapped to the flow around the ellipse on the right. [NOTE: In the text, we confirm that this new approach yields the same answer that we formerly obtained via the method of images.] In this manner, we may derive the flow round an infinite variety of obstacles: choose any analytic mapping $w = f(z)$ which is one-to-one outside C and which behaves like cz for large $|z|$, so that the flow becomes uniform, then $\tilde{\Omega}(w) = f^{-1}(w) + [1/f^{-1}(w)]$ is the flow round an obstacle whose boundary curve B is $f(C)$.

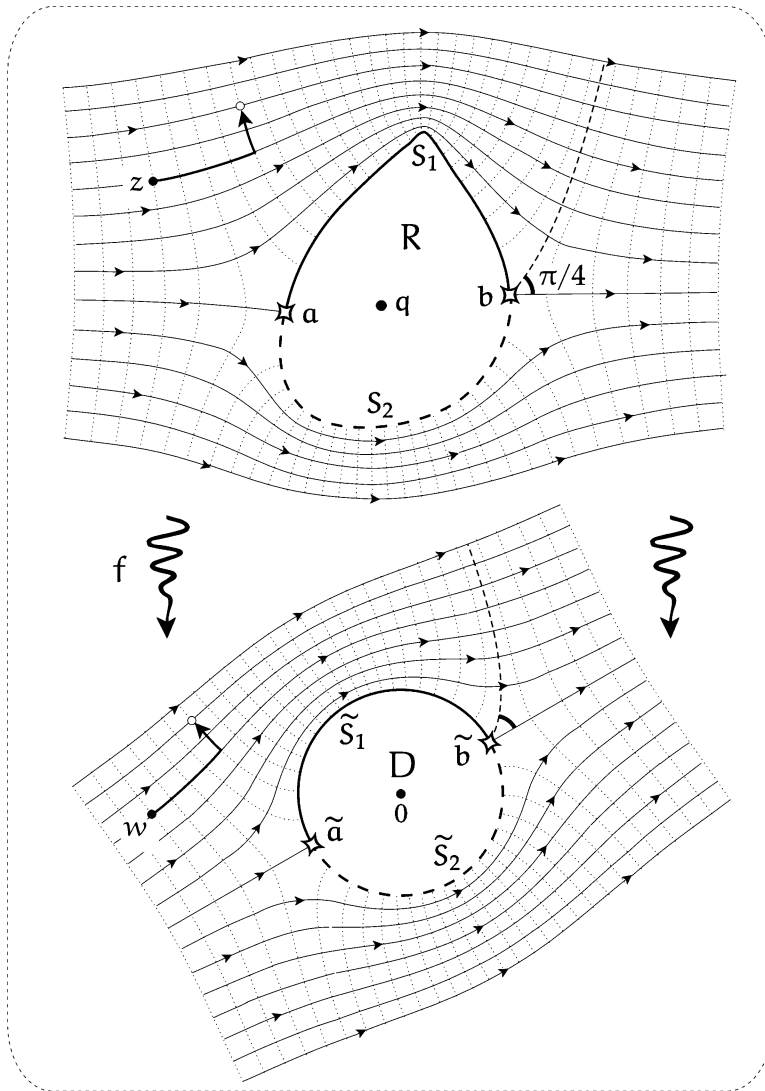
Thus the complete result we seek to understand is this: *there exists a three-parameter family of one-to-one, conformal mappings between $\mathbb{R}$ and $\mathbb{D}$.*

There are at least two standard proofs of this fundamental result, and most advanced books on complex analysis include one of these. Despite the fact that one of the arguments (due to Koebe) is constructive in nature—and therefore, in principle, comprehensible—we have not yet found a way to present it in a manner consistent with the aims of this book. However, the interested reader will find an excellent description of the idea underlying Koebe’s proof in [Hilbert \(1952\)](#), and a clear description of the technical details in [Nehari \(1952\)](#) or [Nevanlinna and Paatero \(2007\)](#). To our knowledge, the deepest investigation of this idea is that given by [Henrici \(1991\)](#).

On a brighter note, the above ideas on flows *will* enable us to gain considerable insight into both the existence of such mappings and the fact that they have three degrees of freedom. We shall do so by reversing the idea of conformally mapping one flow onto another. That is, by resorting to physical experiment one may obtain flows, and these may then be used to construct conformal mappings.

12.6.2 Exterior Mappings and Flows Round Obstacles

Consider [12.25]. The top part depicts an obstacle R that has been inserted into uniform flow having velocity 1. If we do not introduce any circulation round R



[12.25] Constraint on the mapping between flows. Divide the flow around R and the flow around the unit disc D into small, square k -cells. If the illustrated point z is mapped by f to $w = f(z)$, then if z moves four squares downstream and two squares “up” the equipotentials, w must do the same. But the sought-after conformal mapping $w = f(z)$ has a constraint: it must map the stagnation points a and b to the stagnation points $\tilde{a}$ and $\tilde{b}$. This is because the geometric signature of the critical point b is the angle of $(\pi/4)$ between the streamline and equipotential through b , and this property is preserved by a conformal mapping.

then, as we have previously remarked, the flow is unique. Next, we have arbitrarily chosen a point [not shown] from which to measure circulation and flux, i.e., a point at which the potential Φ and stream function Ψ both vanish. With a small value of k , we have then constructed the k -flux tubes and k -work tubes, thereby dividing the exterior of R into small, approximately square, k -cells. As usual, we may imagine shrinking the value of k to zero, so that ultimately the k -cells are square.

As illustrated, let a and b be the two stagnation points on the boundary of R (ordered so that the flow passes from a to b), and let S_1 and S_2 be the two segments of boundary streamline connecting them. Since the flow is totally irrotational, the circulations along S_1 and S_2 must be equal to each other, the common value being the potential difference $[\Phi]$ between a and b :

$$[\Phi] = \Phi(b) - \Phi(a).$$

Put geometrically, this says that the number of squares abutting S_1 and S_2 must be equal to each other, this number being given by $[\Phi]/k$.

Now insert the unit disc D into a uniform flow with velocity $v e^{i\phi}$ and divide up the resulting flow into k -cells, using the same value of k as before. Let us employ tildes to denote corresponding entities in the new flow: the stagnation points are $\tilde{a}$ and $\tilde{b}$, the segments of streamline connecting them are $\tilde{S}_1$ and $\tilde{S}_2$, and the potential and stream functions (relative to an arbitrarily chosen point) are $\tilde{\Phi}$ and $\tilde{\Psi}$.

The method of conformally mapping points z in the exterior of R to points $w = f(z)$ in the exterior of D now seems clear: identify “corresponding” squares in the two grids! Let us control our excitement and think this through. Presumably, by “corresponding” we mean that once we know the image w of one grid point z then the image of any other is determined by the pair of grids: if z moves four squares downstream and two squares “up” the equipotentials, then w does the same. See [12.25].

However, we cannot arbitrarily choose to map a particular z to a particular w , for the sought-after conformal mapping $w = f(z)$ must map the stagnation points a and b to the stagnation points $\tilde{a}$ and $\tilde{b}$. This is because the geometric signature of the critical point b is the angle of $(\pi/4)$ between the streamline and equipotential through b , and this property is preserved by a conformal mapping.

Now the snag is this: defining the image of a to be $\tilde{a}$ (as we must), b will map to $\tilde{b}$ if and only if the number of squares abutting $\tilde{S}_1$ is equal to the number abutting S_1 . Looking closely, we see this is not the case in [12.25]: there are fewer squares along $\tilde{S}_1$ than S_1 . To correct the situation we must slightly increase the speed v of the flow round the disc. More precisely, v must be chosen so that the potential difference $[\tilde{\Phi}]$ across D equals the potential difference $[\Phi]$ across R . Since we know that $[\tilde{\Phi}] = 4v$, we deduce that

A one-to-one, conformal mapping between the two grids is obtained if and only if the disc is inserted into a flow of speed $v = [\Phi]/4$.

Although we hope our use of grids has helped to make the mapping vivid (and to show that it is both one-to-one and conformal), we should perhaps point out that the grids are in no way essential to the definition of f . All we are doing is identifying points by means of circulation and flux: the image $w = f(z)$ of z is defined by

$$\tilde{\Omega}(w) = \Omega(z) + \left\{ \tilde{\Omega}(\tilde{a}) - \Omega(a) \right\},$$

the constant having been chosen so as to ensure that $\tilde{a} = f(a)$.

Since the choice of the zero point of flux and circulation in each of the flows has no bearing on the construction of f , we may choose these two points to our advantage. Henceforth, *we will take the zero point in the image flow to be the image of the zero point in the original flow*. In [12.25], for example, this convention says that if we choose the zero point for the flow round R to be at a (i.e., $\Omega(a) = 0$), then we must take the zero point for the flow round D to be at $\tilde{a}$ (i.e., $\tilde{\Omega}(\tilde{a}) = 0$). The above equation then simplifies to

$$\tilde{\Omega}(w) = \Omega(z).$$

In other words, w and z correspond if and only if their circulation and flux are equal.

In this case the mapping may be written as $f = \tilde{\Omega}^{-1} \circ \Omega$, which we may interpret as follows. Refer to [12.25] and [12.15]. The complex potentials $\Omega(z)$ and $\tilde{\Omega}(w)$ map points lying strictly outside R and, respectively, D to points in a plane that is slit along the real axis from 0 to $[\Phi]$. Thus f may be thought of as first mapping the exterior of R to the slit plane by means of Ω , then mapping the slit plane to the exterior of D by means of $\tilde{\Omega}^{-1}$.

We now return to the mappings themselves and ask, how “many” of them do we obtain by means of this construction? We begin by explaining the illusory nature of some of the apparent freedoms in the construction. First, why not insert R into a uniform flow of arbitrary speed? Of course we can, but nothing new results from doing so. For example, if we double this speed then we must also double v , because we must maintain equality between the number of k -cells along S_1 and $\tilde{S}_1$. But this yields the same mapping f as before. Second, in constructing the grid outside D , why not use a different value of k , say $\tilde{k}$, from that used outside R ? In order to maintain equality between the number of squares along S_1 and $\tilde{S}_1$ we would then have to change v to $\tilde{k}[\Phi]/4k$ [why?], and this would produce the same grid and mapping as before.

Clearly, however, we do obtain new mappings by varying the direction ϕ of the flow round D . If F denotes the particular mapping f corresponding to $\phi = 0$, then the mapping F_ϕ corresponding to a general value of ϕ is $F_\phi(z) = e^{i\phi}F(z)$, namely, F followed by a rotation.

It would seem plausible that still other mappings could be obtained by varying the direction—let us call it θ —of the flow into which R is inserted. Not so. This follows from the fact that the flow round R in this case may be obtained by applying F^{-1} to the flow round the disc when *it* is inserted into a uniform flow in the direction θ . Clarify this in your own mind by referring to figures [12.23] and [12.24]. Thus the mapping between the two flows is only sensitive to the angle *between* their directions: inserting R and C into flows with directions θ and ϕ , respectively, yields the mapping $F_{(\phi-\theta)}$.

Since ϕ is the only genuine degree of freedom, the mappings we have constructed belong to a one-parameter family. We are therefore missing two degrees of freedom. Though we will explain this mystery shortly, you may care to think about it on your own before reading further.

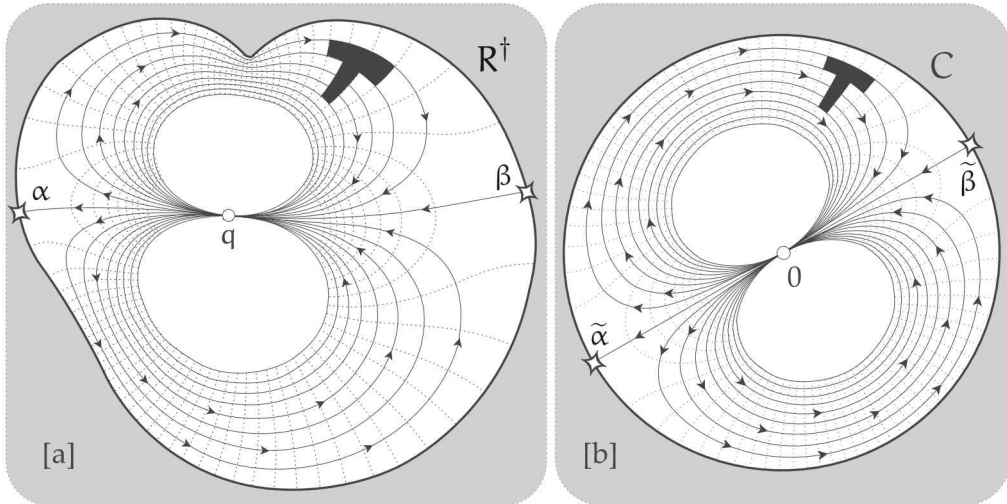
Granted the generosity of Nature in providing such flows round obstacles, we now have a physical method of determining f , but we lack a mathematical procedure for doing so. This is a very hard problem. However, there does exist an explicit formula for f in the case that R is bounded by a polygon. This is called the *Schwarz-Christoffel formula*; see [Nehari \(1952\)](#) or [Pólya and Latta \(1974\)](#) for good discussions of the result. We shall not enter into this here, except to say that the advent of high-speed computers has opened the way to approximating f by approximating the boundary of a given R with a polygon; see [Trefethen \(1986\)](#) and [Henrici \(1991\)](#) for this and other algorithmic approaches to the problem.

12.6.3 Interior Mappings and Dipoles

In [12.25], suppose that the speed of the flow round D has been adjusted to $[\Phi]/4$, yielding a conformal mapping between the exteriors of R and D . If we now perform inversion in the boundary of D , we obtain the flow [12.26b] inside the unit circle C . This is the familiar flow of a dipole inserted into the centre of a circular pool of fluid.

Similarly, if we choose the arbitrarily selected interior point q of R shown in the previous figure, then perform an inversion⁴ in a circle of unit radius centred there, we obtain [12.26a]. Use a computer, or your Peaucellier linkage [see p. 205], to verify that the boundary of R does indeed invert to the illustrated curve $R^\dagger$. In [12.25], the streamlines far from R are parallel lines, and the inversion therefore sends these to small, mutually tangent circles through q . Thus, as illustrated, the inverted flow in [12.26a] represents a dipole which has been inserted at q into the $R^\dagger$ -shaped pool

⁴ Ideas related to those which now follow may be found in [Bak and Newman \(2010\)](#), [Siegel \(1969, p. 148\)](#), and especially [Courant \(1950\)](#). Although Riemann himself employed physical reasoning, the idea of relating his mapping theorem to dipoles seems to have originated with [Hilbert \(1965, Vol. 3, pp. 73–80\)](#).



[12.26] Interior mappings and dipoles. Referring to the previous figure, inversion of R in a circle centred at the arbitrarily selected interior point q yields [a], while inversion of D in the unit circle yields [b]. In both cases, we obtain dipoles. In this way, the one-to-one, conformal mapping between the exteriors of R and C now yields another such mapping between the interior dipole flows of $R^\dagger$ and C : for example, the black T-shape is mapped to the black T-shape.

of fluid. In fact [exercise] since the original uniform flow had unit speed, the dipole at q has unit strength.

In this way, the one-to-one, conformal mapping between the exteriors of R and C now yields another such mapping between the interiors of $R^\dagger$ and C . For example, the stagnation point α maps to the stagnation point $\tilde{\alpha}$, q maps to 0 , and the black T-shape inside $R^\dagger$ maps to the one inside C . Although the speed of the flow grows arbitrarily large as we approach q , there is nothing dramatic about the behaviour of the mapping near q . For example, imagine sliding the T-shape along the streamlines towards q , and consider its image.

We may look at this construction rather differently. Suppose that $R^\dagger$ is a *given*, fixed curve whose interior we wish to map to the unit disc. We may obtain such a mapping f as follows:

- (i) Insert a horizontal dipole of unit strength into the interior of $R^\dagger$ at an arbitrary interior point p . Divide the interior into k -cells of the flow, and let N denote the number of them abutting $R^\dagger$.
- (ii) At the centre 0 of C , insert a dipole of strength d , and direction ϕ . Divide the interior of C into the k -cells of this flow, and adjust the strength d until the number of them abutting C is equal to N . If $[\Phi]$ again denotes the circulation

along a segment of boundary streamline connecting the stagnation points α and β , this condition is met by $d = [\Phi]/4$.

- (iii) Except for the fact that the dipole inside $R^\dagger$ is now located at a general point p (instead of the particular point q), we are now back to the situation depicted in [12.26], and the mapping f is completely determined.

This one-to-one, conformal mapping f from the interior of $R^\dagger$ to the unit disc is, in fact, the most general such mapping. This may be seen from the fact that the construction has the full *three* degrees of freedom: *two* for the location p of the dipole in $R^\dagger$, and *one* for the direction ϕ of the dipole in C . It is not hard to see the geometric significance for f of these physical degrees of freedom: p is the point which is mapped to the centre of the disc, i.e., $f(p) = 0$; and ϕ is clearly the twist of f at p , i.e., $\phi = \arg[f'(p)]$.

Note that while the twist of f at p is freely specifiable, the amplification $|f'(p)|$ is not. In fact [exercise, or read on] $|f'(p)|$ is simply the strength d of the dipole in C , and we have seen how the latter is fixed in the course of the above construction. Put differently, $|f'(p)| = [\Phi]/4$.

We can now return to the construction in [12.25] and see why it is missing two degrees of freedom. To obtain the general mapping between the exteriors of R and D , we may take this newly constructed *general* mapping between the interiors of $R^\dagger$ and C , then reverse (i.e., repeat) the inversions which took us from [12.25] to [12.26].

In the above process of generalization the fundamental step was moving the dipole at q to an arbitrary interior point p . Since the flow in [12.26b] did not undergo any fundamental change, inversion returns it to a flow round D like that shown in [12.25]. However, to send $R^\dagger$ back to the boundary of R we must invert in the circle of unit radius centred at q (which we now take to be the origin), and *the flow inside $R^\dagger$ of the dipole at p inverts to the flow outside R of a dipole placed at $(1/\bar{p})$* . Figure [12.17] illustrates such a flow when R is a disc.

The construction in [12.25] is now recognizable as the special $p = q$ in which the dipole outside R has been placed at ∞ rather than at a general point. Here are our two missing degrees of freedom: there was nothing wrong with choosing to place one of the dipoles (the one outside D) at ∞ , but we then (unnecessarily) insisted on placing the other dipole there too. In terms of the resulting mapping between the exteriors, this amounted to insisting that ∞ map to ∞ , whereas the general construction allows us to map any point outside R to ∞ by placing a dipole there.

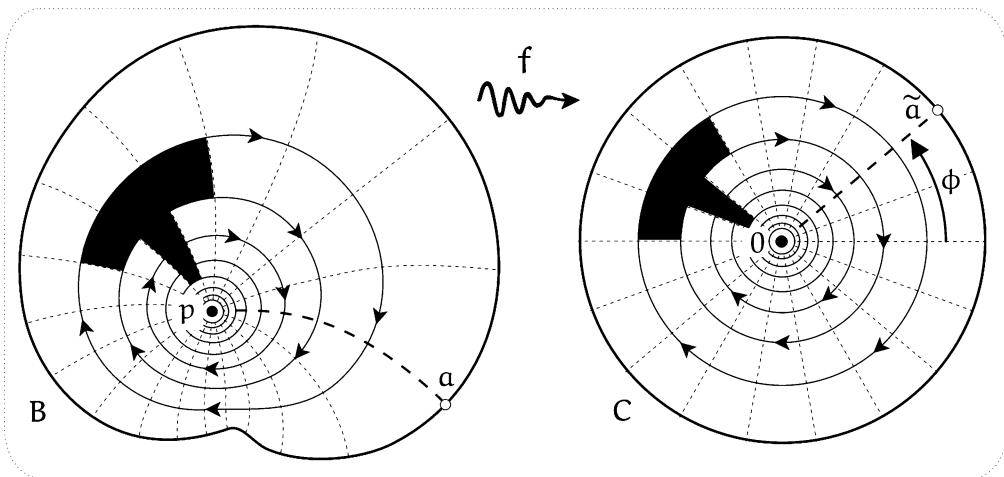
12.6.4 Interior Mappings, Vortices, and Sources

Having had the inspiration to construct mappings by means of multipoles, it is natural to wonder if we may simplify this dipole method by employing the most

primitive of all multipoles, the source. However, if we continue to think in terms of fluid flows then this idea cannot work. For if we place a source inside the region we wish to map, the fluid flowing out of the source and into the region has nowhere to go! The only way out is to also insert a sink of equal strength⁵, thereby creating a doublet. But this is really no improvement over our earlier dipole construction, for a dipole is merely a limiting form of a doublet.

However, as we shall explain shortly, the failure of this attempted use of sources is merely a consequence of our thinking in terms of fluid flowing within a strangely-shaped pool; by thinking in terms of electric fields or heat flows, we *can* use sources to construct mappings.

Continuing with the fluid interpretation for the time being, there does exist a simple alternative to the dipole method, but in place of sources we must use *vortices*. Let B be the boundary curve of the simply connected region which we wish to map to the interior of the unit circle C . To construct the mapping, insert a vortex into B at an arbitrary interior point p , and insert another vortex into C at 0 . See [12.27]. Dividing each flow into k -cells, as before, conformal mappings leap from the page. Let's examine the details.



[12.27] Mapping interiors with vortices instead of dipoles. Place a source of strength S at p inside B , and a source of strength $\tilde{S}$ at the centre of C . Dividing each flow into small, square k -cells, we readily obtain a conformal mapping from one interior to the other, with p mapping to 0 . But this is only possible if the number N of k -cells abutting B equals the number $\tilde{N}$ abutting C . This implies that **the two vortices must be of equal strength**: $S = Nk = \tilde{N}k = \tilde{S}$. Finally, we may specify which boundary point $\tilde{a}$ of C corresponds to the boundary point a of B . Now the mapping is completely tied down, and, for example, the black T-shape is mapped to the black T-shape.

⁵ Of course we could insert several sinks whose strengths summed to that of the source, but that would be messier still.

Firstly, in defining the correspondence between the grids we are forced to map p to 0 . Secondly, having chosen the strength S of the vortex in B , the strength $\tilde{S}$ of the one in C must be chosen so that (as illustrated) the number $\tilde{N}$ of k -cells abutting C is equal to the number N abutting B . This is the same idea as in the dipole construction, but the answer in this case is much simpler: put $\tilde{S}$ equal to S .

The explanation is as simple as the answer. Concentrate on one of the vortices, say the one in B . Its strength S is, by definition, the circulation along any simple loop round p , and we may conveniently take this to be one of the streamlines. The equipotentials cut this streamline into N segments, each having circulation k . Thus $S = Nk$, and likewise $\tilde{S} = \tilde{N}k$. The condition $\tilde{S} = S$ follows immediately.

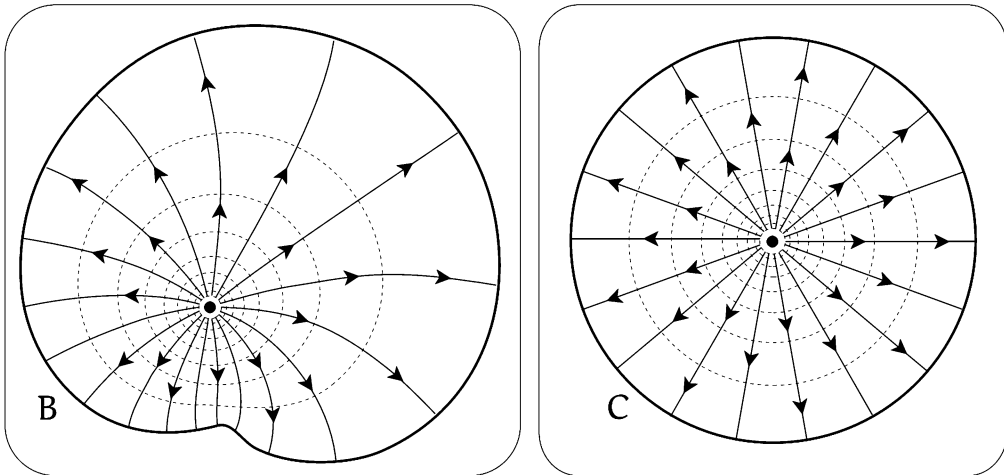
Finally, observe that we have not yet pinned down the mapping. In both [12.25] and [12.26] there were stagnation points which we were forced to identify, thereby tying down the mapping. However, here there are none, and we must do the job by hand. A common procedure goes like this. We know that the streamline B maps to the streamline C , and we may now insist that a particular point a on the first maps to a particular point $\tilde{a}$ on the second.

If we wished to be more definite, we might do the following. Consider the heavily-dashed equipotential in B which exits p travelling due east, and choose a to be its intersection with B . See [12.27]. If we now choose $\tilde{a} = e^{i\phi}$ then the mapping f between the two grids is completely tied down. For example, the black T-shape in B maps to the black T-shape in C . The only advantage of specifying a and $\tilde{a}$ in this particular way is that ϕ then has a simple interpretation in terms of f , for it is clearly the twist of f at p : $\phi = \arg[f'(p)]$.

As with the dipole method, we have obtained the full three-parameter family of mappings: *two* for the point p that f maps to the centre of C , and *one* for the twist of f at p .

Now let us turn to other physical interpretations of this construction. Taking the dual of the flows in [12.27], we obtain [12.28]. The equipotentials have become streamlines, and the streamlines (B and C in particular) have become equipotentials. By the same reasoning as before, the strengths of the two sources manifest themselves geometrically as k times the number of streamlines emanating from each, and these must be set equal to each other in order to construct the conformal mapping between the two grids.

Let us digress briefly. Since the strength of a source is measured by the number of k -flux tubes emanating from it, a conformal mapping sends a source to another source of equal strength. Obviously, the same goes for a sink. Returning to [12.26], we can understand why the strength d of the dipole inside C is the amplification of f at p . The dipole of unit strength at p may be thought of as a doublet in which the source and sink are an infinitesimal distance ϵ apart, and each has strength $(1/\epsilon)$. The conformal mapping f sends this to another doublet at 0 : the strengths of



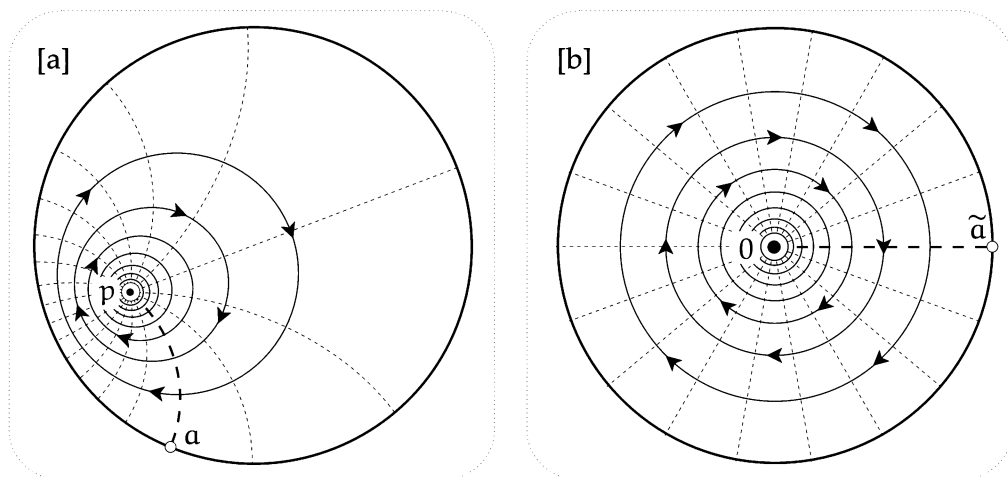
[12.28] The source construction dual to the vortex construction of the previous figure. In order to make physical sense, here we must replace the fluid flow with an electrostatic field, the sources being electric charges. By the same reasoning as before, the strengths of the two charges manifest themselves geometrically as k times the number of electric field lines emanating from each, and these must be set equal to each other in order to construct the one-to-one conformal mapping between the two grids.

the source and sink are preserved, but their separation (and hence the strength of dipole) is amplified by $|f'(p)|$, as was to be shown.

Returning to the physical interpretation of [12.28], we have already observed that such pictures of sources make no sense when thought of as fluid flows. However, they do make perfect sense when thought of as electrostatic fields. Imagine that the dark regions in [12.28] represent cross-sections of blocks of copper through which we have bored “cylindrical” holes with cross-sections given by B and C. Now imagine that p and 0 are the cross-sections of two long, very thin, uniformly (and equally) charged wires running down these holes. The electrostatic fields generated by these wires automatically have B and C as equipotentials, so [12.28] faithfully represents these fields, with the streamlines being the electric field lines.

We may also interpret [12.28] in terms of heat flows. Imagine that the white shapes bounded by B and C have been cut from a sheet of heat-conducting metal, and imagine that the dark regions are filled with ice, thereby maintaining B and C at constant temperature (i.e., potential). If heat is introduced at a steady rate at p and at 0 then the flow of heat will eventually settle down to the one in [12.28], with the dashed equipotentials being the isotherms.

It must be observed, however, that a point source of heat is a much less physical concept than its electrostatic analogue, which may be realized (as we have said) by a very thin charged wire. The reason is that the potential function Φ becomes



[12.29] Using flows to construct the most general automorphism of the disc. Returning to [12.27], suppose we choose $B = C$ to be the unit circle. Then the vortex flow shown here in [a] may be viewed as a one-to-one conformal mapping to the flow in [b], i.e., it is an automorphism of the disc. Recall that we previously used the Symmetry Principle to find the formula for the most general such automorphism: (3.52), on page 201. We can now rederive this formula by finding the flow in [a], which we accomplish by means of the method of images, in the next figure.

arbitrarily large as we approach a source. In electrostatics this does not present any difficulties, but in the heat-flow interpretation $-\Phi$ represents temperature, so the metal in the vicinity of such an imagined source of heat would *vaporize*! For an excellent discussion of such physical distinctions, see [Maxwell \(1881\)](#), p. 51 onwards.

While many readers may be unfamiliar with electrostatics, few will be unfamiliar with heat. Thus, ignoring the above objection, we shall persist in expressing ideas mainly in the language of heat rather than electrostatics.

12.6.5 An Example: Automorphisms of the Disc

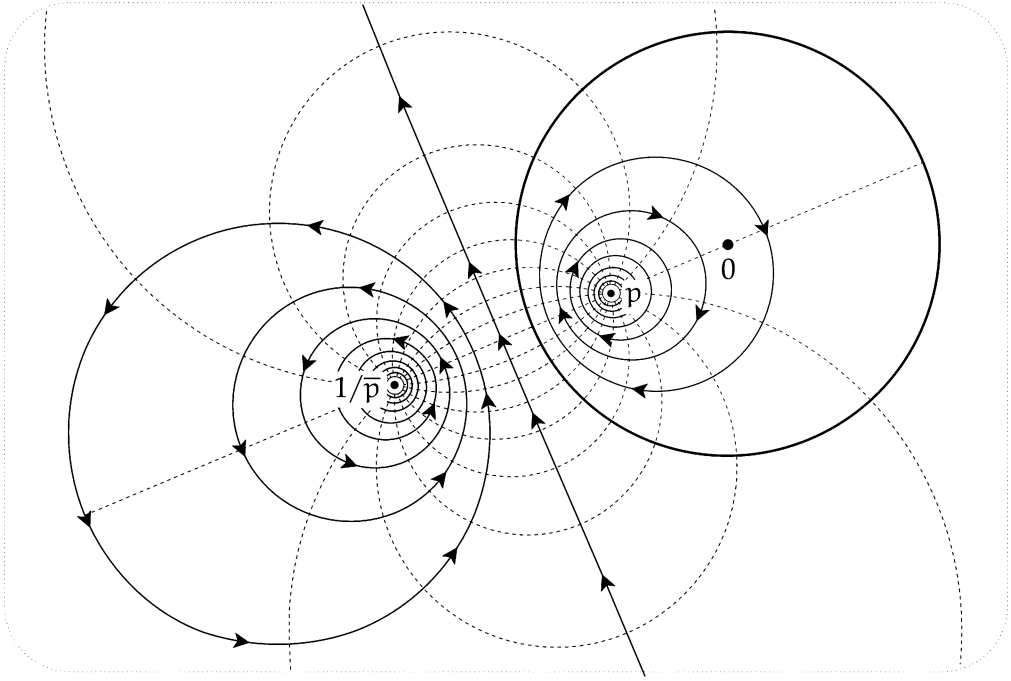
Let us explicitly carry out the construction in [12.27] for the case $B = C$, and thereby re-obtain the automorphisms of the unit disc from a fresh point of view.

Figure [12.29] illustrates the construction in this case. Previously we did not specify the strength S of the two vortices because the choice had no effect on the resulting mapping, but let us now choose $S = -2\pi$. Taking the zero of flux and circulation in [12.29b] to be at 1, the complex potential for [12.29b] is then

$$\tilde{\Omega}(w) = i \log w.$$

Now consider the flow in [12.29a]. We may find its complex potential by the method of images⁶. That is, we superpose the real vortex at p with its reflection

⁶ However, note that we are *not* using (12.18). See Ex. 14 to understand why.



[12.30] Using the method of images to construct the most general automorphism of the disc. To obtain the flow in [12.29a], we superpose a vortex at p with an equal and opposite vortex at its reflection, $1/\bar{p}$.

in $\mathbb{C}$, namely, a fictitious vortex of equal and opposite strength at $(1/\bar{p})$. See [12.30]. The complex potential of [12.29a] is thus

$$\begin{aligned}\Omega(z) &= i \log(z - p) - i \log(z - 1/\bar{p}) - \gamma \\ &= i \log \left[\frac{z - p}{\bar{p}z - 1} \right] - \delta\end{aligned}\quad (12.20)$$

where γ and δ are constants.

In [12.29b] we chose to measure circulation and flux from a boundary point, and we now choose to do the same in [12.29a]. In terms of the above equation this is equivalent [exercise] to demanding that the constant δ be an arbitrary *real* number.

To pin down the mapping we must now choose a and $\tilde{a}$. Instead of doing this as we did in [12.27], let us this time choose them to be the two boundary points from which we have elected to measure circulation and flux in the two flows. As previously discussed, the mapping is then effected by equating circulation and flux:

$$\tilde{\Omega}(w) = \Omega(z).$$

Solving for w , we do indeed recover the familiar automorphisms of the disc:

$$w = f(z) = e^{i\delta} \left[\frac{z - p}{\bar{p}z - 1} \right]. \quad (12.21)$$

Of course we have not done anything really new, for we have used the method of images, and this is merely a disguised form of the Symmetry Principle by means of which the result was *originally* obtained. Nevertheless, we hope you have found it instructive—and delightful!—to be able to understand these automorphisms from a new, physical point of view.

12.6.6 Green's Function

We now return to [12.28] and to the heat-flow approach to the construction of conformal mappings.

Figure [12.31] is essentially a copy of [12.28], but with a few added details which we will need shortly. We supply heat at the constant rate 2π to the point p of the region R while holding the temperature all around the boundary B at the constant value 0. After the heat flow has settled down, the temperature in R will be a well defined (except at p) harmonic function $\mathcal{G}_p(z)$ called the *Green's function* of R with *pole* at p . Note the new, special sense of the word “pole”.

As an example, let us obtain the Green's function for the unit disc. In [12.29a] we considered the flow of a vortex of strength -2π inserted at p . And because we chose a point on C as the zero of circulation and flux, the stream function vanished on the boundary streamline. The dual of this flow is therefore precisely what we are after: a source of strength 2π at p , with the boundary at zero potential. By (12.20), the complex potential is

$$\Omega(z) = \Phi(z) + i\Psi(z) = \log \left[\frac{z-p}{\bar{p}z-1} \right] + i\delta,$$

where δ is a real constant. Thus the temperature in the disc is given by

$$\mathcal{G}_p(z) = -\Phi(z) = -\ln \left| \frac{z-p}{\bar{p}z-1} \right|. \quad (12.22)$$

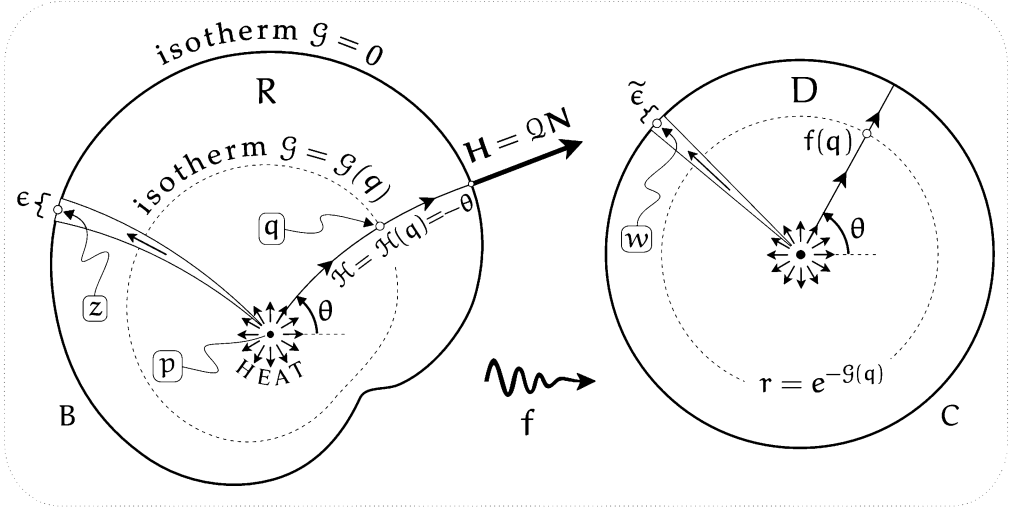
Note that $\mathcal{G}_p(z) = -\ln|f(z)|$, where $f(z)$ is any of the one-parameter family of mappings to the unit disc such that $f(p) = 0$. We shall see that this is true quite generally.

Also note that this Green's function has a very interesting symmetry property:

$$\mathcal{G}_p(q) = \mathcal{G}_q(p).$$

Thus, with the boundary packed with ice, the steady-state temperature at q due to a point source of heat at p is the same as the temperature at p when the source is instead at q . Remarkably, we shall see that this symmetry holds true for the Green's function in a region of *arbitrary* shape!

The Green's function is a powerful tool in several areas of mathematics and physics. For the time being, we will concern ourselves primarily with its relationship to the conformal mappings $w = f(z)$ from the interior of B to the unit disc. In the next section we shall discuss another important application.



[12.31] The Green's Function $\mathcal{G}_p(z)$ of a region R is defined to be the harmonic equilibrium temperature distribution when heat is supplied at constant rate 2π to the point p while holding the temperature all around the boundary B at the constant value 0. We may then construct a harmonic dual $\mathcal{H}_p(z)$, that is, a harmonic function whose level curves are the paths along which the heat flows (orthogonally through the isotherms $\mathcal{G}_p = \text{const.}$) from p to the boundary. Thus knowledge of $\mathcal{G}_p$ is sufficient for the construction of the complete complex potential, $\Omega(z) = -[\mathcal{G}_p(z) + i\mathcal{H}_p(z)]$, and the conformal mapping of R to D is then explicitly given by $w = f(z) = e^{\Omega(z)} = e^{-\mathcal{G}} e^{-i\mathcal{H}}$.

Returning to the general case in [12.31], suppose $\mathcal{G}_p$ is known. As previously described, we may then construct a harmonic dual $\mathcal{H}_p(z)$, that is, a harmonic function whose level curves are the paths along which the heat flows (orthogonally through the isotherms $\mathcal{G}_p = \text{const.}$) from p to the boundary. Thus knowledge of $\mathcal{G}_p$ is sufficient for the construction of the complete complex potential, $\Omega(z) = -[\mathcal{G}_p(z) + i\mathcal{H}_p(z)]$.

Consider the behaviour of $\mathcal{G}_p$ in the immediate vicinity of p . Physical intuition leads us to expect, irrespective of the temperatures assigned to B , that the flow out of p will almost be like that of an isolated source, so that $\Omega(z) \approx \log(z - p)$. Thus if $z = p + \rho e^{i\theta}$ then

$$\mathcal{G}_p(z) \approx -\ln \rho, \quad (12.23)$$

for small values of ρ .

Similarly, $\mathcal{H}_p(z) \approx -(\theta + \phi)$, where ϕ is a constant. The freedom in choosing ϕ is equivalent, as we shall see, to the freedom of choosing which point of B maps to a particular point of C . If we choose $\phi = 0$ then the value of $\mathcal{H}_p$ at a typical point q has a particularly simple interpretation. See [12.31]. Following the flow of heat back from q to p , the angle at which it enters p is $-\mathcal{H}_p(q)$.

Returning to (12.23), the precise version is that $\mathcal{G}_p$ differs from $-\ln(\rho)$ by a function $g_p(z)$ which is harmonic throughout R :

$$\mathcal{G}_p(z) = -\ln \rho + g_p(z).$$

Since $\mathcal{G}_p$ vanishes on the boundary, the values of g_p on B are determined directly by the shape of B and the location of p within it:

$$g_p(z) = \ln \rho.$$

The problem of constructing $\mathcal{G}_p$ is therefore equivalent to the problem of finding the function g_p that takes these values on B and is harmonic throughout the interior. This is an example of the type of “Dirichlet problem” to be discussed in the next section. The solution of this problem also gives us $\mathcal{H}_p$, for we may construct a harmonic dual h_p of g_p , and then $\mathcal{H}_p = -\theta + h_p$.

To construct a conformal mapping $w = f(z)$ to the unit disc D such that $f(p) = 0$, we may (as previously explained) equate the complex potential $\Omega(z)$ of the flow inside R with the complex potential $\tilde{\Omega}(w) = \log w$ in D of the heat source at 0, with the boundary at zero temperature. Thus

$$w = f(z) = e^{\Omega(z)} = e^{-\mathcal{G}} e^{-i\mathcal{H}}. \quad (12.24)$$

As illustrated in [12.31], the dashed isotherm at temperature $\mathcal{G}(q)$ is mapped to the dashed circle of radius $e^{-\mathcal{G}(q)}$, and the streamline $\mathcal{H} = \mathcal{H}(q) = -\theta$ entering p at angle θ maps to the ray entering 0 at angle θ . Thus f has zero twist at p . This is the significance for f of our previously choosing $\phi = 0$; in general $\arg[f'(p)] = \phi$.

Now that we possess the mapping f , any harmonic temperature distribution $T(z)$ on R may be conformally transplanted to a harmonic function $\tilde{T}(w)$ on D (and vice versa) by assigning equal temperatures to corresponding points of the two regions: $\tilde{T}[f(z)] \equiv T(z)$. In particular, the values of T on B are transplanted to C .

We can use this to understand the following. If we know a conformal mapping $w = f(z)$ from R to D such that $f(p) = 0$, then the Green’s function of R with pole at p is

$$\mathcal{G}_p(z) = -\ln |f(z)|.$$

To see this, consider the temperature distribution $\tilde{\mathcal{G}}_p(w)$ in D obtained by transplanting $\mathcal{G}_p$ with f . We know that this conformal transplantation preserves the harmonicity of the original, that it sends the source at p to an equal source at $f(p) = 0$, and that the vanishing temperature on B is transplanted to vanishing temperature on C . Thus this temperature distribution $\tilde{\mathcal{G}}_p(w) = \mathcal{G}_p[f^{-1}(w)]$ in D must be the Green’s function with pole at 0, and we already know that this is $-\ln |w|$. Done.

Exactly the same reasoning yields the following generalization. Let $J(z)$ be a one-to-one conformal mapping of R to some other simply connected region S with

boundary Y . Then J conformally transplants the Green's function $\mathcal{G}_a(z)$ of R with pole at a to the Green's function of S with pole at $J(a)$. In particular, the streamlines of the flow in R map to the streamlines of the flow in S . In this sense, *the concept of the Green's function is conformally invariant*.

This result immediately yields the following generalization of (12.6.6) to the case where the pole of the Green's function of R is an arbitrary point s , rather than $f^{-1}(0)$. From (12.22), we know the formula for the Green's function in the disc when the pole is moved from 0 to $f(s)$. Conformal transplantation by means of f^{-1} carries this pole to the desired point s of R , so

$$\mathcal{G}_s(z) = -\ln \left| \frac{f(z) - f(s)}{f(s)f(z) - 1} \right|.$$

As a bonus, notice that this general formula establishes the previously claimed "symmetry property" of the Green's function:

$$\mathcal{G}_s(z) = \mathcal{G}_z(s).$$

For a more common approach to the symmetry property, see Ex. 15.

We end this section with a result which we will need later. The analogue for heat flows of the velocity of a fluid flow is the *heat flow vector* H ; in the present case, $H = -\nabla \mathcal{G}$. Let us call its magnitude $Q \equiv |H|$ the *local heat flux*; this is the analogue of fluid speed, and it represents the heat flux (per unit length) across a short line-segment at right angles to the flow. Since B is an isotherm, H is orthogonal to B , and so the local heat flux at a boundary point z may be expressed as

$$Q(z) = -\frac{\partial \mathcal{G}}{\partial n}, \quad (12.25)$$

where n measures distance in the direction of N , the outward unit normal vector to B (see [12.31]). We can now state the result:

$$\text{At a boundary point } z, \text{ the local heat flux is equal to the amplification of } f : Q(z) = |f'(z)|. \quad (12.26)$$

For example, using (12.21), the result predicts that the local heat flux at the boundary of the unit disc is given by [exercise]

$$Q(z) = |f'(z)| = \frac{1 - |p|^2}{|z - p|^2} = \frac{1 - |p|^2}{\rho^2}. \quad (12.27)$$

This formula will play a central role in the next section. Of course $Q(z)$ may instead be calculated directly by substituting (12.22) into (12.25), but this is a little easier said than done [exercise].

The general result (12.26) can be understood very intuitively. See [12.31]. With an infinitesimal value of k , consider the shaded k -flux tube emanating from p which hits B at z , and let its width there be ϵ . Its image under f is a k -flux tube emanating

from 0 and hitting C at $w = f(z)$. [Remember, f was originally *defined* to have this property!] Let $\tilde{\epsilon}$ be the width of this image tube at w . Since the segment of B at z of length ϵ is amplitwisted by $f'(z)$ to the segment of C at w of length $\tilde{\epsilon}$,

$$|f'(z)| = \frac{\tilde{\epsilon}}{\epsilon}.$$

Next, recall that the width of a k -flux tube at any given point is equal to k divided by the local heat flux [previously fluid speed] at that point. Since the local heat flux is constant on C , its value at w is simply the ratio of the strength of the source at 0 to the perimeter of C , and by construction this ratio is $(2\pi/2\pi) = 1$. Since the local heat flux at z is $\mathcal{Q}(z)$, we see that

$$\tilde{\epsilon} = k/1 \quad \text{and} \quad \epsilon = k/\mathcal{Q}.$$

Thus

$$|f'(z)| = \frac{\tilde{\epsilon}}{\epsilon} = \frac{k}{k/\mathcal{Q}} = \mathcal{Q}(z),$$

as was to be shown.

12.7 Dirichlet's Problem

12.7.1 Introduction

Consider a steady heat flow within a metal plate whose faces are insulated. Other than at singularities, the temperature $T(z)$ is then a harmonic function, and the (locally) sourceless heat flow vector field is $H = -\nabla T$.

Let us measure the temperature around the circumference C of a circle of radius R , the interior of which is free of sources and sinks, and the centre of which we may conveniently choose to be the origin. As $z = R e^{i\theta}$ moves round C , we may express the measured temperature as a function of the angle: $T = T(\theta)$. We hope that it may seem physically plausible that these values actually determine the temperature at any interior point a . Indeed, if $a = 0$ then we know [from Gauss's Mean Value Theorem] that the temperature at the centre of C is simply the average of the temperatures on C :

$$T(0) = \langle T \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} T(\theta) d\theta. \quad (12.28)$$

Eventually we will discover that the generalization of this result to the case $a \neq 0$ is given by

$$T(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\frac{R^2 - |a|^2}{|z - a|^2} \right] T(\theta) d\theta. \quad (12.29)$$

Writing $a = r e^{i\alpha}$ ($r < R$), and appealing to the cosine formula [exercise], this is usually written as

$$T(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\frac{R^2 - r^2}{R^2 + r^2 - 2Rr \cos(\theta - \alpha)} \right] T(\theta) d\theta.$$

This is called *Poisson's formula*, and the quantity in square brackets is called the *Poisson kernel*, which we shall write as $\mathcal{P}_a(z)$.

Formula (12.29) says that $T(a)$ is a *weighted* average of T on C , the temperature of each element of C contributing to $T(a)$ in proportion to its weight $\mathcal{P}_a(z)$. Notice that $\mathcal{P}_a(z)$ dies away inversely with the square of the distance between a and the element of C , so that if one element is twice as far from a as another, its influence on the temperature at a is only one quarter as great. If $a = 0$ then all parts of C have equal influence (for all are equally far from a) and you can see that we do recover (12.28).

Poisson's formula is connected with the following important and difficult issue. Instead of dealing with a pre-existing harmonic function, *Dirichlet's problem* demands that we arbitrarily (but piecewise continuously) assign values to the boundary of a simply connected region R and then seek a continuous harmonic function in R which takes on these values as the boundary is approached.

In the case of the disc, H. A. Schwarz demonstrated that not only does the solution to Dirichlet's problem exist, but it is explicitly given by (12.29). If we are handed the piecewise continuous values $T(\theta)$ on C then we may construct a function $T(a)$ in the interior according to Poisson's recipe. Schwarz's solution then amounted to showing that $T(a)$ is automatically harmonic, and that as a approaches a boundary point at which $T(\theta)$ is continuous, $T(a)$ approaches the given value $T(\theta)$. Let us begin to explain all this.⁷

12.7.2 Schwarz's Interpretation

In 1890 Schwarz⁸ discovered a lovely geometric interpretation of formula (12.29) which deserves to be far better known than it is:

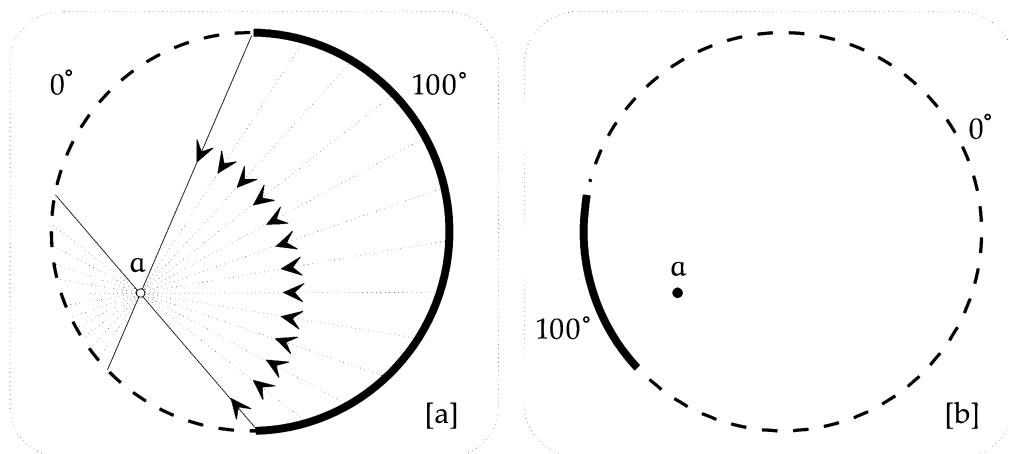
To find the temperature at a , transplant each temperature on C to the point directly opposite to it as seen from a , then take the average of the new temperature distribution on C . (12.30)

Schwarz deduced this from Poisson's formula, itself derived by computation. We shall instead demonstrate his result directly and geometrically, only then producing the Poisson formula as a corollary.

The example in [12.32] illustrates the beauty of (12.30). In [12.32a] half of C is kept at 100 degrees with steam, while the other half is kept at 0 degrees with ice.

⁷ Much of the following material previously appeared in Needham (1994).

⁸ Schwarz (1972b).



[12.32] In 1890 Schwarz discovered a beautiful geometrical interpretation of Poisson's formula. [a] Transplant each temperature on C to the point directly opposite to it as seen from a , thereby obtaining the new temperature distribution shown in [b]. Then the temperature at a is simply the average of the new temperature distribution on C .

Being close to the cold side, we would expect a to be cool. Figure [12.32b] shows the new temperature distribution obtained by projection through a . It is now vividly clear how the distant hot semicircle is 'focused' through a onto a much smaller arc, yielding a low average temperature on C and hence a low temperature at a itself.

To begin to establish (12.30), recall the conformal invariance of harmonic functions: if $T(z)$ is any harmonic function and $h(z)$ any conformal mapping, then $T(h(z))$ is automatically harmonic, with $z^* = h(z)$.

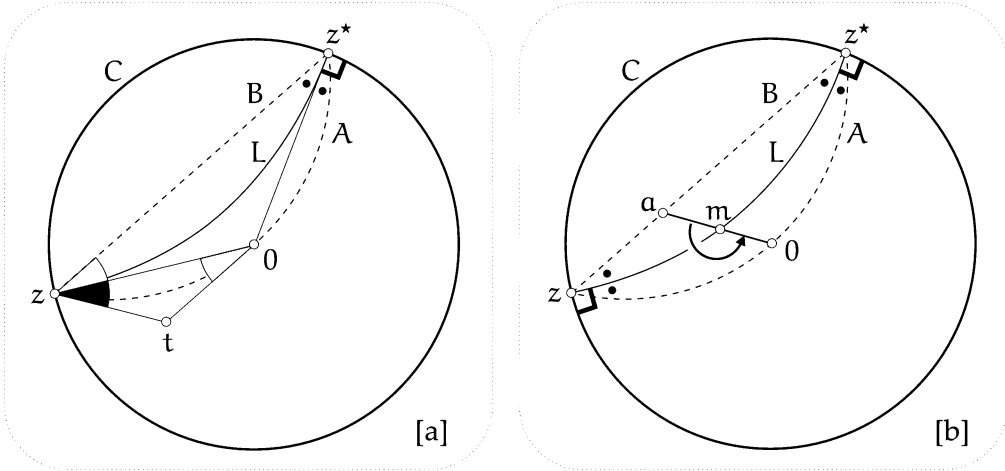
Suppose now that $h(z)$ maps the disc to *itself*. If $z = R e^{i\theta}$ lies on C then so does $z^* = R e^{i\theta^*}$, and since we suppose that we have measured the temperature all round C , we therefore know the temperature $T(\theta^*)$ at z^* . Having the values of $T(\theta^*)$, we may now compute the integral in (12.28) for the harmonic function $T[h(z)]$ to obtain

$$T(0^*) = \frac{1}{2\pi} \int_{-\pi}^{\pi} T(\theta^*) d\theta, \quad (12.31)$$

in which it should be stressed that the averaging is still taking place with respect to the angle of z , not its image z^* .

We may interpret (12.31) as follows: the temperature at 0^* is the average of the new temperature distribution on C obtained by transplanting the temperature measured at each z to the new location z^* . We are now half way to Schwarz's result. To find the temperature at a we must find a conformal mapping of the disc to itself such that 0 is sent to a , then take the average of the new temperature distribution.

But viewing the disc as the Beltrami–Poincaré model of the hyperbolic plane, we are already very familiar with such mappings! Peek at [12.33b], to which we



[12.33] Hyperbolic geometry proof of Schwarz's interpretation of Poisson's formula. According to (12.31), if we can find a conformal mapping of the disc to itself that sends 0 to α , sending boundary points z to z^* , then we may find the temperature at α by averaging these transplanted temperatures at z^* . But [b] reminds us that we have already met such a transformation in hyperbolic geometry! In the Beltrami–Poincaré disc, let m be the midpoint (in the hyperbolic sense) of the line-segment 0α , then the half-turn $z \mapsto z^* = M_\alpha(z)$ of the hyperbolic plane about m interchanges 0 and α : $0^* = \alpha$ (as we desire) and $\alpha^* = 0$. As explained in the text, [a] allows us to prove that in fact z^* lies at the end of the Euclidean chord through α and z , as illustrated, thereby proving Schwarz's result.

shall return in a moment. If m is the midpoint (in the hyperbolic sense) of the line-segment 0α , then the half-turn⁹ $z \mapsto z^* = M_\alpha(z)$ of the hyperbolic plane about m interchanges 0 and α : $0^* = \alpha$ (as we desire) and $\alpha^* = 0$. Thus to establish (12.30) we need only demonstrate the illustrated fact that if z lies on C then z^* lies at the end of the (Euclidean) chord B passing from z through α . In Chapter 3 we derived a formula for $M_\alpha(z)$, so we could easily obtain this result by calculation; however, we prefer a direct geometric approach.

First we need a simple result which is explained in [12.33a]. Consider the family

$$\mathcal{N} \equiv \{\text{circular arcs passing from } z \text{ to } z^*\},$$

where for the moment z^* may be thought of as any given point of C . The figure shows three members of $\mathcal{N}$: the arc A through 0, the Euclidean chord B , and the hyperbolic line L . [Recall that in terms of hyperbolic geometry the members of $\mathcal{N}$ consist of the equidistant curves of L .] The result we need is this:

The Euclidean chord B is the unique member of $\mathcal{N}$ such that L bisects the angle contained by A and B . (12.32)

Since each member of $\mathcal{N}$ is uniquely determined by the direction in which it emerges from z , and since the radius $z0$ is tangent to L at z , this is equivalent to

⁹ See (3.53), p. 201.

the following: if tz and $t0$ are tangent to A at z and 0 , respectively, then the black angle $tz0$ and the shaded angle z^*z0 are equal. The proof is immediate from the figure and is left to the reader.

Now turn your attention to [12.33b], in which z^* is the image of z after a half-turn about m . To finally establish Schwarz's result we must prove the illustrated fact that a lies on the chord B . To do so, consider the image A^* of A . Since the half-turn interchanges the pair z and z^* and the pair 0 and a , A^* must be a member of $\mathcal{N}$ passing through a . But since $L^* = L$, the conformality of the mapping also says that the angle between A^* and L must equal the angle between A and L . We conclude from (12.32) that this arc A^* through z , a , and z^* is none other than B . Done¹⁰.

12.7.3 Dirichlet's Problem for the Disc

Our example in [12.32] was a trifle hasty. For the moment, Schwarz's result merely says how the interior values of a given harmonic function in the disc may be found from the values on C . But in [12.32] we blithely assumed that we could also use it to *construct* such a function in the disc, given arbitrary piecewise continuous boundary values. In other words we assumed Schwarz's solution of Dirichlet's problem for the disc, as outlined in the introduction. We now justify this.

Figure [12.34a] shows a approaching a boundary point z ; also shown are the images (C_1^* and C_2^*) under projection through a of the two small arcs (C_1 and C_2) adjacent to z . If the given boundary values are continuous at z then T is essentially constant on $C_1 \cup C_2$, and so the new temperature distribution is likewise almost constant on $C_1^* \cup C_2^*$. As required, the constructed function $T(a)$ therefore *does* approach $T(z)$ as a approaches z .

Although Dirichlet's problem makes no demands on the behaviour of $T(a)$ as a approaches a boundary point at which T is *discontinuous*, it is easy to see (though not to calculate!) what actually happens. Suppose that the boundary temperature jumps from T_1 to T_2 as we pass from C_1 to C_2 . If a arrives at z while travelling in a direction making an angle $\beta\pi$ with C , then [exercise] $T(a)$ approaches $[\beta T_1 + (1 - \beta)T_2]$. This result is relevant to the representation of discontinuous functions by Fourier series.

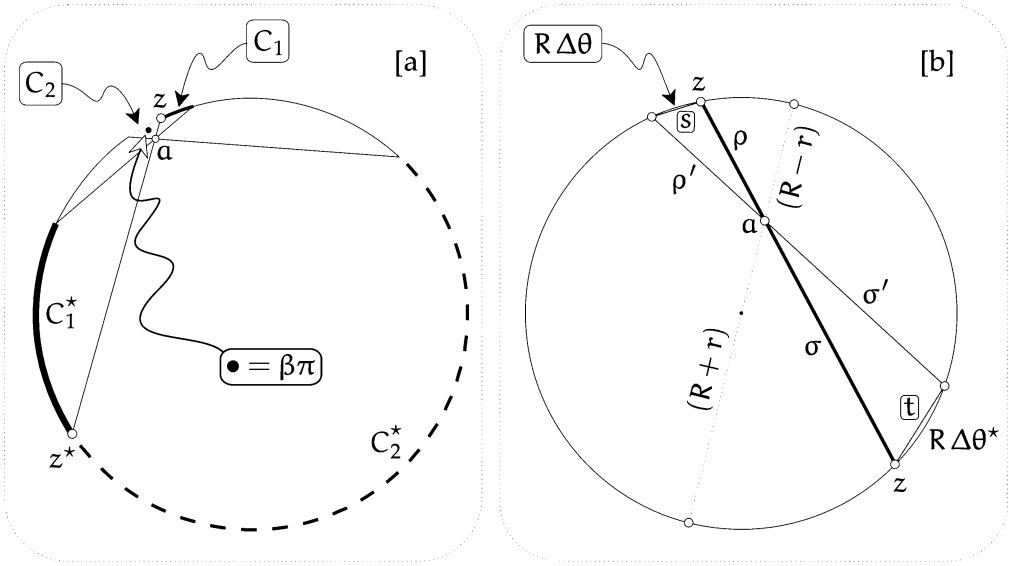
It now only remains to show that the constructed function is indeed harmonic. First we shall pause to recover Poisson's formula in its classical form. We begin by noting that (12.31) may be re-expressed [why?] as

$$T(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} T(\theta) d\theta^*. \quad (12.33)$$

In order to put this into the same form as (12.29), we now require $d\theta^*$ in terms of $d\theta$.

Consider [12.34b], which shows the movement $R\Delta\theta^*$ of z^* resulting from a movement $R\Delta\theta$ of z . These arcs are ultimately equal to the chords t and s , so

¹⁰ This argument is perhaps conceptually clearer than the more elementary one in Needham (1994).



[12.34] Solving Dirichlet's Problem with Schwarz's geometric construction. [a] Suppose that the boundary temperature jumps from T_1 to T_2 as we pass from C_1 to C_2 . If a arrives at z while travelling in a direction making an angle $\beta\pi$ with C , then $T(a)$ approaches $[\beta T_1 + (1 - \beta) T_2]$. [b] To derive Poisson's formula from (12.33), consider the movement $R\Delta\theta^* \asymp t$ of z^* resulting from a movement $R\Delta\theta \asymp s$ of z . So $(\Delta\theta^*/\Delta\theta) \asymp (t/s)$. But t and s are corresponding sides of two similar triangles [shaded], so $(t/s) = (\sigma'/\rho)$. And since $(\sigma'/\rho) \asymp (\sigma/\rho)$, we obtain $\frac{d\theta^*}{d\theta} = \left[\frac{\sigma}{\rho}\right]$. Finally, $\rho\sigma = \rho'\sigma' = (R+r)(R-r) = (R^2 - r^2)$, so $\frac{d\theta^*}{d\theta} = \left[\frac{\sigma}{\rho}\right] = \left[\frac{R^2 - r^2}{\rho^2}\right] = \mathcal{P}_a(z)$. Done.

that $(\Delta\theta^*/\Delta\theta)$ is ultimately equal to (t/s) . But t and s are corresponding sides of two similar triangles [shaded], so $(t/s) = (\sigma'/\rho)$. Finally, since (σ'/ρ) is ultimately equal to (σ/ρ) , we obtain

$$\frac{d\theta^*}{d\theta} = \left[\frac{\sigma}{\rho}\right].$$

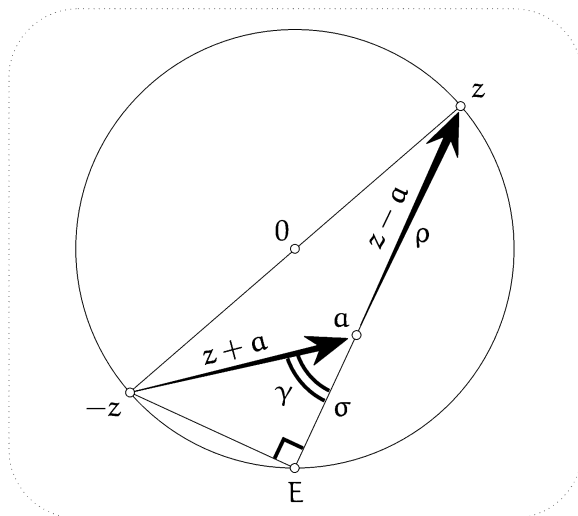
Thus (12.33) becomes

$$T(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\frac{\sigma}{\rho}\right] T(\theta) d\theta. \quad (12.34)$$

Consequently, to derive Poisson's formula we need only show that $[\sigma/\rho]$ is the Poisson kernel $\mathcal{P}_a(z)$. This was precisely how Schwarz, working in the opposite direction, originally deduced his result from Poisson's formula.

Since $\rho\sigma = \rho'\sigma'$ is constant, we may evaluate it for the dotted diameter through a to obtain $\rho\sigma = (R^2 - r^2)$. Thus we do indeed find that

$$\left[\frac{\sigma}{\rho}\right] = \left[\frac{R^2 - r^2}{\rho^2}\right] = \mathcal{P}_a(z).$$



[12.35] Geometric proof that $\mathcal{P}_a(z)$ is harmonic, and the derivation of Schwarz's formula. Since the angle at E is a right angle, we have $\mathcal{P}_a(z) = \left[\frac{\sigma}{\rho} \right] = \frac{|z+a| \cos \gamma}{|z-a|} = \operatorname{Re} \left(\frac{z+a}{z-a} \right)$. Since $\mathcal{P}_a(z)$ is the real part of an analytic function, it is indeed harmonic. Let S be a harmonic dual of T , so that $f = T + iS$ is an analytic function. This function is uniquely defined (up to a constant), and so it must be given by **Schwarz's formula**: $f(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{z+a}{z-a} \right) T(\theta) d\theta$, for this is analytic and has $T(a)$ as its real part. Thus Schwarz was able to resurrect the complete analytic function f everywhere inside C just from the ashes of its real part on C !

As an interesting consequence of the geometric interpretation of the Poisson kernel, we see that (with z fixed) the level curves of $\mathcal{P}_a$ are the circles which are tangent to C at z (i.e., horocycles), with $\mathcal{P}_a = 0$ being C itself.

Returning to the issue of harmonicity, we see that if we permit ourselves differentiation under the integral sign of (12.34), then it is sufficient to show that $[\sigma/\rho]$ is a harmonic function of a . To see that it is, consider [12.35]. Since the angle at E is a right angle, we have

$$\left[\frac{\sigma}{\rho} \right] = \frac{|z+a| \cos \gamma}{|z-a|} = \operatorname{Re} \left(\frac{z+a}{z-a} \right).$$

Because it is the real part of an analytic function of a , $[\sigma/\rho]$ is automatically harmonic, and we are done.

This line of reasoning yields a bonus result. Let S be a harmonic dual of T , so that $f = T + iS = -(\text{complex potential})$ is an analytic function. This function is uniquely defined (up to an additive imaginary constant) and so it must be given by

$$f(a) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{z+a}{z-a} \right) T(\theta) d\theta,$$

for this is analytic and has $T(\alpha)$ as its real part. This result is called *Schwarz's formula*, and it enables us to resurrect the complete analytic function f everywhere *inside* C just from the ashes of its real part *on* C .

12.7.4 The Interpretations of Neumann and Bôcher

If we specify arbitrary piecewise continuous temperatures $T(x)$ along the edge (the real axis) of the upper half-plane, then there is another formula due to Poisson that yields the temperature at any point $\alpha = X + iY$ ($Y > 0$):

$$T(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} \left[\frac{Y}{(X-x)^2 + Y^2} \right] T(x) dx. \quad (12.35)$$

We shall explain this result by reinterpreting (12.33) in terms of elementary hyperbolic geometry. The transition from (12.29) to (12.35) will then be seen as nothing more than a transition between the Beltrami–Poincaré disc and upper half-plane models of the hyperbolic plane. First, however, let us obtain still another geometric interpretation of Poisson's formula.

For simplicity, let us employ the *unit* circle. Consider [12.36]. Let the arc K be heated to unit temperature while the rest of C is kept at zero degrees. By Schwarz's result, the temperature at α is $T(\alpha) = (K^*/2\pi)$, while the temperature at the centre of the circle is $T(0) = (K/2\pi)$.

Next, imagine yourself standing at α looking out at a vast number of thermometers placed at equal intervals along the circle. As you turn your head through a full revolution—remembering to turn your feet!—let $\langle T \rangle_\alpha$ denote the average (over all directions) of the temperatures you see. For example, the average $\langle T \rangle$ occurring in Gauss's Mean Value Theorem is $\langle T \rangle_0$.

In our case $\langle T \rangle_\alpha = (\lambda/2\pi)$, where λ is the angle subtended by K at α . But we see from the figure that¹¹

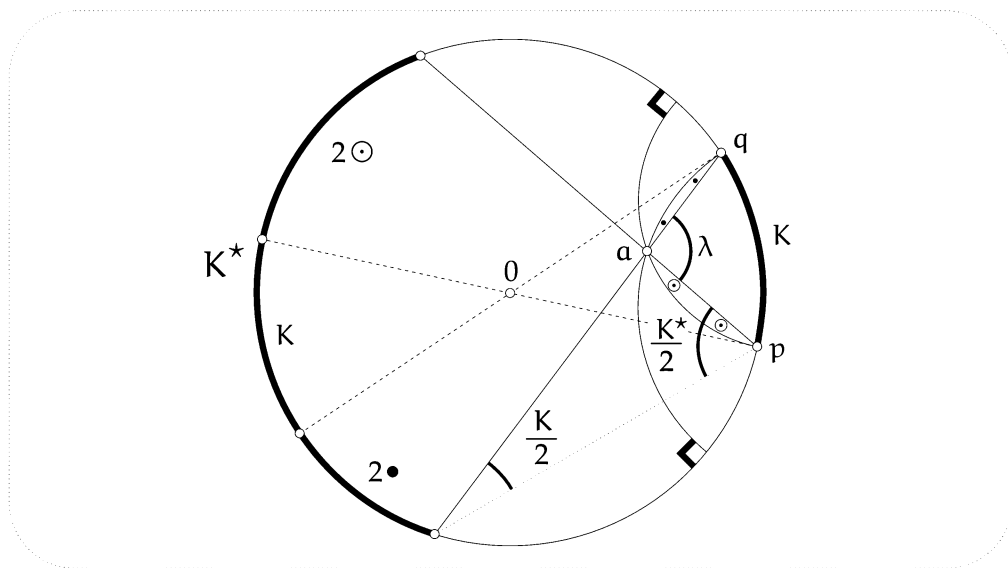
$$\lambda = \frac{1}{2} (K^* + K),$$

so $\langle T \rangle_\alpha = \frac{1}{2} [T(\alpha) + T(0)]$: *the average of the boundary temperatures as they appear to you is equal to the average of the temperature where you are and the temperature at the centre*. It is then easy to see that this is still true if we instead have many arcs at different temperatures, and ultimately a general piecewise-continuous temperature distribution. Thus Poisson's formula may be re-expressed as what we shall call *Neumann's formula*:

$$T(\alpha) = 2 \langle T \rangle_\alpha - T(0).$$

This result is due to [Neumann \(1884\)](#); we merely rediscovered it, as did [Duffin \(1957\)](#) from another point of view. For an interesting generalization, see [Perkins \(1928\)](#).

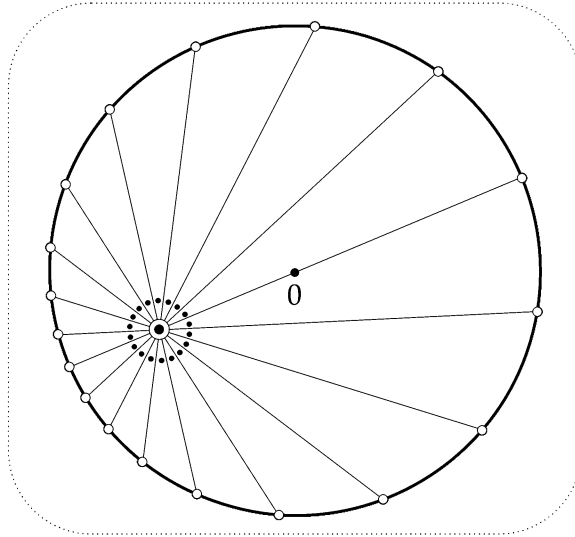
¹¹ Incidentally, this means that the isotherms are the arcs of circles through p and q .



[12.36] Neumann's geometrical interpretation of Poisson's formula. Let the arc K be heated to unit temperature while the rest of C is kept at zero degrees. By Schwarz's result, the temperature at a is $T(a) = (K^*/2\pi)$, while the temperature at the centre of the circle is $T(0) = (K/2\pi)$. Standing at a , turn your head through a full revolution, and let $\langle\langle T \rangle\rangle_a$ denote the average (over all directions of your head) of the temperatures you see. In our case $\langle\langle T \rangle\rangle_a = (\lambda/2\pi)$, where λ is the angle subtended by K at a . But we see from the figure that $\lambda = \frac{1}{2}(K^* + K)$, so $\langle\langle T \rangle\rangle_a = \frac{1}{2}(T(a) + T(0))$: **the average of the boundary temperatures as they appear to you is equal to the average of the temperature where you stand and the temperature at the centre.** This is still true if we instead have many arcs at different temperatures, and ultimately a general piecewise-continuous temperature distribution. Thus Poisson's formula may be re-expressed as what we shall call **Neumann's formula**: $T(a) = 2 \langle\langle T \rangle\rangle_a - T(0)$.

Figure [12.37] is intended to make this result vivid. Turning one's head successively through the same small angle marked $\bullet$ one would see the thermometers located at the white dots on the boundary. The average of their temperatures is then a good approximation [exact as $\bullet \rightarrow 0$] to $\langle\langle T \rangle\rangle_a$, and hence to the average of the temperature where we stand and the temperature at 0. Note how the white dots become crowded together on the part of the boundary nearest us. As anticipated, this part of the boundary therefore has the greatest influence on the temperature where we stand.

To obtain our third and final interpretation of Poisson's formula, imagine that the disc is the Beltrami–Poincaré model of the hyperbolic plane, and that you are once again standing at the point a looking out to K , which is now infinitely far away on the horizon. How big does K appear to you in this distorted geometry? To a Godlike observer looking down on this model of the hyperbolic plane, the straight



[12.37] Visualizing Neumann's geometrical interpretation of Poisson's formula. Turning one's head successively through the same small angle marked $\bullet$ one would see the thermometers located at the white dots on the boundary. The average of their temperatures is then a good approximation [exact as $\bullet \rightarrow 0$] to $\langle\langle T \rangle\rangle_a$, and hence to the average of the temperature where we stand and the temperature at 0. Note how the white dots become crowded together on the part of the boundary nearest us. As anticipated, this part of the boundary therefore has the greatest influence on the temperature where we stand.

lines along which light travels to you now appear to be arcs of circles orthogonal to C , and so *you* see the angular size of K as being

$$\text{hyperbolic angle} = \lambda + (\bullet + \odot).$$

But we see in the figure that

$$(\bullet + \odot) = \frac{1}{2} (K^* - K),$$

and hence we obtain the following remarkable fact:

$$\begin{aligned} \text{hyperbolic angle} &= \frac{1}{2} (K^* + K) + \frac{1}{2} (K^* - K) \\ &= K^* \\ &= 2\pi T(a). \end{aligned}$$

The temperature where you are is simply proportional to how big K looks! [The result can also be obtained directly by appealing to the conformal and circle-preserving nature of the hyperbolic half-turn $M_a(z)$ considered earlier.]

Reinterpreting (12.33), we now see that $d\theta^*$ is simply the hyperbolic angle subtended at a by the element of C : *the temperature of each element of C contributes to*

the temperature at an interior point in proportion to its hyperbolic size as seen from that point. Much as we did in the Euclidean case, let $\prec T \succ_a$ denote the average of the temperatures you see on the horizon of the hyperbolic plane as you turn your head through a full revolution while standing at a . We have thus found what we shall call *Bôcher's formula*:

$$T(a) = \prec T \succ_a . \quad (12.36)$$

This result (exceeding even the beauty of Schwarz's) is due to [Bôcher \(1898\)](#), [Bôcher \(1906\)](#). We have chosen to present (12.36) as a consequence of Schwarz's result, but at the end of the section we shall see that it can be understood in a much simpler way.

The analogue of [12.37] is now [12.38]. Standing at the same point as before, and again turning one's head successively through the angle $\bullet$, the figure shows the new locations of the thermometers we see on the boundary. The average of their temperatures is then a good approximation [exact as $\bullet \rightarrow 0$] to $\prec T \succ_a$, and hence to the temperature where we stand. Note how the white dots again become crowded together on the part of the boundary nearest us, so that this part of the boundary has the greatest influence on the temperature where we stand.

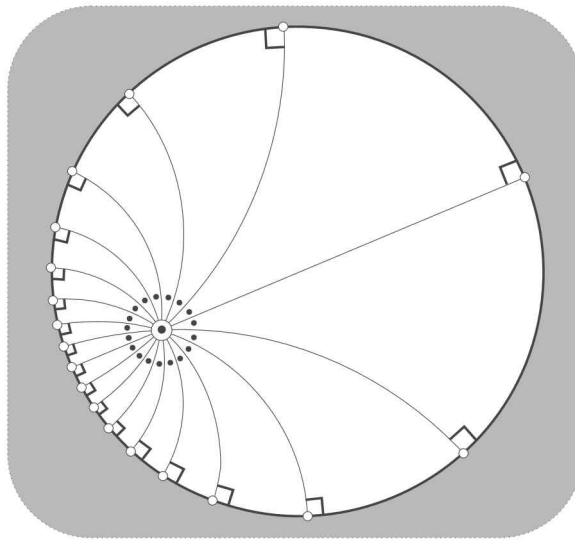
From the vantage point of (12.36), the distinction between (12.28) and (12.33) evaporates. Every point of the hyperbolic plane is on an equal footing with every other, it is merely that the hyperbolic angle $d\theta^*$ happens to coincide with the more familiar Euclidean angle $d\theta$ when $a = 0$.

Formulated in this way, we may carry the result over to the upper half-plane model for hyperbolic geometry. [The full justification for this transition will be explained at the end of the section.] The horizon is now the real axis and 'straight lines' are now (for our Godlike observer) semicircles meeting the real axis at right angles. The temperature where you stand is now the average (as $\bullet \rightarrow 0$) of the temperatures at the white boundary points in [12.39].

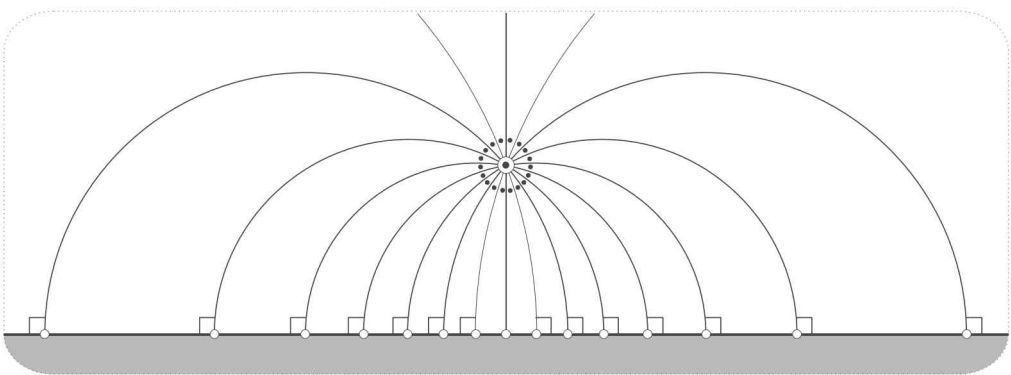
Figure [12.40] analyses this in greater detail. It shows both the hyperbolic angle $\Delta\theta^*$ and the Euclidean angle $\Delta\theta$ subtended at a by the element Δx of the horizon. Thinking of Δx as sufficiently small that $T(x)$ is essentially constant on it, the contribution to the temperature at a is $(1/2\pi) T(x) \Delta\theta^*$. Integrating along the entire horizon we obtain

$$T(a) = \frac{1}{2\pi} \int_{x=-\infty}^{x=\infty} T(x) d\theta^* . \quad (12.37)$$

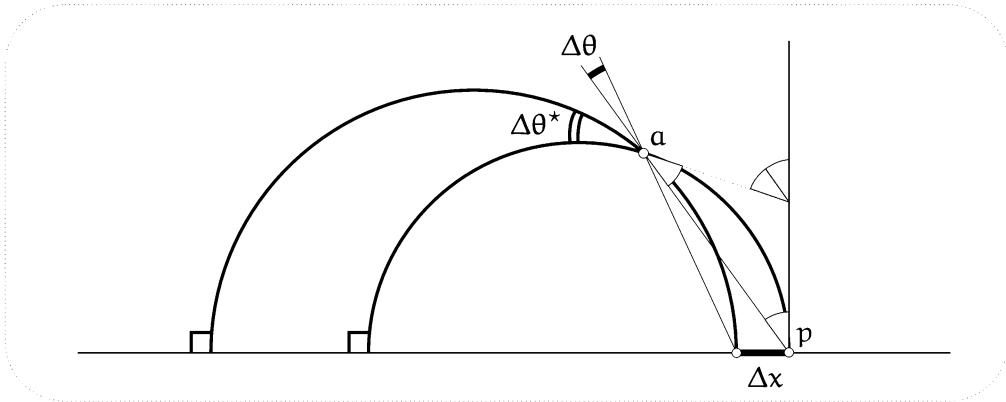
In order to put this into precisely the same form as (12.35), we need to find $(d\theta^*/dx)$. We shall do this via an attractive and rather surprising fact: *The non-Euclidean angle $\Delta\theta^*$ is exactly double the Euclidean one $\Delta\theta$, even if Δx is not small.* To see this, concentrate on the semicircle meeting the axis at p . The angle between the dotted tangent at a and the vertical is clearly double that between the chord ap and the vertical. The result then follows immediately.



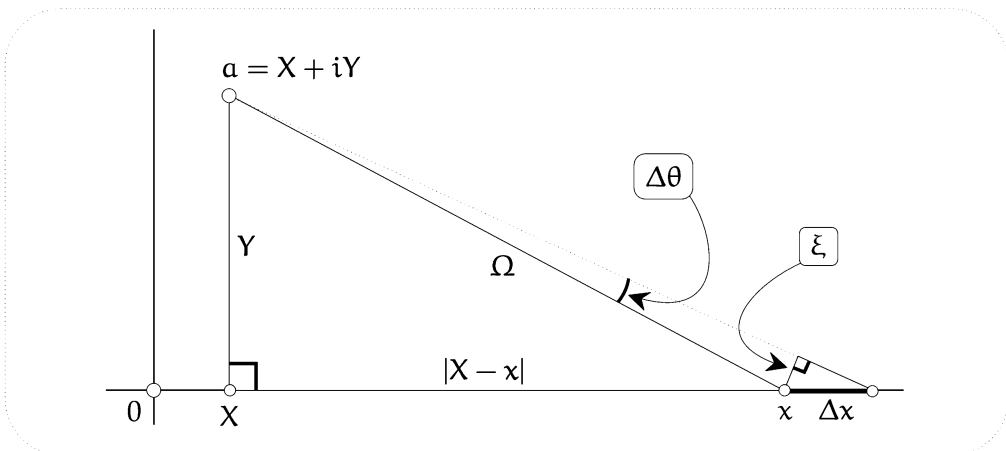
[12.38] **Bôcher's hyperbolic interpretation of Poisson's formula.** Let $\langle T \rangle_a$ denote the average of the temperatures you see on the horizon of the hyperbolic plane as you turn your head through a full revolution while standing at a . Then we have what we shall call **Bôcher's formula**: $T(a) = \langle T \rangle_a$. This figure visualizes this average: turning your head successively through the small angle $\bullet$, the white dots show the locations of the thermometers you see on the boundary as light travels along hyperbolic straight lines to your eye. The average of their temperatures is then a good approximation [exact as $\bullet \rightarrow 0$] to $\langle T \rangle_a$, and hence to the temperature where you stand. Note how the white dots again become crowded together on the part of the boundary nearest to you, so that this part of the boundary has the greatest influence on the temperature where you stand.



[12.39] **Bôcher's hyperbolic interpretation of Poisson's formula in the upper half-plane.** The hyperbolic interpretation of the previous figure carries over without change to the Beltrami–Poincaré upper half-plane model of the hyperbolic plane. The horizon is now the real axis and hyperbolic straight lines are now semicircles meeting the real axis at right angles. The temperature where you stand is now the average (as $\bullet \rightarrow 0$) of the temperatures at the white boundary points.



[12.40] Geometric proof that Bôcher's formula is equivalent to Poisson's half-plane formula. To establish that (12.37) is equivalent to (12.35), we must find $(d\theta^*/dx) \asymp (\Delta\theta^*/\Delta x)$. The first step in evaluating this is to recognize that even if Δx is not small (as we shall ultimately need it to be), the non-Euclidean angle $\Delta\theta^*$ subtended at a by Δx is exactly double the Euclidean one, $\Delta\theta$. To see this, concentrate on the semicircle meeting the axis at p . The angle between the dotted tangent at a and the vertical is clearly double that between the chord ap and the vertical. The result then follows immediately.



[12.41] Conclusion of the geometric proof that Bôcher's formula is equivalent to Poisson's. The small shaded triangle is constructed to be right-angled, and it is thus ultimately similar to the large shaded triangle as $\Delta\theta \rightarrow 0$. Thus $(\xi/\Delta x) \asymp (Y/\Omega)$. Also, since ξ is ultimately an arc of a circle of radius Ω , it is ultimately equal to $\Omega \Delta\theta$. Thus as $\Delta\theta \rightarrow 0$, $\frac{\Omega \Delta\theta}{\Delta x} \asymp \frac{\xi}{\Delta x} \asymp \frac{Y}{\Omega}$. We can now combine this with the result of the previous figure to obtain $\frac{d\theta^*}{dx} = 2 \left[\frac{d\theta}{dx} \right] = 2 \left[\frac{Y}{\Omega^2} \right] = 2 \left[\frac{Y}{(x-X)^2 + Y^2} \right]$. Inserting this into (12.37), we do indeed obtain (12.35).

Now consider [12.41]. The small shaded triangle is constructed to be right-angled, and it is thus ultimately similar to the large shaded triangle as $\Delta\theta \rightarrow 0$. Thus $(\xi/\Delta x)$ is ultimately equal to (Y/Ω) . Also, since ξ is ultimately an arc of a circle of radius Ω , it is ultimately equal to $\Omega \Delta\theta$. Thus as $\Delta\theta \rightarrow 0$,

$$\frac{\Omega \Delta \theta}{\Delta x} \asymp \frac{\xi}{\Delta x} \asymp \frac{Y}{\Omega}.$$

We can now combine this with the result of the previous figure to obtain

$$\frac{d\theta^*}{dx} = 2 \left[\frac{d\theta}{dx} \right] = 2 \left[\frac{Y}{\Omega^2} \right] = 2 \left[\frac{Y}{(X-x)^2 + Y^2} \right].$$

Inserting this into (12.37), we obtain (12.35).

While the precise form of the above argument may be new, the basic idea of transferring Bôcher's result from the disc to the half plane was given by Osgood (1928). For a different approach to (12.35), see Lange and Walsh (1985). For more on all three of the interpretations thus far obtained, see Perkins (1928).

12.7.5 Green's General Formula

If R is a simply connected region of *arbitrary* shape there exists a generalization of Poisson's formula (due to Green) for finding the temperature at any point a inside R in terms of the values $T(z)$ on the boundary B . As before, let $\mathcal{G}_a(z)$ be the Green's function of the region when the heat source is placed at the point a , so that local heat flux at the boundary point z is given by

$$\mathcal{Q}_a(z) = -\frac{\partial \mathcal{G}_a}{\partial n}.$$

With the aid of $\mathcal{Q}_a$ we may now determine $T(a)$. Here is the remarkable *Green's formula*:

$$T(a) = \frac{1}{2\pi} \oint_B \mathcal{Q}_a(z) T(z) ds, \quad (12.38)$$

where ds is an element of arc length along B . Thus $\mathcal{Q}_a$ now plays the same role as the Poisson kernel did in (12.29). Indeed, we previously calculated $\mathcal{Q}_a$ for the unit disc, and we now recognize the result (12.27) as the Poisson kernel $\mathcal{P}_a$.

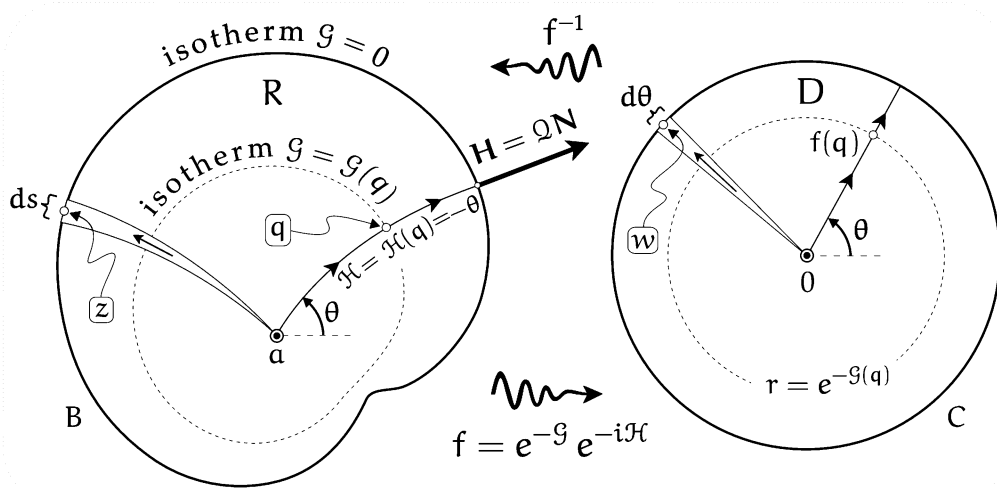
Although formula (12.38) is valuable both in theory and practice, we should point out that it is less explicit than Poisson's formula, for to find $\mathcal{Q}_a$ we must first find the Green's function. But as we previously explained, the problem of finding $\mathcal{G}_a$ is *itself* a Dirichlet problem: to construct

$$\mathcal{G}_a(z) = -\ln \rho + g_a(z),$$

we must find the harmonic function g_a with boundary values $g_a(z) = \ln \rho$. Formula (12.38) says that if we can just solve this particular boundary value problem then we can solve them all.

To begin with, imagine that $T(z)$ is a given harmonic function in R whose value $T(a)$ at an interior a we wish to determine from the boundary values. The idea behind our explanation of (12.38) is very simple¹². The Green's function $\mathcal{G}_a$ enables

¹² For a beautiful *physical* explanation of (12.38) in terms of electrostatic energy, see Maxwell (1873) or, better still, Maxwell (1881, Ch. III).



[12.42] Geometric proof of Green's formula. If we know the Green's function of R , then, as previously explained, we can use it to conformally map R to the disc D : $z \mapsto w = f(z) = e^{-g} e^{-i\mathcal{H}}$, and $f(a) = 0$. The harmonic function $T(z)$ in R may be transplanted to the harmonic function $\tilde{T}(w) \equiv T[f^{-1}(w)]$ in D , the boundary values on B becoming the boundary values on C . But the average of these boundary temperatures on C is the temperature at the centre, and this is precisely what we want, because $\tilde{T}(0) = T(a)$. Therefore, $T(a) = \frac{1}{2\pi} \oint_C \tilde{T}(w) d\theta = \frac{1}{2\pi} \oint_B T(z) |f'(z)| ds$. Finally, recall the result (12.26): the amplification $|f'(z)|$ equals the local heat flux $\mathcal{Q}_a(z)$. This concludes the derivation of Green's formula, (12.38).

us to construct a conformal mapping f (and an inverse mapping f^{-1}) between R and the unit disc D such that a corresponds to 0 . Figure [12.42] essentially reproduces [12.31] in which this was explained.

Using the mapping $z \mapsto w = f(z)$, the harmonic function $T(z)$ in R may be transplanted to the harmonic function $\tilde{T}(w) \equiv T[f^{-1}(w)]$ in D , the boundary values on B becoming the boundary values on C . But the average of these boundary temperatures on C is the temperature at the centre, and this is precisely what we were after, because $\tilde{T}(0) = T(a)$.

Expressing this idea symbolically, we have

$$T(a) = \frac{1}{2\pi} \oint_C \tilde{T}(w) d\theta.$$

If the element $d\theta$ of C at $w = f(z)$ is the image of the element ds of B at z then

$$d\theta = |f'(z)| ds,$$

so

$$T(a) = \frac{1}{2\pi} \oint_B T(z) |f'(z)| ds.$$

Finally, recall the result (12.26): the amplification $|f'(z)|$ equals the local heat flux $\mathcal{Q}_a(z)$. This concludes the derivation of formula (12.38).

This argument also explains the stronger result that (12.38) solves Dirichlet's problem for R . Using f to conformally transplant the given boundary values from B to C , we know that Poisson's formula allows us to construct the solution to Dirichlet's problem in D . Transferring this solution back from D to R with f^{-1} , we have found the harmonic function T in R , and its value at a must then be given by (12.38). You can now understand why we lavished so much attention on the special case of the disc.

We end this section with the observation that Green's formula (12.38) possesses the beautiful geometric interpretation shown on the LHS of [12.43]. Just as in [12.38], one imagines standing at a and turning one's head successively through the small angle $\bullet$. But now *suppose that light travels along the illustrated streamlines of the heat flow* $H = -\nabla \mathcal{G}_a$ *associated with the Green's function*. We would then see the thermometers at the illustrated points on the boundary. The general formula (12.38) says that the average of these temperatures (as $\bullet \rightarrow 0$) is the temperature where one stands! Bôcher's interpretation is clearly just a special case¹³.

The explanation essentially reiterates the derivation of (12.26). Let z_θ be the boundary point we see when we look in the direction θ . Green's formula says that the temperature $T(z_\theta)$ of the element ds contributes to the temperature at a in proportion to $\mathcal{Q}_a(z_\theta) ds$, which is the flux of H through ds . Now follow the shaded flux tube back to the source at a , and let $d\theta$ be its angular width there. Since 2π of flux emerges symmetrically from a , the flux $\mathcal{Q}_a(z_\theta) ds$ emitted into our tube is equal to $d\theta$. Thus (12.38) may be re-expressed as

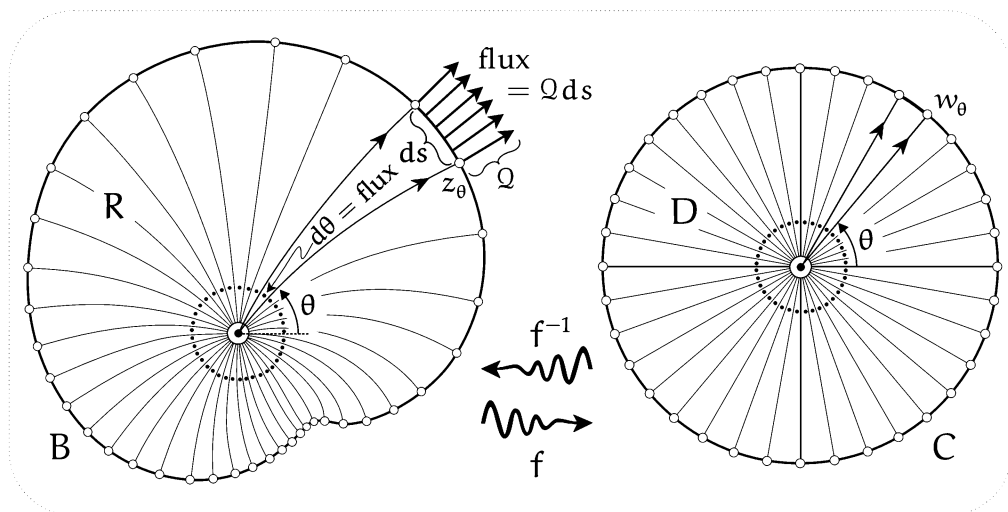
$$T(a) = \frac{1}{2\pi} \oint_B T(z_\theta) d\theta, \quad (12.39)$$

namely, as the average of the boundary temperatures $T(z_\theta)$ over all directions. Done.

We have presented [12.43] as a geometric interpretation of (12.38), but we may instead use it to simplify and illuminate our derivation of that formula. The key observation is that (even without passing to the limit of vanishing $\bullet$) *the average in [12.43] of the observed temperatures on B is conformally invariant*. As before, let $J(z)$ be a one-to-one conformal mapping of R to some other simply connected region S with boundary Y . Just as we did with f , let us choose J so that the directions of curves through a are preserved (i.e., $\arg [J'(a)] = 0$). Let $w_\theta \equiv J(z_\theta)$ be the image on Y of z_θ on B .

By the conformal invariance of the Green's function, the image of the streamline leaving a at angle θ is the streamline leaving $J(a)$ at the same angle. Thus w_θ is not

¹³ If we define the distance between two infinitesimally separated points of R to be the hyperbolic distance between their images in D , then R becomes a (non-standard) conformal model of the hyperbolic plane, and the geodesics emanating from a are the streamlines of [12.43]. The two results may then be viewed as identical.



[12.43] The geometric meaning of Green's formula. Green's formula (12.38) possesses the beautiful geometric interpretation shown on the left. Just as in [12.38], one imagines standing at a and turning one's head successively through the small angle $\bullet$. But now suppose that light travels along the illustrated streamlines of the heat flow $H = -\nabla \mathcal{G}_a$ associated with the Green's function. We would then see the thermometers at the illustrated points on the boundary. The general formula (12.38) says that the average of these temperatures (as $\bullet \rightarrow 0$) is the temperature where one stands! Bôcher's interpretation is clearly just a special case. Indeed, if we define the distance between two infinitesimally separated points of R to be the hyperbolic distance between their images in D , then R becomes a (non-standard) conformal model of the hyperbolic plane, and the geodesics emanating from a are the streamlines. The two results may then be viewed as identical.

only the image of z_θ , it is also the boundary point which an observer at $J(a)$ sees when he looks in the direction θ . But, by definition, the temperature at each point z_θ on B is transplanted to w_θ on Y , so the observer at $J(a)$ sees exactly the same temperatures on Y as the original observer at a saw on B . Done.

Passing to the limit of vanishing $\bullet$, the conformal invariance of this average may be expressed as

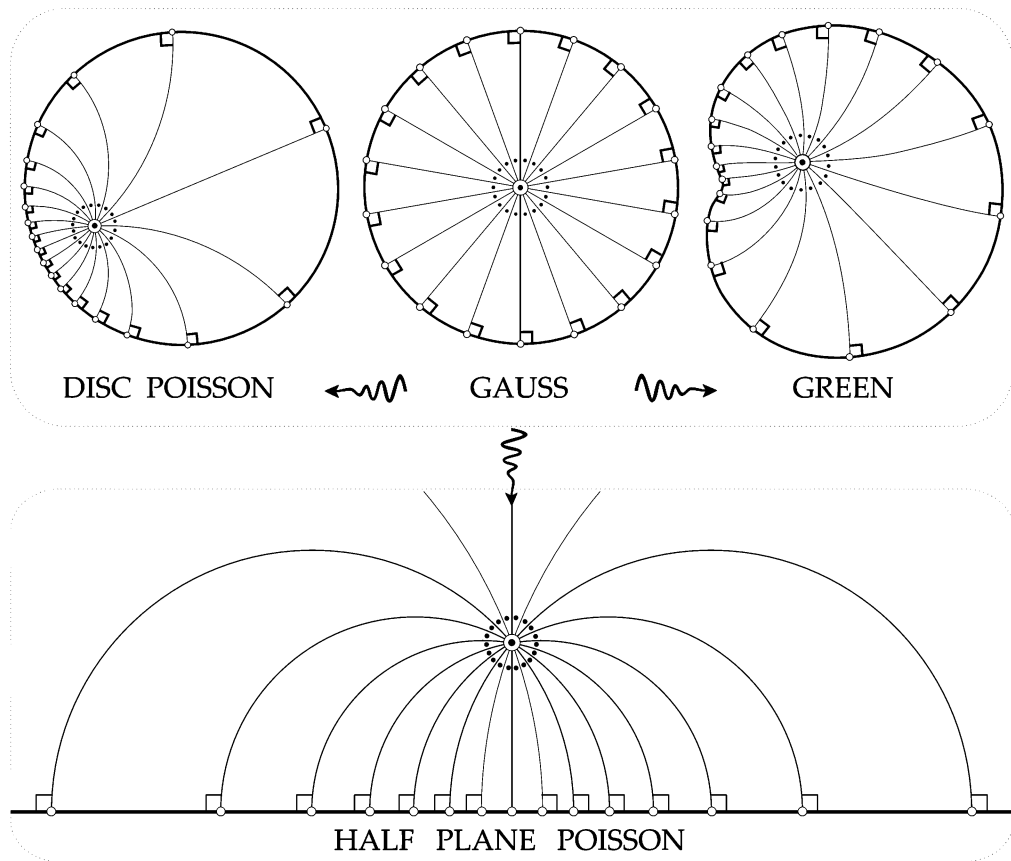
$$\frac{1}{2\pi} \oint_B T(z_\theta) d\theta = \frac{1}{2\pi} \oint_Y \tilde{T}(w_\theta) d\theta.$$

Figure [12.43] illustrates the particular case where $J = f$ is the previously constructed function which maps R to D and a to 0 . The virtue of this special case is that the conformally invariant average may now be *evaluated*. By Gauss's Mean Value Theorem, the average of the transplanted temperatures $\tilde{T}(w_\theta) \equiv T(z_\theta)$ on C is the temperature $\tilde{T}(0) \equiv T(a)$ at the centre:

$$\frac{1}{2\pi} \oint_B T(z_\theta) d\theta = \frac{1}{2\pi} \oint_C \tilde{T}(w_\theta) d\theta = T(a).$$

Thus, returning to R and passing to the limit of vanishing $\bullet$, the average of the observed temperatures is the temperature where you stand. Finally, the argument associated with [12.43] shows that (12.39) is equivalent to (12.38).

As illustrated in [12.44], this idea of a conformally invariant hyperbolic average lends unity to much of what we have done. Top centre is a depiction of Gauss's



[12.44] Hyperbolic geometry unifies all methods of solving Dirichlet's Problem. For the Beltrami–Poincarites who inhabit the hyperbolic plane, the conformally invariant average that Gauss calls for at the centre of the disc [see top centre] is utterly indistinguishable from the Poisson disc-formula [top left] at an arbitrary point. In both cases, the Beltrami–Poincarite turns her head successively through the small angle $\bullet$, and both figures show the locations of the thermometers she sees on the boundary as light travels along hyperbolic straight lines (geodesics) to her eye. The average of their temperatures is then a good approximation [exact as $\bullet \rightarrow 0$] to the temperature where she stands. But she could also not detect any change if we transplanted her and the boundary temperatures to the Beltrami–Poincaré upper half-plane: the temperature would still be the average over all directions. Finally, if we conformally transplant both the boundary temperatures and the hyperbolic metric to a general region R [top right], then it becomes a (non-standard) conformal model of the hyperbolic plane. All four figures may then be viewed as hyperbolically indistinguishable.

theorem: as $\bullet \rightarrow 0$ the average of the temperatures at the white boundary points is equal to the temperature at the centre of the disc. Applying an automorphism to this picture yields the visual form of Poisson's formula for the disc; applying a Möbius transformation from the Beltrami–Poincaré disc model to the half-plane model yields the visual form of Poisson's formula for the half-plane; and applying a more general conformal mapping yields the visual form of Green's formula.

12.8 Exercises

- 1 (i) Show that the dual of a dipole is a dipole.
 (ii) Think of a dipole as the limiting form of a doublet [a source and sink of equal strength]. Sketch the dual of a doublet, and thereby make geometric sense of part (i).
- 2 If Φ is a real harmonic function, and $\widehat{\Phi}$ is its harmonic dual, show that $\Phi\widehat{\Phi}$ is also harmonic. [Hint: Consider the complex function $(\Phi + i\widehat{\Phi})$.]
- 3 Use Gauss's Mean Value Theorem to show that a harmonic function cannot have a local maximum.
- 4 Find the generalization of (12.7) in three dimensional space.
- 5 (i) If f is analytic, show that $\nabla|f| = (f\overline{f'})/|f|$, and explain how this agrees with part (i) of Ex. 19, p. 425.
 (ii) Show that if Δ is the Laplacian, then $\Delta|f|^2 = 4|f'|^2$. Try deriving this by brute force.
- 6 Let $f(z)$ be analytic. By applying $\overline{\nabla}$ in turn to $(f + \overline{f})$ and to $(f\overline{f})$, show that if either the real part or the modulus of f is constant, then f itself is constant.
- 7 (i) By thinking of z and $\overline{z}$ as functions of x and y , use the chain rule of partial differentiation to show (at least formally) that

$$\partial_z = \frac{1}{2} \overline{\nabla} \quad \text{and} \quad \partial_{\overline{z}} = \frac{1}{2} \nabla.$$

- (ii) Deduce (at least formally) that *an analytic function f depends on z but not on $\overline{z}$:*

$$\partial_z f = f' \quad \text{and} \quad \partial_{\overline{z}} f = 0.$$

- 8 Let J denote the Jacobian of a transformation $z \mapsto w$. Referring to the previous question, show that the determinant is given by $\det(J) = |\partial_z w|^2 - |\partial_{\overline{z}} w|^2$.
- 9 From Ex. 7(i) we see that $\Delta = \overline{\nabla}\nabla = 4\partial_z\partial_{\overline{z}}$. Use this fact to solve the following problems:
 - (i) Show that $\Phi = [1 - (x^2 + y^2)]^{-1}$ satisfies $\Delta\Phi = 4\Phi^2(2\Phi - 1)$.
 - (ii) Solve $\Delta F = e^z$ for F by formally integrating with respect to z and then with respect to $\overline{z}$. Deduce that $R = \frac{1}{4}e^x(x\cos y + y\sin y)$ is a solution of $\Delta R = e^x\cos y$. Verify this by calculating ΔR explicitly.
 - (iii) Show that if f is the most general harmonic function in the plane, then $f(z, \overline{z}) = p(z) + q(\overline{z})$, where p and q are arbitrary analytic functions.

- 10 As usual, let $\widehat{\xi}$ be the unit tangent to a curve of curvature κ , and let $\widetilde{\kappa}$ be the curvature of the image under the analytic mapping $f(z)$. Also let s and $\widetilde{s}$ denote arc length of the original and image curves. Finally, let $\Psi = (1/|f'|)$ be the stream function of the complex curvature, $\mathcal{K} = -i\nabla\Psi$.

- (i) Use (5.31), p. 272 to show that $\partial_{\widetilde{s}} \widetilde{\kappa} = \Psi \partial_s [\kappa\Psi + \mathcal{K} \cdot \widehat{\xi}]$.
- (ii) Show that $\partial_s \widehat{\xi} = i\kappa\widehat{\xi}$ and $\partial_s \Psi = \widehat{\xi} \cdot (i\mathcal{K})$.
- (iii) Deduce that

$$\partial_{\widetilde{s}} \widetilde{\kappa} = \Psi^2 \partial_s \kappa + \Psi (\partial_s \mathcal{K}) \cdot \widehat{\xi}.$$

[Hint: Remember (or prove) that $(i\mathbf{a}) \cdot (\mathbf{b}) + (\mathbf{a}) \cdot (i\mathbf{b}) = 0$.]

- (iv) Recall Ex. 18, p. 296, in which we saw how Newton attempted to define the “angle” Θ between two touching curves as the difference of their curvatures: $\Theta \equiv (\kappa_1 - \kappa_2)$. Although this is not quite conformally invariant, show that $[\Theta^2/\partial_s \Theta]$ is conformally invariant. This geometrically meaningful generalization of the concept of angle is called *Kasner’s Invariant*.
- 11 With the same notation as in the previous exercise, let p measure distance in the direction $i\widehat{\xi}$ perpendicular to the curve. By substituting $\mathcal{K} = -i\nabla\Psi$ into (5.31), p. 272, show that the image curvature is given by the tidy formula,

$$\widetilde{\kappa} = \partial_p \Psi + \kappa \Psi.$$

- 12 Consider the image under an analytic mapping f of a source of strength S located at p .
- (i) Show geometrically, then algebraically, that if p is not a critical point of f (i.e., $f'(p) \neq 0$) then the image is another source of strength S at $f(p)$.
 - (ii) Show geometrically, then algebraically, that if p is a critical point of f of order $(n-1)$ then the image is again a source, but now of strength (S/n) .
- 13 Repeat the investigation of the previous question in the case of a dipole located at p .
- 14 Show that applying the Milne-Thomson formula (12.18) to a vortex at a point p inside the unit circle yields *two* new vortices: at $1/\overline{p}$ and at 0. Explain this with the aid of a picture of the flow on the Riemann sphere.
- 15 (i) Show that if u and v are harmonic then $X \equiv (u\nabla v - v\nabla u)$ is sourceless.
- (ii) Prove (12.38) by taking $u = T$ and $v = \mathcal{G}_a$, then equating the flux of X out of B with the flux out of an infinitesimal circle centred at a .
 - (iii) By taking $u = \mathcal{G}_a$ and $v = \mathcal{G}_b$, prove the symmetry property of the Green’s function: $\mathcal{G}_a(b) = \mathcal{G}_b(a)$.

- 16 Let $T(z)$ be the temperature distribution in the unit disc if the top semicircle ($\text{Im}(z) > 0$) is kept at temperature $+(\pi/2)$ and the bottom semicircle is kept at temperature $-(\pi/2)$. Show that

$$T(z) = \text{Arg} \left[\frac{1+z}{1-z} \right].$$

- 17 Let $T(z)$ be a *non-negative* temperature distribution in the disc $|z| \leq R$. Writing $|a| = r$, use Poisson's formula (12.29) to derive *Harnack's inequality*:

$$\left(\frac{R-r}{R+r} \right) T(0) \leq T(a) \leq \left(\frac{R+r}{R-r} \right) T(0).$$

- 18 Use Harnack's inequality [previous exercise] to prove the following analogue of Liouville's Theorem: *If T is harmonic in the whole plane and is bounded from above (or below), then T is a constant.*
- 19 Substitute the disc Green's function (12.22) into (12.25), thereby confirming the formula (12.27) for the heat flux at the boundary of the disc.
- 20 (i) Use the method of images to find the Green's function for the upper half-plane.
 (ii) Use this to show that Green's general formula (12.38) does indeed yield the Poisson half-plane formula (12.35).
- 21 Use the idea behind the method of images to show that if $0 < \text{Re}(p) < 1$ then the Green's function of the half-disc $\text{Re}(z) \geq 0, |z| \leq 1$ is

$$\mathcal{G}_p(z) = -\ln \left| \frac{z-p}{\bar{p}z-1} \right| + \ln \left| \frac{z+\bar{p}}{pz+1} \right|.$$

Check this by getting the computer to draw the level curves of $\mathcal{G}_p(z)$.

